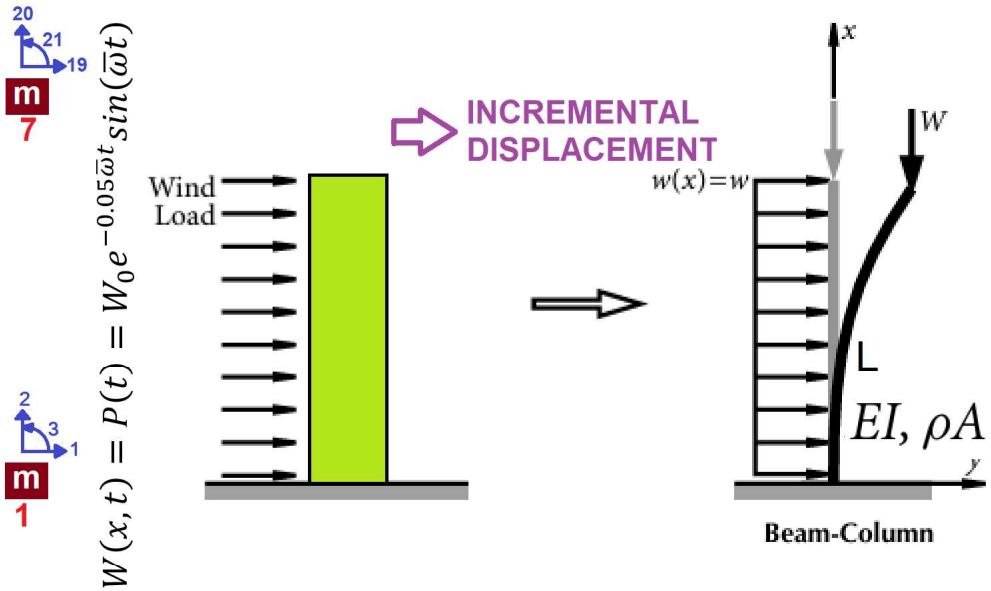
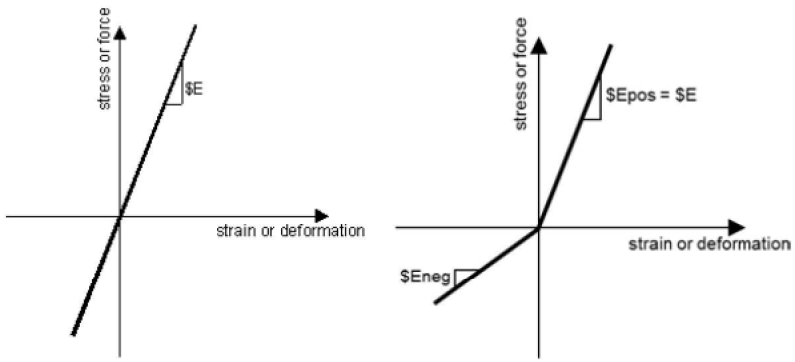
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE COLUMN: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

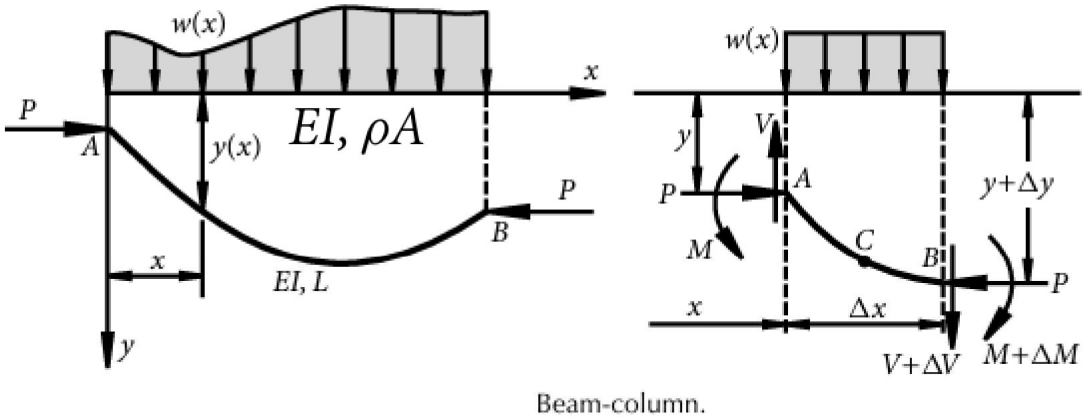
THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



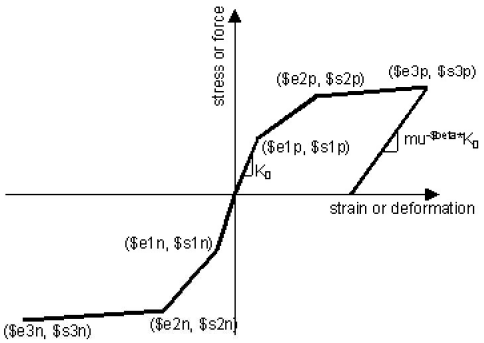


ELEMENT ELASTIC MATERIAL

$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$



Beam-column.



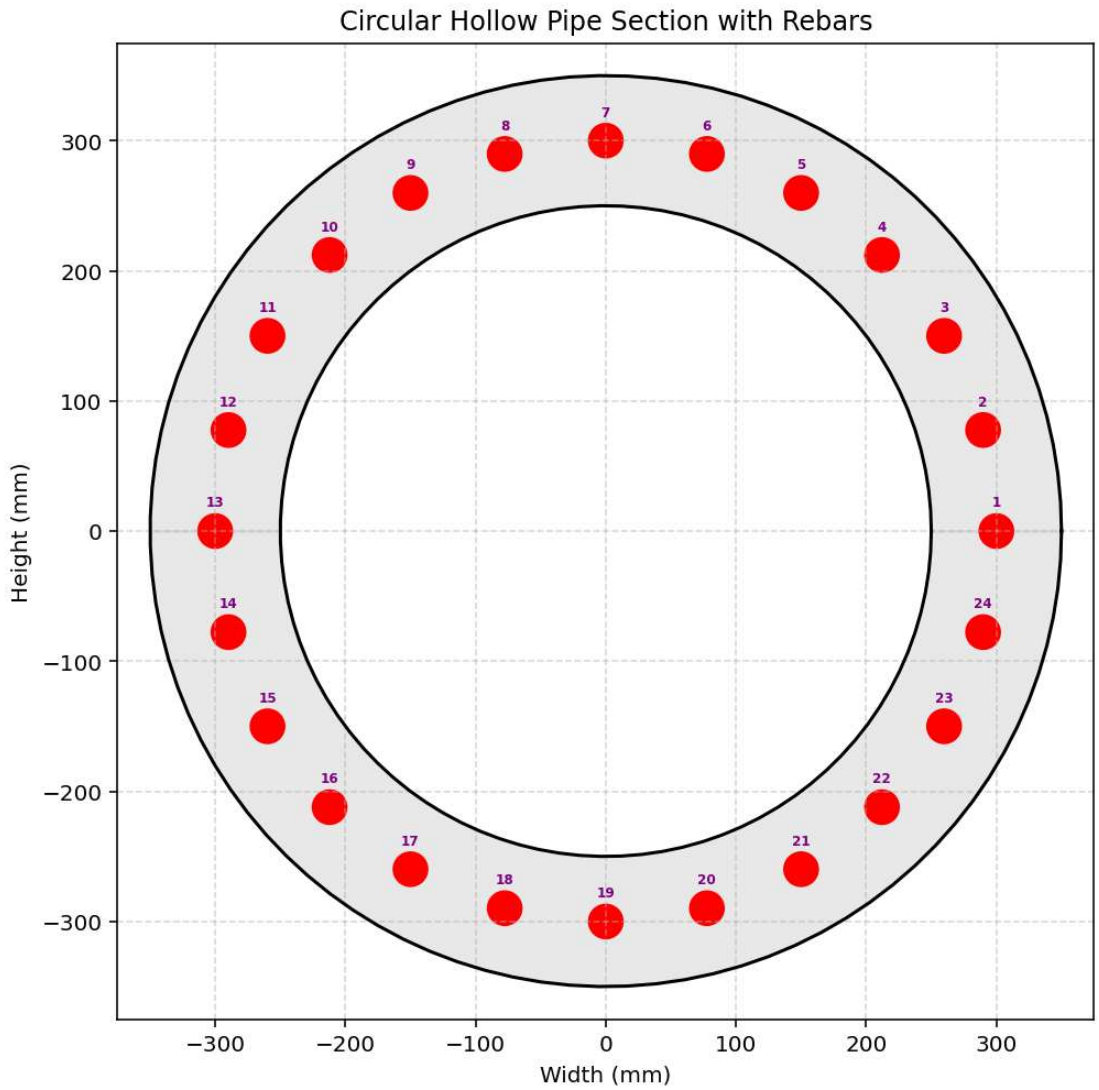
ELEMENT INELASTIC MATERIAL

$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\rho A \frac{\partial^2 v}{\partial t^2} + EI \frac{\partial^4 v}{\partial x^4} = w(x,t).$$


Circular Hollow Pipe Section with Rebars
Height (mm)
Width (mm)
300
200
100
0
−100
−200
−300
−300
−200
−100
0
100
200
300
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES

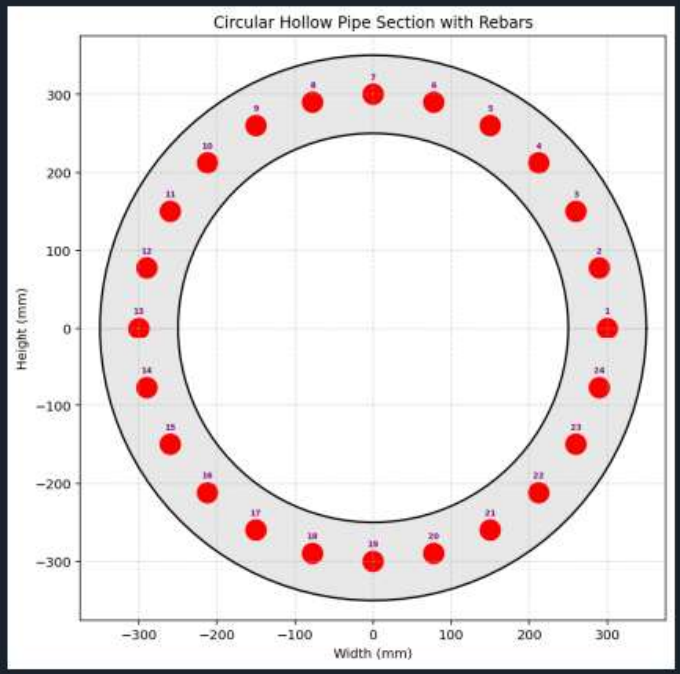
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CONCRETE_CIRCULALR_HOLLOW_PIPE_SECTION_FUN_TWO.py

W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

```python
def CONCRETE_CIRCULAR_HOLLOW_PIPE_SECTION_FUN(
    secTag, STEEL_TYPE, Do, t,
    numSubdivCirc, numSubdivRad,
    fc, Kfc,
    CONCRETE_DENSITY,
    nRebars=68,
    rebar_dia=25,
    plot=True
    ):
    """
    Create a circular hollow confined-concrete fiber section for OpenSees,
    with automatic placement of reinforcing bars along the mid-thickness circle.

    Parameters
    ----------
    secTag          : int   - section identifier
    STEEL_TYPE      : str   - 'ELASTIC' or 'INELASTIC'
    Do              : float - outer diameter (mm)
    t               : float - wall thickness (mm)
    numSubdivCirc   : int   - number of circumferential subdivisions for concrete patch
    numSubdivRad    : int   - number of radial subdivisions for concrete patch
    fc              : float - unconfined concrete compressive strength (MPa, negative)
    Kfc             : float - ratio of confined to unconfined concrete strength
    CONCRETE_DENSITY: float - density of concrete (kg/m³) - converted to N·s²/mm³ internally
    plot            : bool  - if True, draw a 2D sketch of the section
    nRebars         : int   - total number of reinforcing bars (default = 68)
    rebar_dia       : float - diameter of each rebar (mm), uniform for all bars (default = 25)

    Returns
    -------
    Do              : float - outer diameter (mm)
    ELE_MASS        : float - element mass per unit length (kg/mm)
```



IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1, Col 1 | UTF-8 | CRLF | RW | Mem 49%

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES

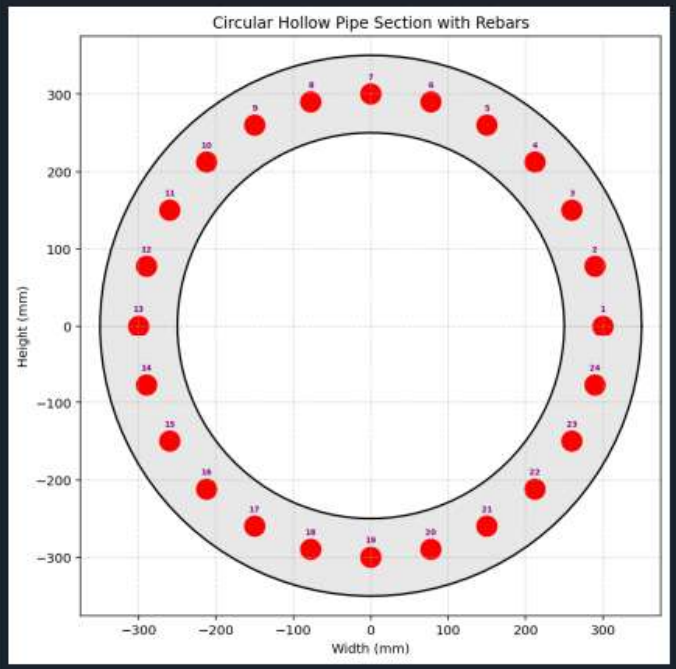
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

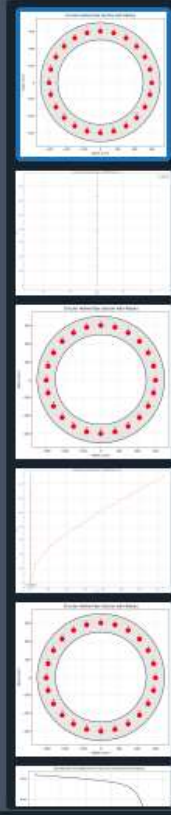
W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

```
#############################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#            COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A CONCRETE COLUMN : AN OPENSEES
#FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HIST
#---------------------------------------------------------------------------------------------
#                EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#---------------------------------------------------------------------------------------------
#          ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacemen
#---------------------------------------------------------------------------------------------
#                                          W(x,t) = W0 sin(wt)
#                                     W(x,t) = W0 exp(-0.05wt) sin(wt)
#---------------------------------------------------------------------------------------------
#         ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
#                                     OF ENERGY DISSIPATION CAPACITY INDEX
#---------------------------------------------------------------------------------------------
#                     THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                EMAIL: salar.d.ghashghaei@gmail.com
#############################################################################################
"""
Nonlinear Seismic Performance Assessment of a concrete column:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear

This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a concrete
 for performance-based earthquake engineering. The 2D model incorporates both material nonlin
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity to capture P-delta effects
 and large displacements—essential for collapse assessment.

Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
 and plastic mechanism identification
```





42 %

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STATIC ANALYSIS

Spyder (Python 3.12)

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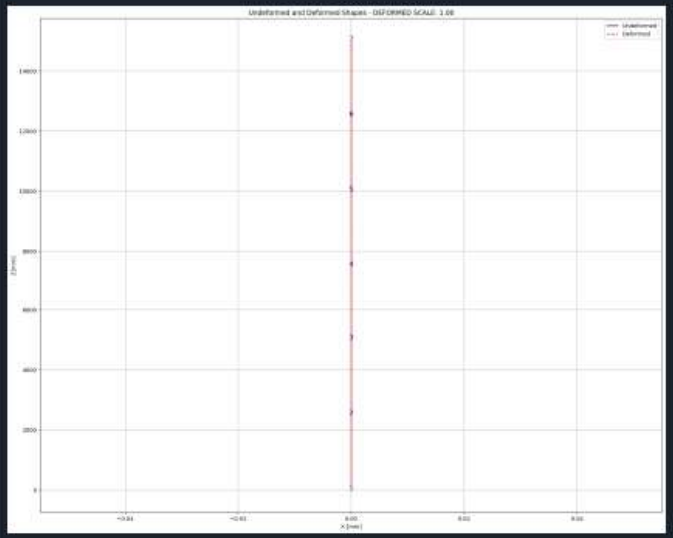
...NERGY_DISSIPATION_CAPACITY_INDEX\14_CONCRETE_COLUMN_8_ANA

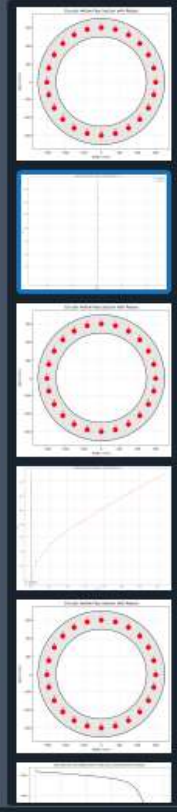
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

```
386
387     if ANAL_TYPE == 'STATIC':
388         ops.integrator('LoadControl', 1.0)
389         ops.analysis('Static')
390         OK = ops.analyze(1)
391         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
392         ops.reactions()
393         reactionX.append(ops.nodeReaction(1, 1))                    # SHEAR BASE REACTION
394         reactionY.append(ops.nodeReaction(1, 2))                    # AXIAL BASE REACTION
395         reactionZ.append(ops.nodeReaction(1, 3))                    # MOMENT BASE REACTIO
396         disp_mid.append(ops.nodeDisp(center_node, 1))               # DISPLACEMENT NODE 0
397         dispX.append(ops.nodeDisp(center_node, 1))                  # DISPLACEMENT NODE 0
398         dispY.append(ops.nodeDisp(center_node, 2))                  # DISPLACEMENT NODE 0
399         # Store axial forces, strain and stress
400         for ele_id in ele_axialforce.keys():
401             ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'force')[0])       # [N] E
402             ele_shearforce[ele_id].append(ops.eleResponse(ele_id, 'force')[1])       # [N] E
403             ele_momentforce[ele_id].append(ops.eleResponse(ele_id, 'force')[2])      # [N] E
404         # Store displacements
405         for node_id in node_displacementsX.keys():
406             node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
407             node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
408         else:
409             print('\n\nSTATIC ANALYSIS DONE.\n\n')
410
411         DATA = (reactionX, reactionY, reactionZ, disp_mid,
412                 ele_axialforce, ele_shearforce, ele_momentforce,
413                 node_displacementsX, node_displacementsY,
414                 dispX, dispY)
415
416         return  DATA
417
418     if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
419         IDctrlDOF = 1   # 1: Horizontal Displacement ; 2: Vertical Displacement
```

18 %





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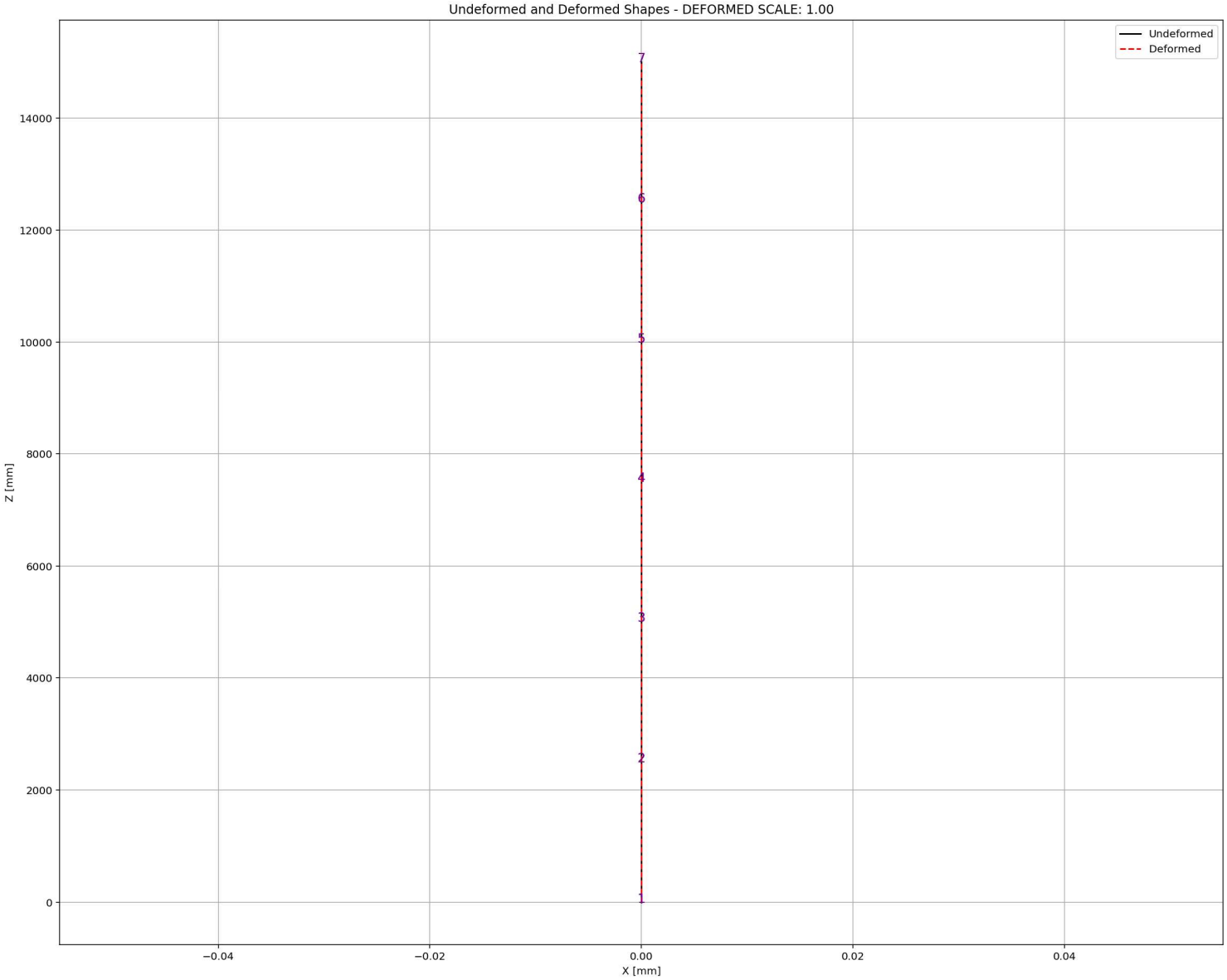
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
7
6
5
4
3
2
1
14000
12000
10000
8000
6000
4000
2000
0
Z [mm]
−0.04
−0.02
0.00
0.02
0.04
X [mm]

PUSHOVER ANALYSIS

Spyder (Python 3.12)

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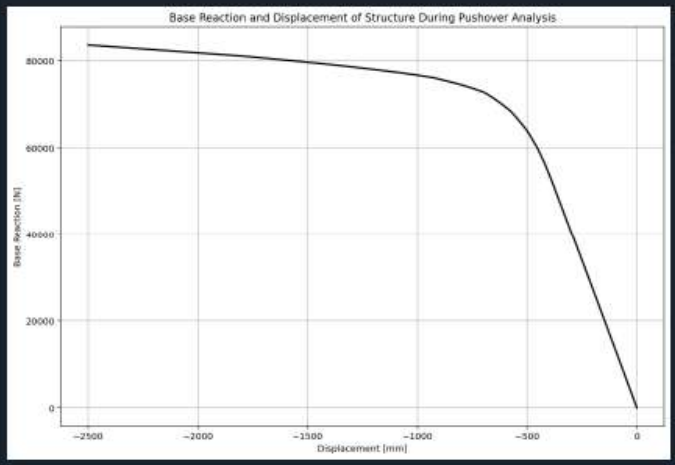
..NERGY_DISSIPATION_CAPACITY_INDEX\14_CONCRETE_COLUMN_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

```
418     if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
419         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
420         DINCR = -0.1    # [mm] Incremental Displacement
421         DMAX = -2500.0  # [mm] Max. Displacement
422         ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
423         ops.analysis('Static')
424         Nsteps =  int(np.abs(DMAX / DINCR))
425         STEP = 0.0
426         for step in range(Nsteps):
427             OK = ops.analyze(1)
428             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
429             ops.reactions()
430             reactionX.append(ops.nodeReaction(1, 1))                    # SHEAR BASE REAC
431             reactionY.append(ops.nodeReaction(1, 2))                    # AXIAL BASE REAC
432             reactionZ.append(ops.nodeReaction(1, 3))                    # MOMENT BASE REA
433             disp_mid.append(ops.nodeDisp(center_node, 1))               # DISPLACEMENT NO
434             dispX.append(ops.nodeDisp(center_node, 1))                  # DISPLACEMENT NO
435             dispY.append(ops.nodeDisp(center_node, 2))                  # DISPLACEMENT NO
436             # Store elements force, strain and stress
437             for ele_id in ele_axialforce.keys():
438                 ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'force')[0])      # [
439                 ele_shearforce[ele_id].append(ops.eleResponse(ele_id, 'force')[1])      # [
440                 ele_momentforce[ele_id].append(ops.eleResponse(ele_id, 'force')[2])     # [
441             # Store displacements
442             for node_id in node_displacementsX.keys():
443                 node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
444                 node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
445             # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
446             #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(3, 0.5*DR, DR, 0, 1)
447             PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(3, PLOT=True)
448             PERIOD_MIN.append(PERIODmin)
449             PERIOD_MAX.append(PERIODmax)
450             if ELE_TYPE == 'nonlinearBeamColumn':
451                 # SECTION STRESS-STRAIN FOR BOTTOM FIBER
```



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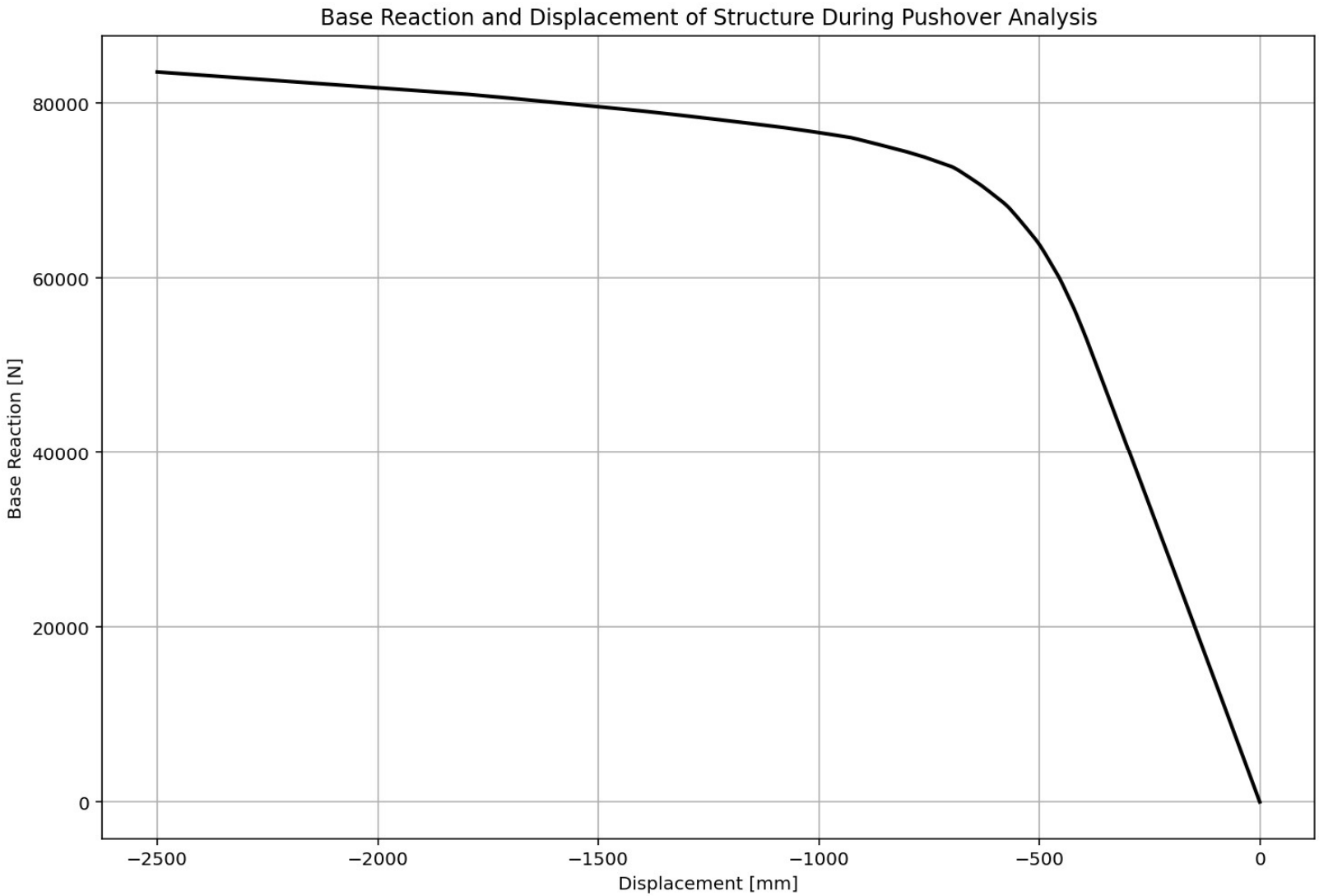
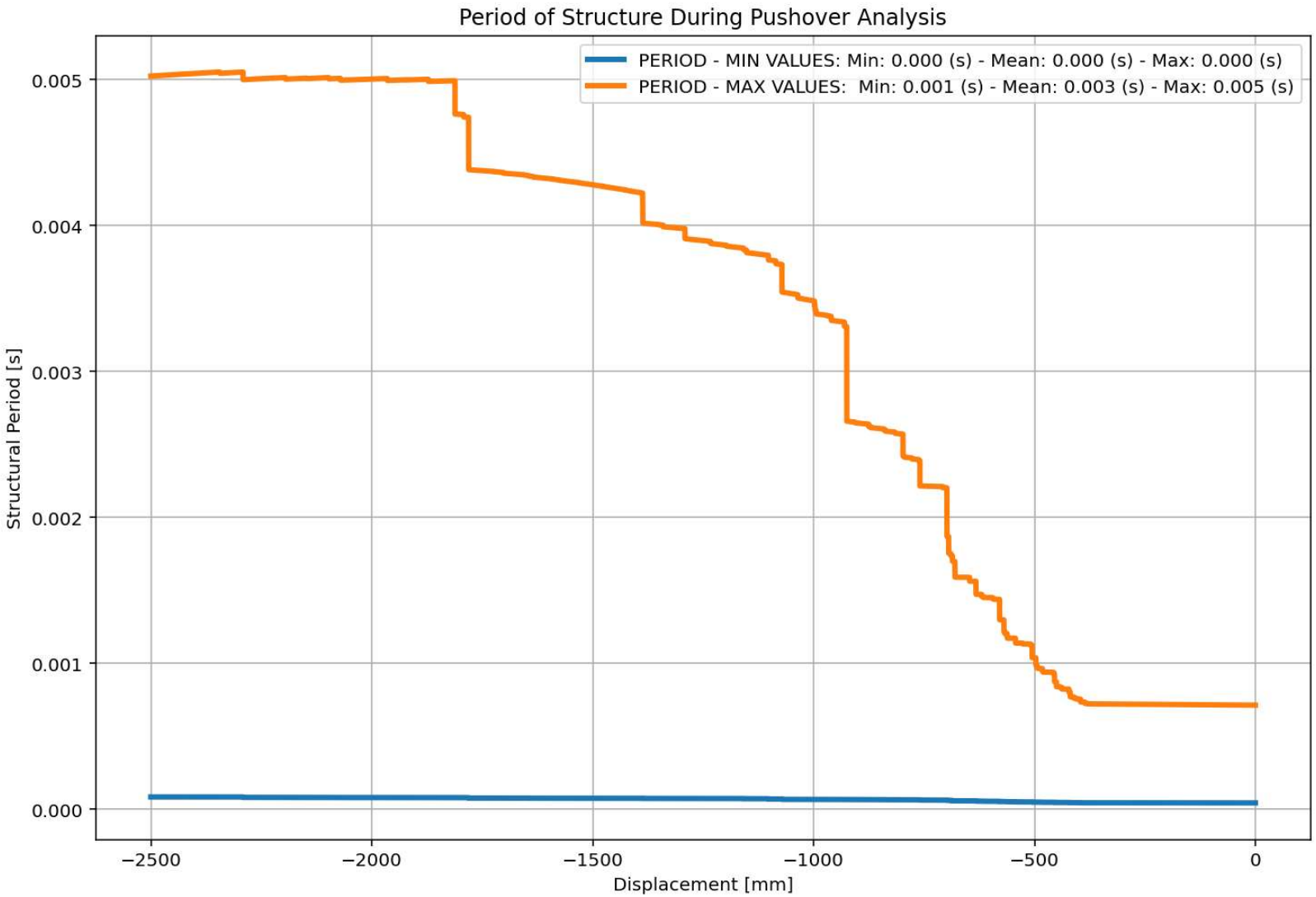
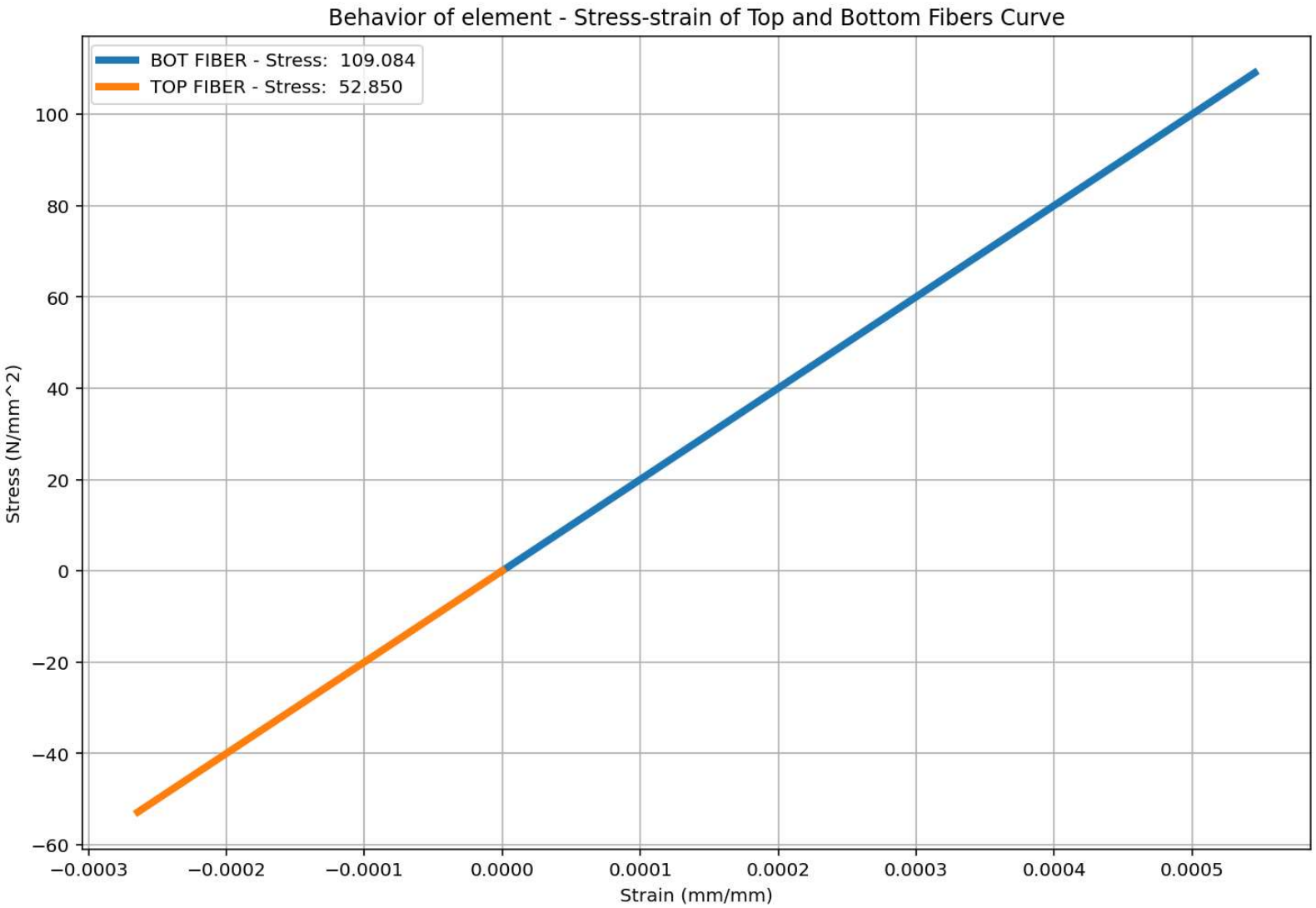
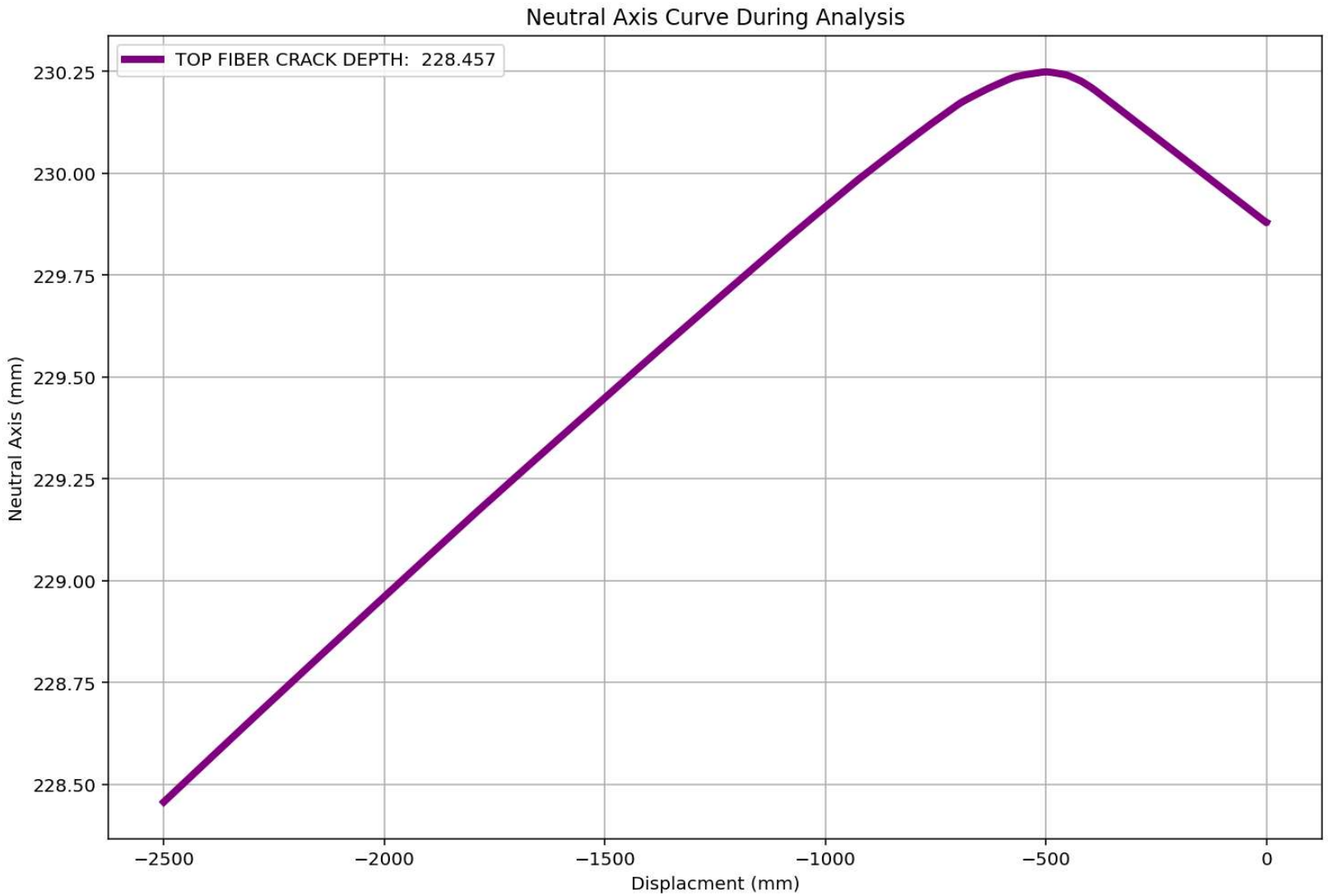
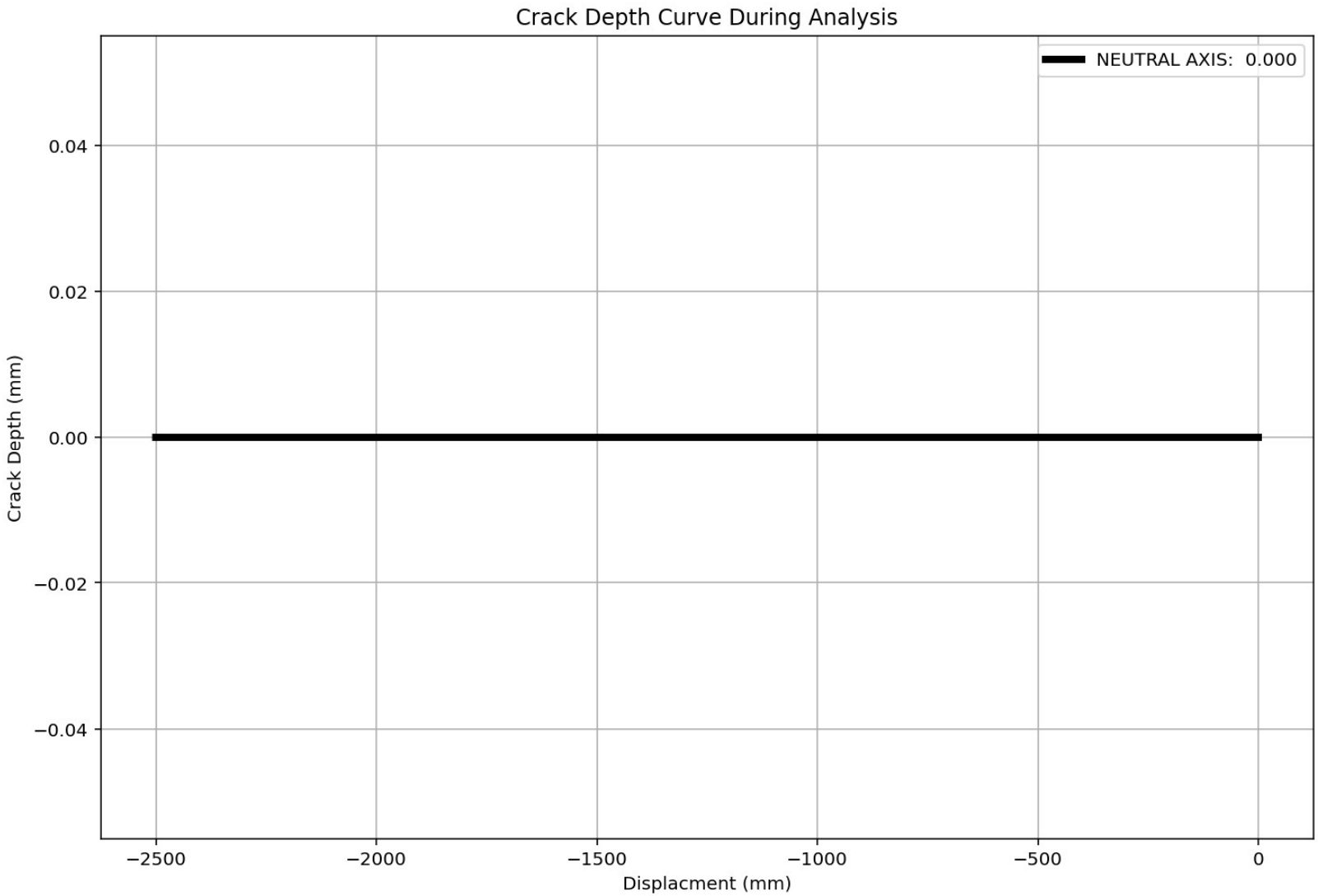
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 112, Col 32 | UTF-8 | CRLF | RW | Mem 54%

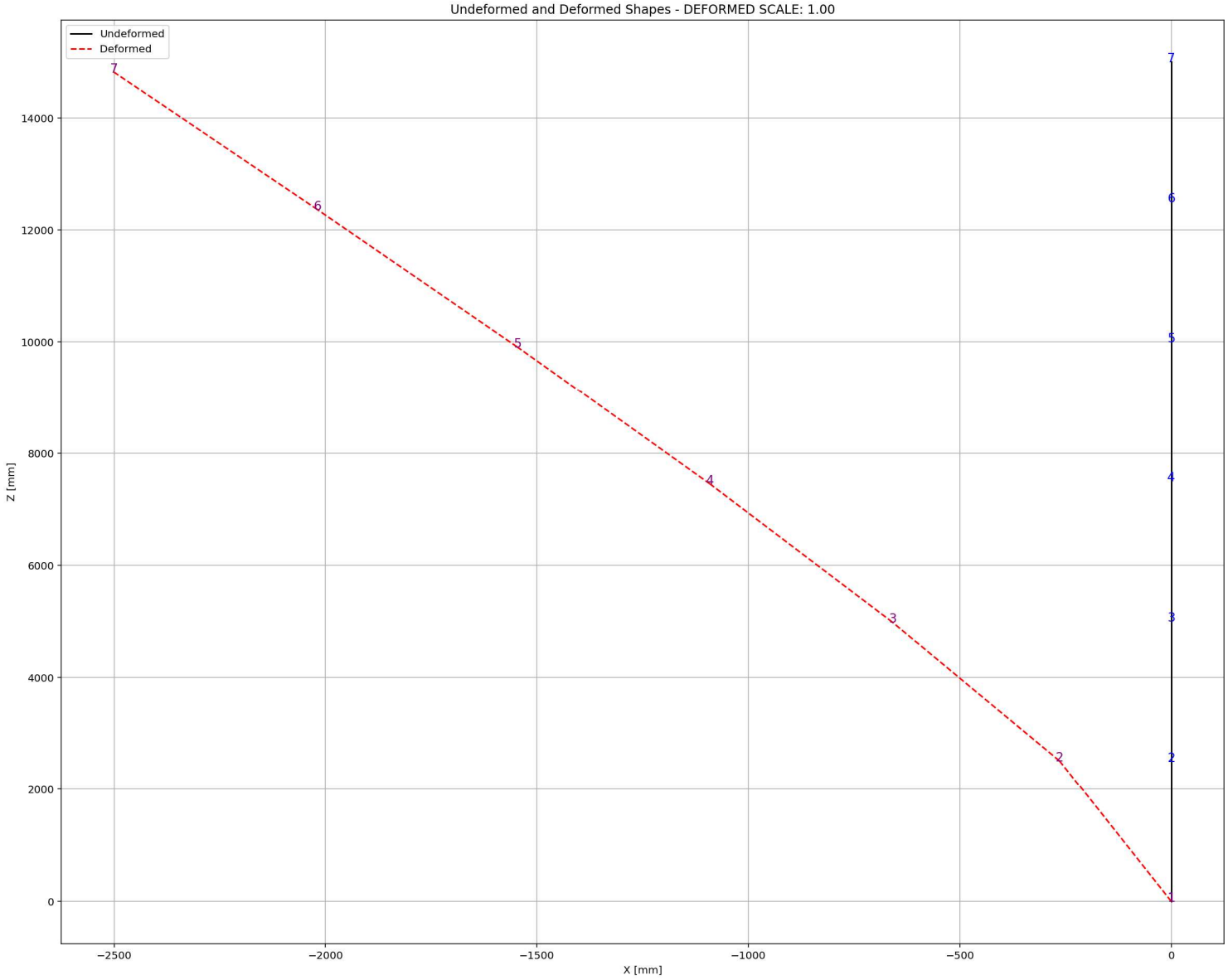

Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
80000
60000
40000
20000
0
−2500
−2000
−1500
−1000
−500
0
Displacement [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: 0.000 (s) - Mean: 0.000 (s) - Max: 0.000 (s)
PERIOD - MAX VALUES: Min: 0.001 (s) - Mean: 0.003 (s) - Max: 0.005 (s)
Structural Period [s]
0.005
0.004
0.003
0.002
0.001
0.000
−2500
−2000
−1500
−1000
−500
0
Displacement [mm]


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 109.084
TOP FIBER - Stress: 52.850
Stress (N/mm^2)
Strain (mm/mm)
100
80
60
40
20
0
−20
−40
−60
−0.0003
−0.0002
−0.0001
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005


Neutral Axis Curve During Analysis
TOP FIBER CRACK DEPTH: 228.457
230.25
230.00
229.75
229.50
229.25
229.00
228.75
228.50
Neutral Axis (mm)
−2500
−2000
−1500
−1000
−500
0
Displacment (mm)


Crack Depth Curve During Analysis
NEUTRAL AXIS: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−2500
−2000
−1500
−1000
−500
0
Displacment (mm)


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
14000
12000
10000
8000
6000
4000
2000
0
−2500
−2000
−1500
−1000
−500
0
7
6
5
4
3
2
1

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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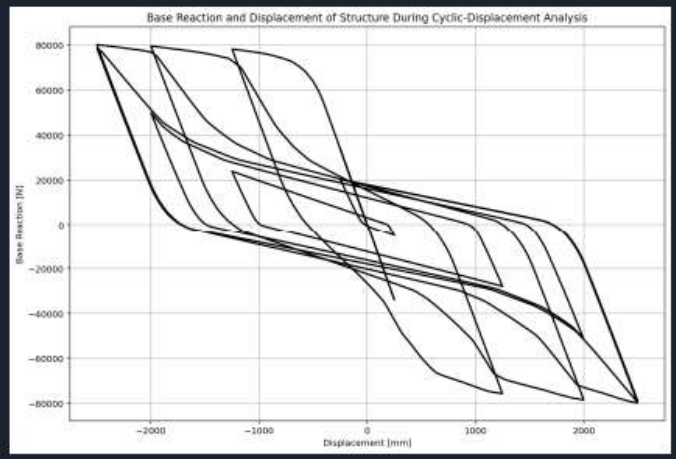
W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

```
491
492     if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
493         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
494         DMAX = -2500.0  # [mm] Max. Displacement
495         n_points = 10000
496         # 4. CYCLIC DISPALCEMENT PROTOCOL
497         # Key strain points (same logic as your protocol)
498         key_disp = np.array([
499             0.0,
500             0.1*DMAX,   -0.1*DMAX,
501             0.5*DMAX,   -0.5*DMAX,
502             0.8*DMAX,   -0.8*DMAX,
503             DMAX,       -DMAX,
504             0.1*DMAX,   -0.1*DMAX,
505             0.5*DMAX,   -0.5*DMAX,
506             0.8*DMAX,   -0.8*DMAX,
507             DMAX,       -DMAX,
508             0.0
509         ])
510
511         # Generate 1000-point displacement protocol
512         disp_protocol = np.interp(
513             np.linspace(0, len(key_disp) - 1, n_points),
514             np.arange(len(key_disp)),
515             key_disp
516         )
517
518         ops.analysis('Static')
519         STEP = 0.0
520         for target_disp in disp_protocol:
521             current_disp = ops.nodeDisp(center_node, 1) # DISPALCEMENT APPLIED IN NODE IN X D
522             dU = target_disp - current_disp
523             ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
524             OK = ops.analyze(1)
```



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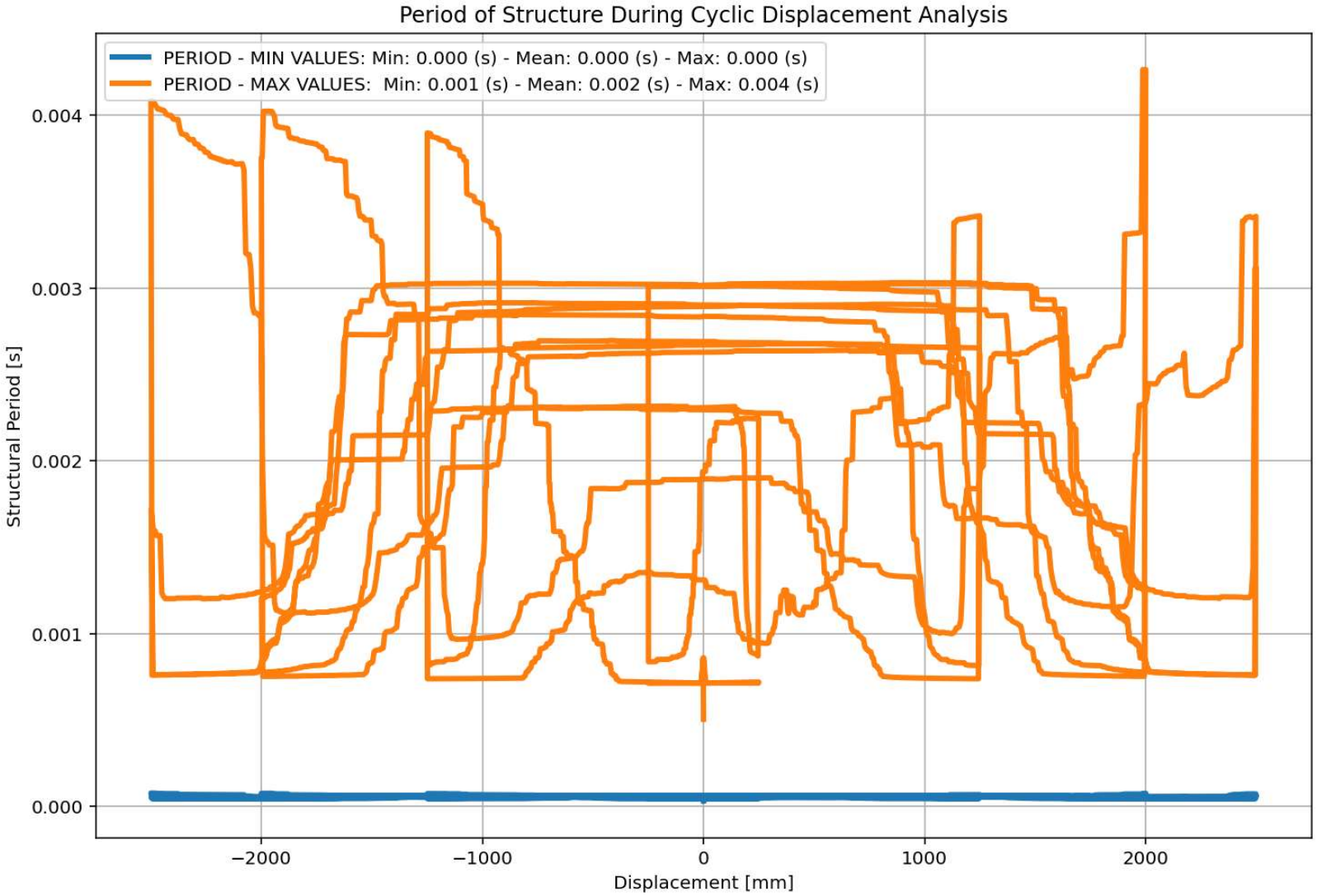
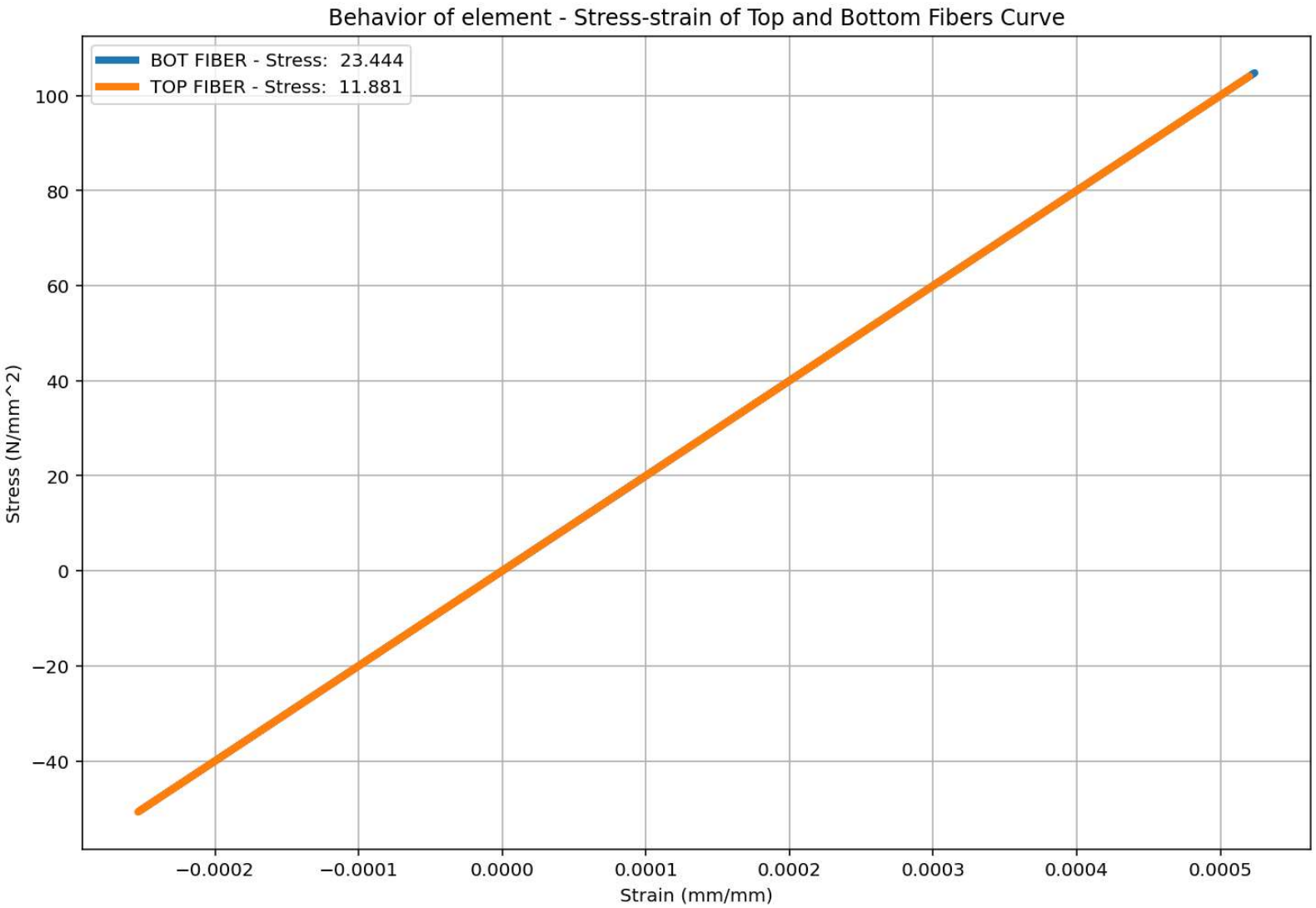
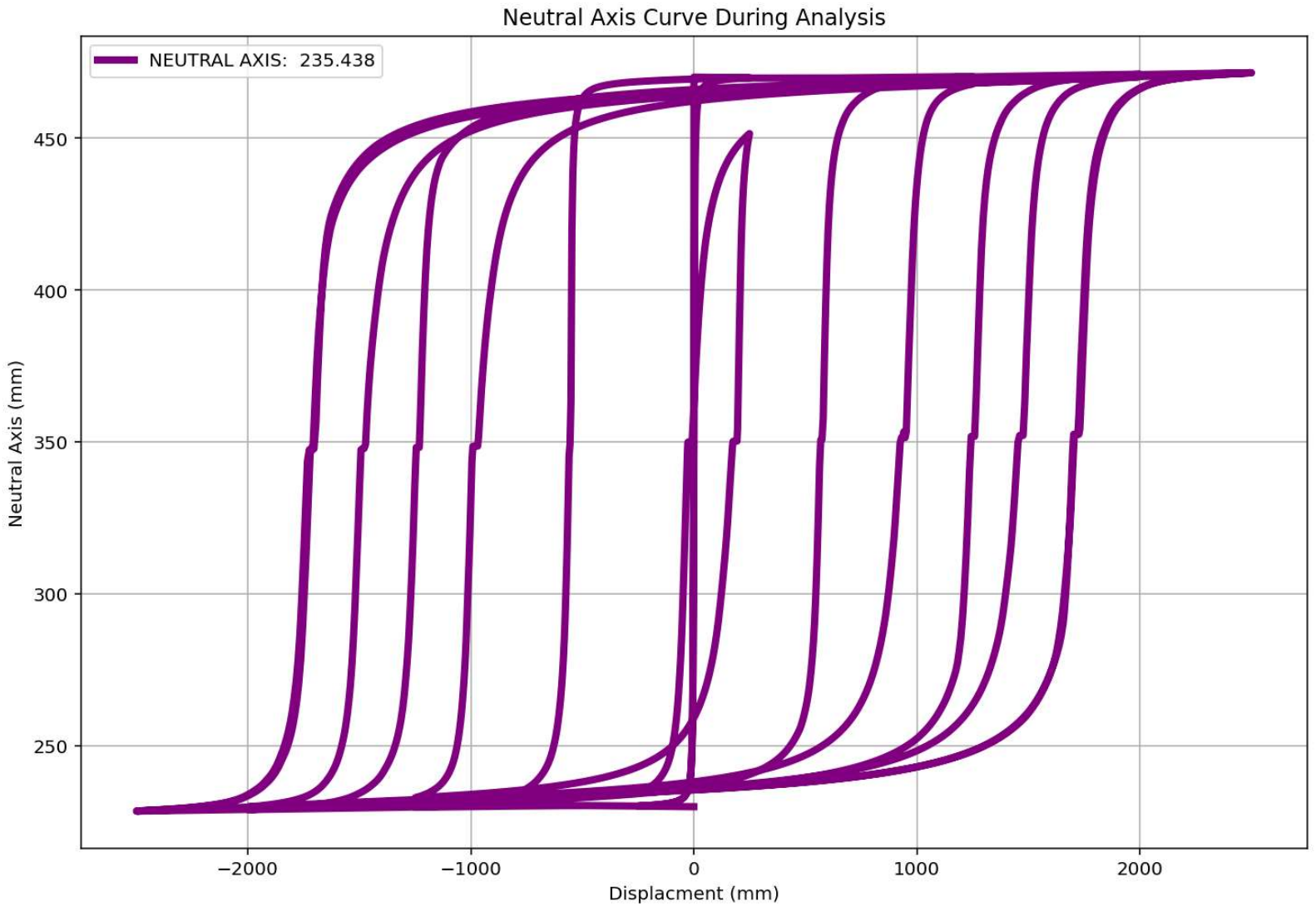
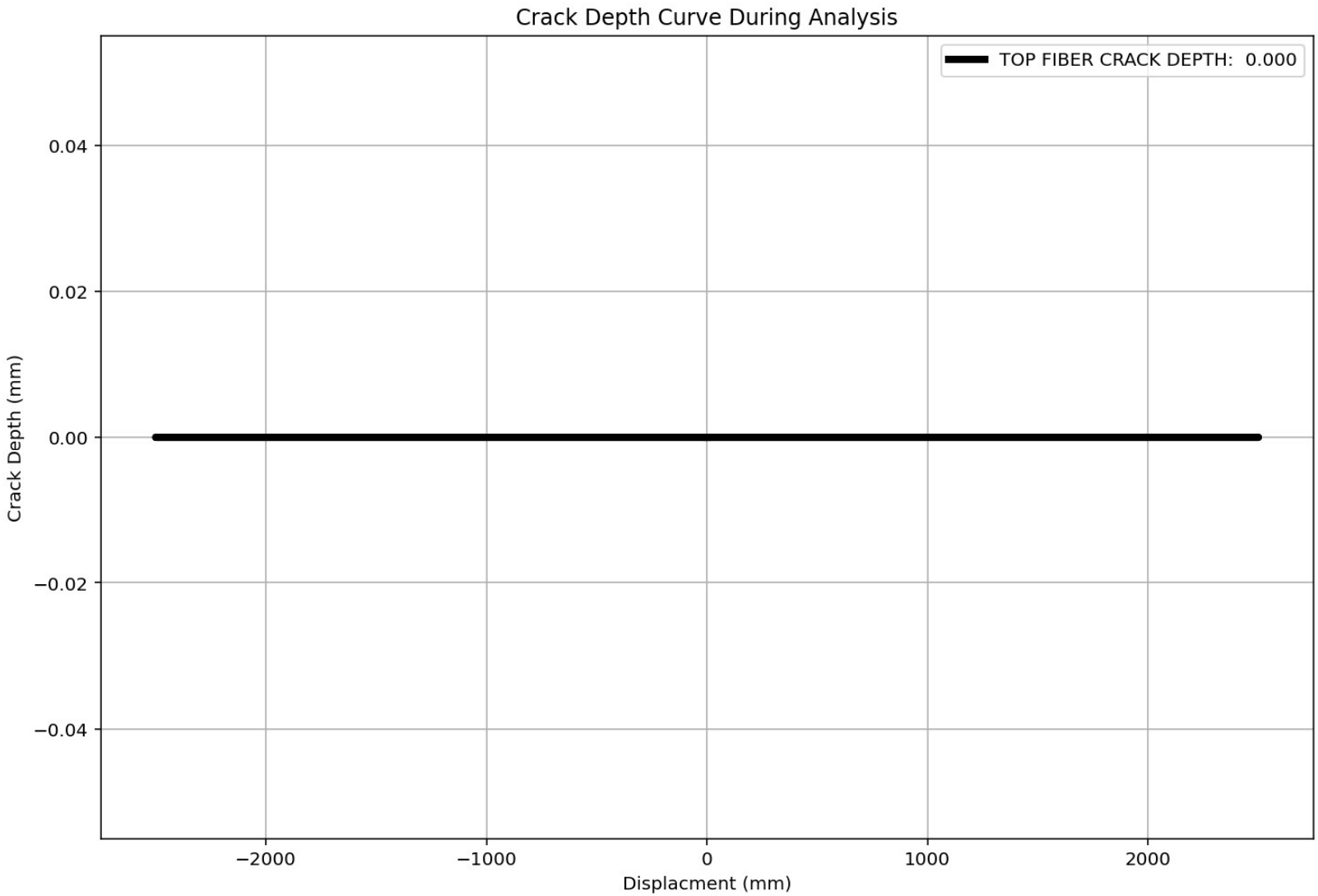
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 112, Col 32 | UTF-8 | CRLF | RW | Mem 55%

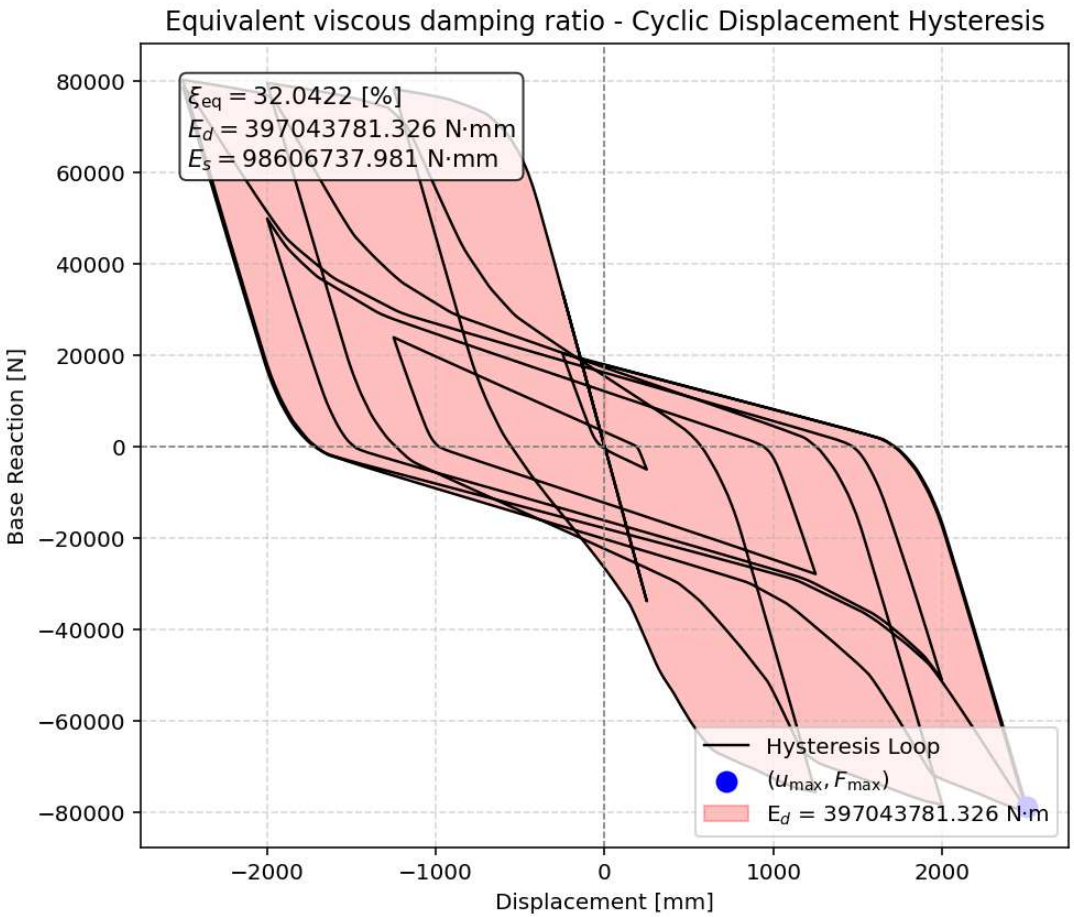

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
80000
60000
40000
20000
0
−20000
−40000
−60000
−80000
−2000
−1000
0
1000
2000
Displacement [mm]

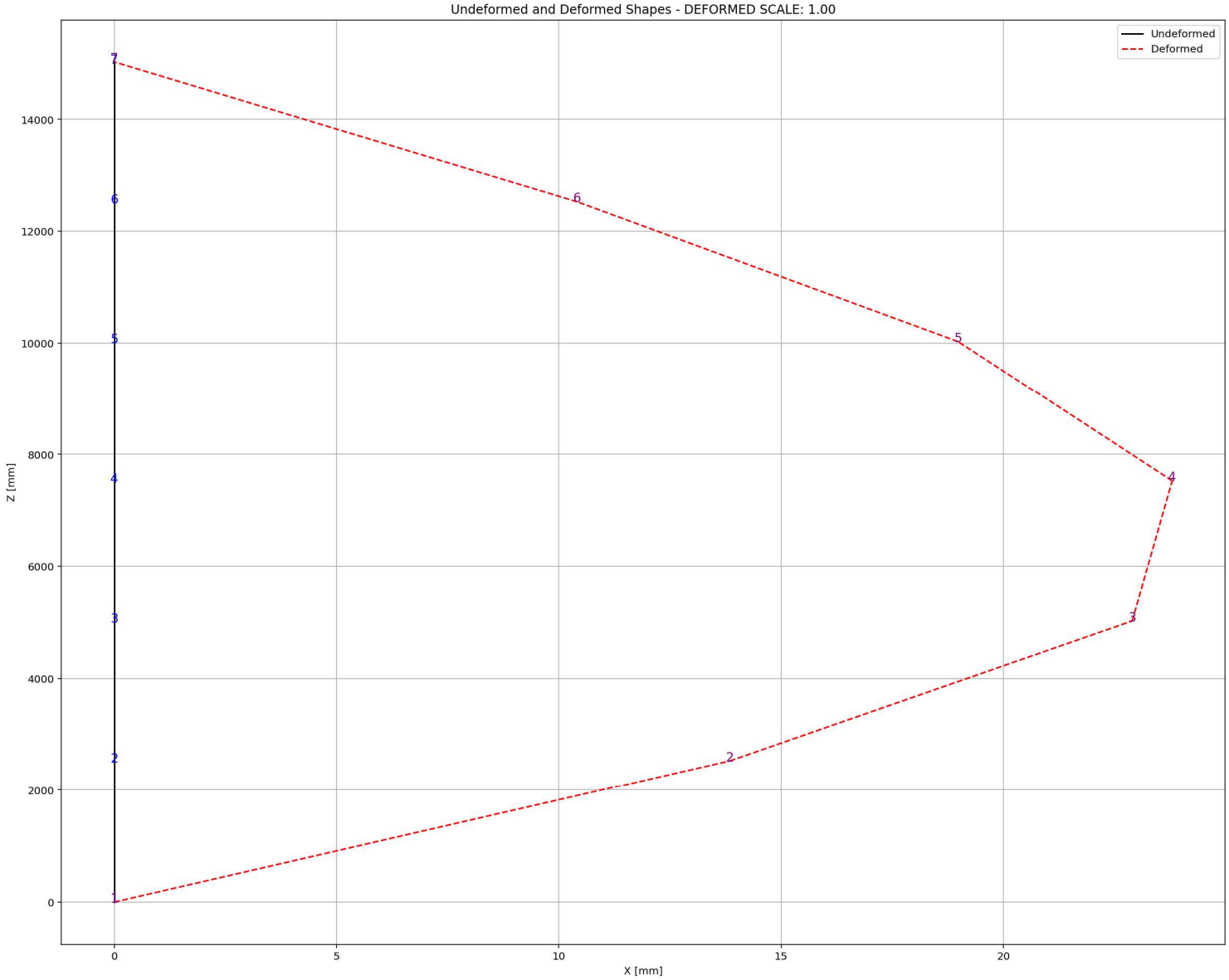

Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.000 (s) - Mean: 0.000 (s) - Max: 0.000 (s)
PERIOD - MAX VALUES: Min: 0.001 (s) - Mean: 0.002 (s) - Max: 0.004 (s)
Structural Period [s]
0.004
0.003
0.002
0.001
0.000
−2000
−1000
0
1000
2000
Displacement [mm]


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 23.444
TOP FIBER - Stress: 11.881
Stress (N/mm^2)
100
80
60
40
20
0
−20
−40
−0.0002
−0.0001
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
Strain (mm/mm)


Neutral Axis Curve During Analysis
NEUTRAL AXIS: 235.438
Neutral Axis (mm)
450
400
350
300
250
−2000
−1000
0
1000
2000
Displacment (mm)


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−2000
−1000
0
1000
2000
Displacment (mm)


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
ξ_{eq} = 32.0422 [%]
E_d = 397043781.326 N·mm
E_s = 98606737.981 N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 397043781.326 N·m
Base Reaction [N]
Displacement [mm]
80000
60000
40000
20000
0
−20000
−40000
−60000
−80000
−2000
−1000
0
1000
2000


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
2000
4000
6000
8000
10000
12000
14000
0
5
10
15
20
1
2
3
4
5
6
7

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

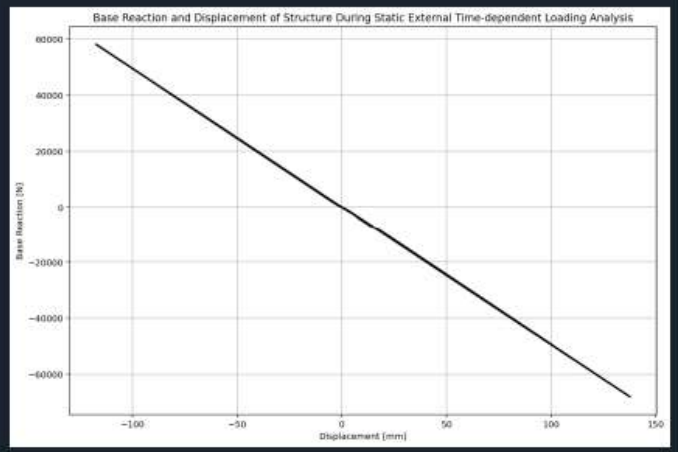
File Edit Search Source Run Debug Consoles Projects Tools View Help

...NERGY_DISSIPATION_CAPACITY_INDEX\14_CONCRETE_COLUMN_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

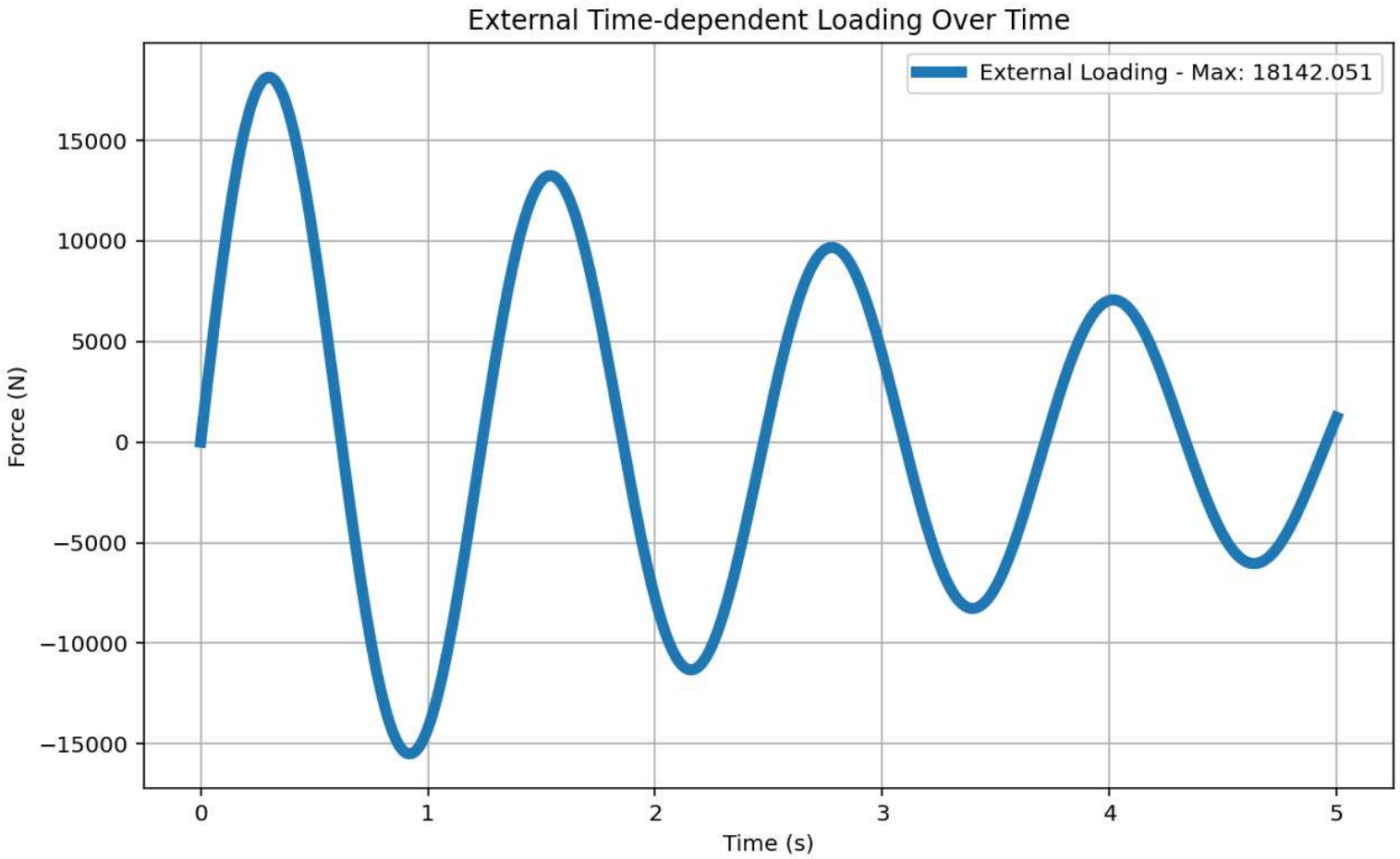
W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

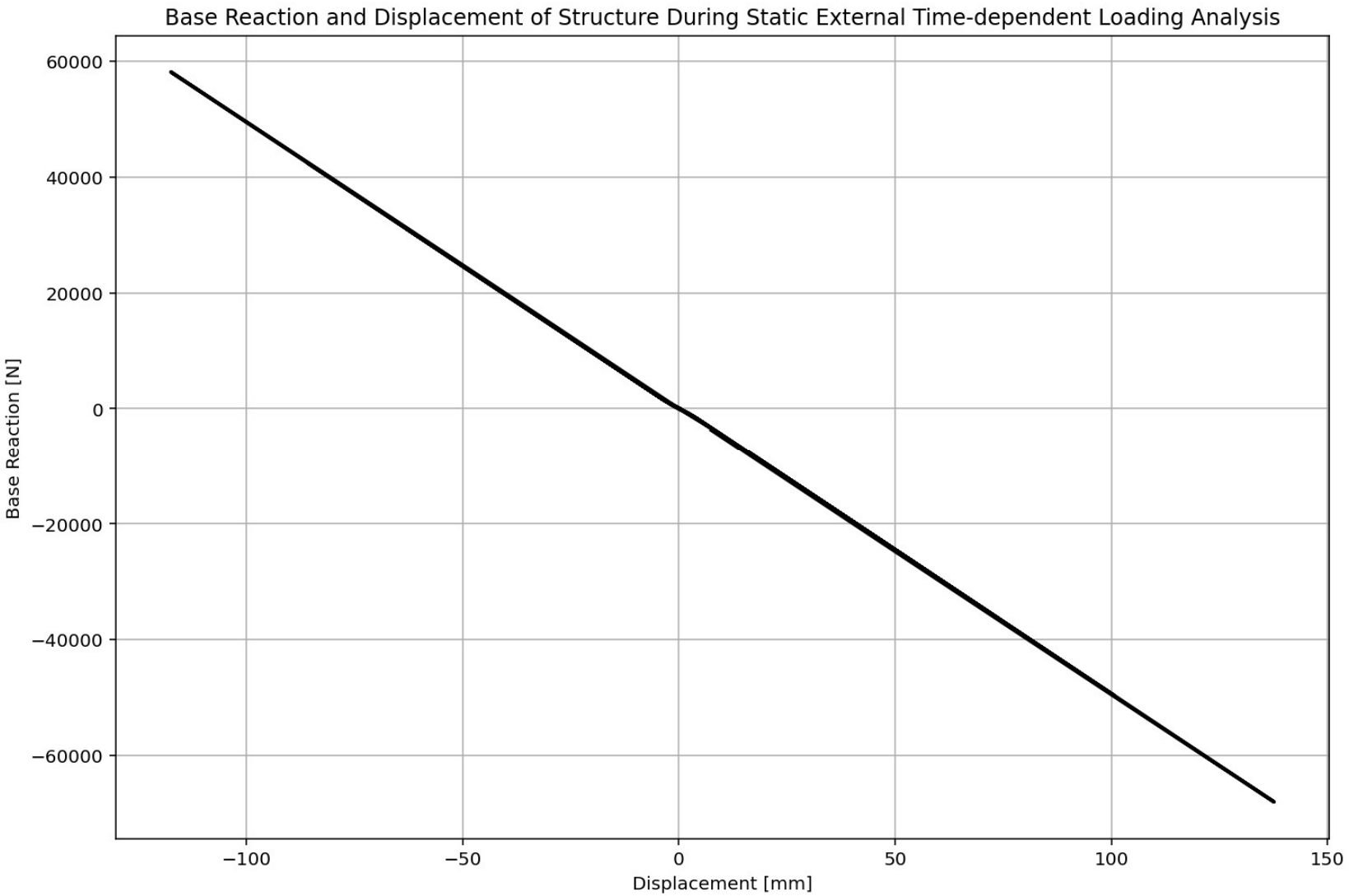
```
588
589     if ANAL_TYPE == 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING': # STATIC TIME-HISTORY ANALYSIS
590         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
591         # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
592         duration = 20.0             # [s] Analysis duration
593         dt = 0.01                   # [s] Time step
594         DT = dt                     # [s] Time step
595         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
596         force_amplitude = 19600.0   # [N] Amplitude Force
597         omega_DT = 5.0715           # [rad/s] Natural angular frequency
598         time_steps = int(duration/dt)
599
600         # Check function
601         def CHECK_FUN(DT_time ,duration):
602             if DT_time > duration:
603                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n\n'
604                 exit()
605             return -1
606
607         CHECK_FUN(DT_time ,duration)
608         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin(
609             import numpy as np
610             import matplotlib.pyplot as plt
611             # External Load Durations
612             num_steps = int(DT_time / DT)
613             load_time = np.linspace(0, DT_time, num_steps)
614             target_frequency = 1.0 * omega_DT  # Target excitation frequency
615             DT_load = force_amplitude * np.sin(target_frequency * load_time)
616             # Plot External Time-dependent Loading
617             plt.figure(figsize=(10, 6))
618             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):.3f
619             plt.title('External Time-dependent Loading Over Time')
620             plt.xlabel('Time (s)')
621             plt.ylabel('Force (N)')
```

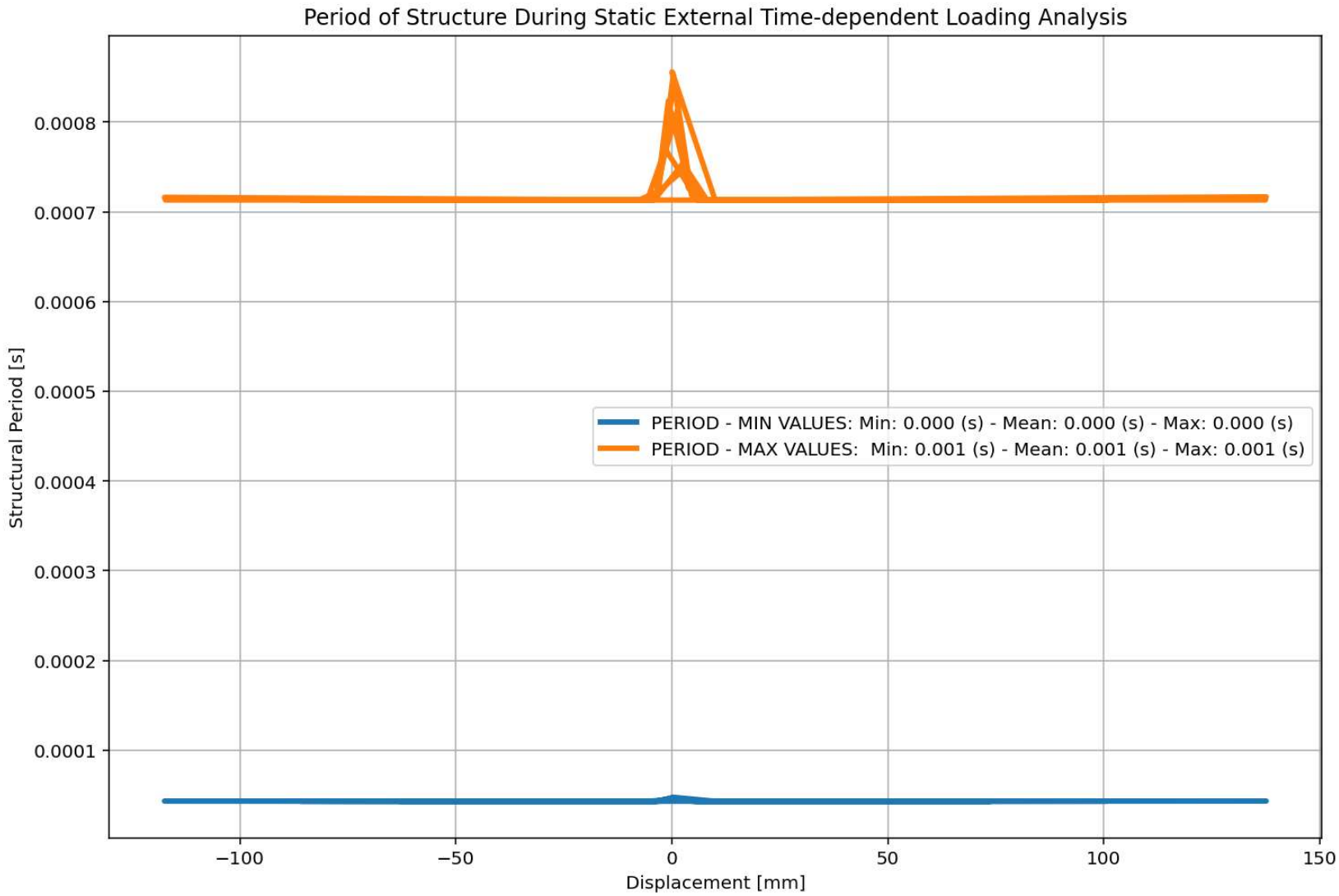


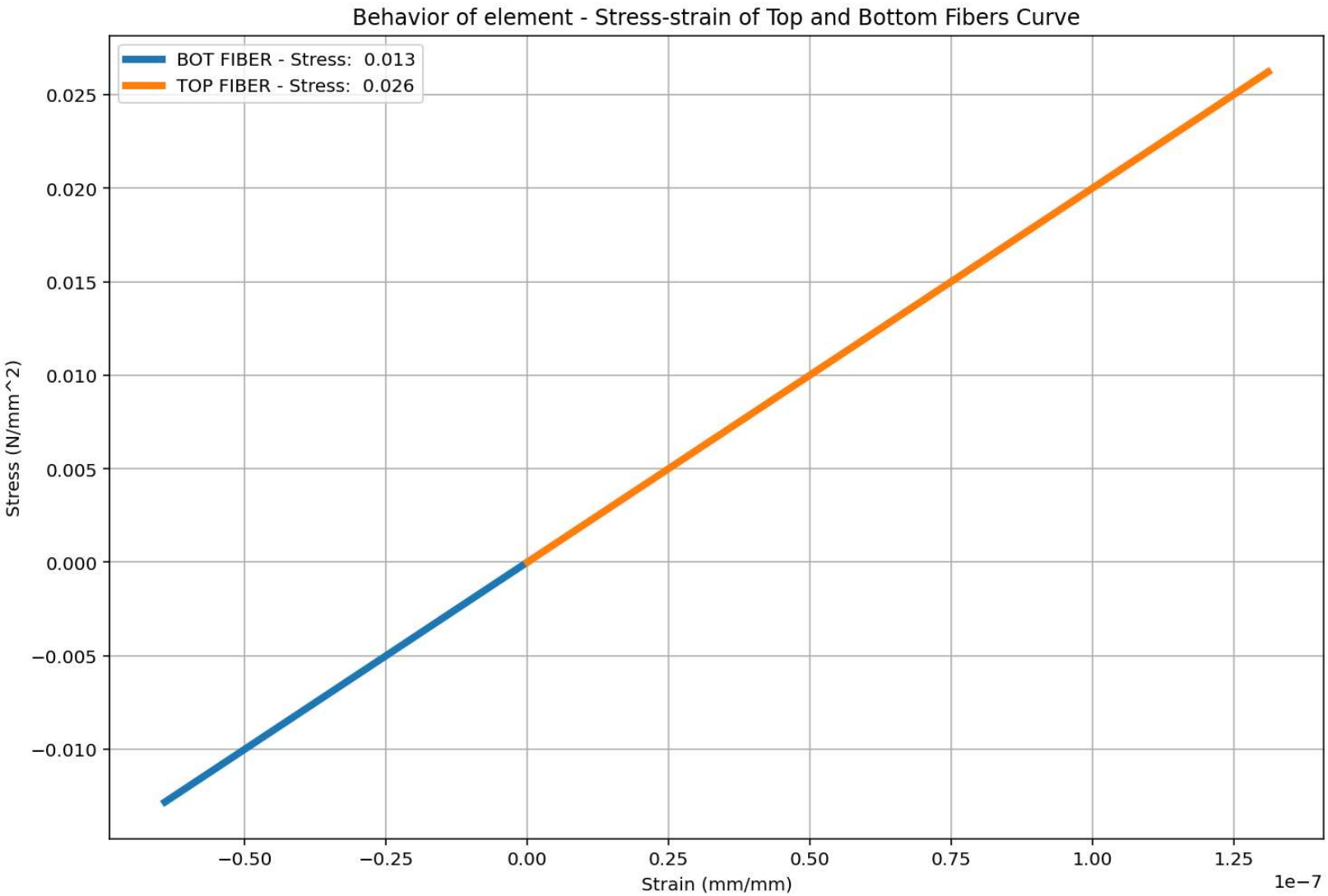
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

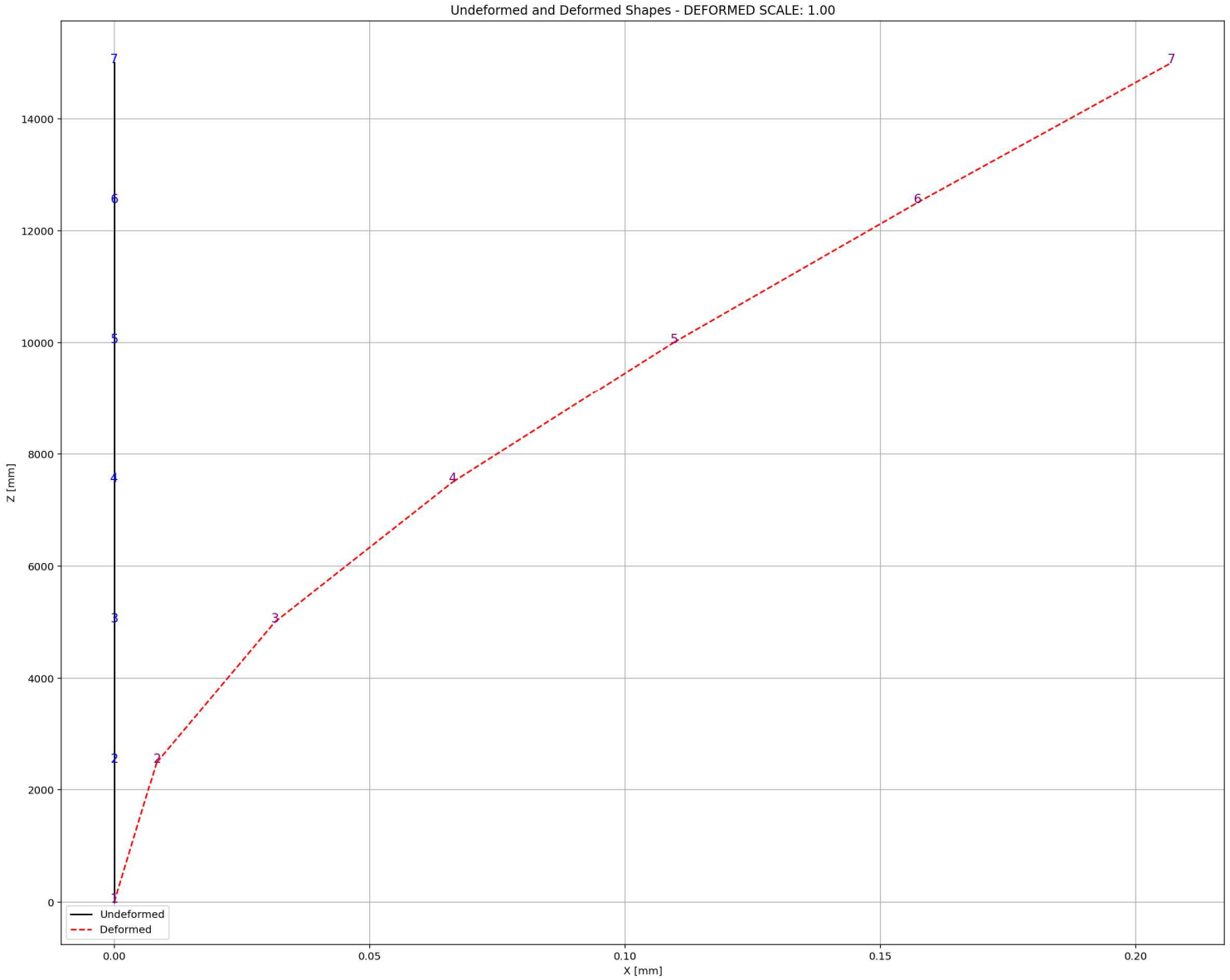
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 112, Col 32 | UTF-8 | CRLF | RW | Mem 55%


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
15000
10000
5000
0
−5000
−10000
−15000
Force (N)
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
60000
40000
20000
0
−20000
−40000
−60000
−100
−50
0
50
100
150
Displacement [mm]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
0.0002
0.0001
PERIOD - MIN VALUES: Min: 0.000 (s) - Mean: 0.000 (s) - Max: 0.000 (s)
PERIOD - MAX VALUES: Min: 0.001 (s) - Mean: 0.001 (s) - Max: 0.001 (s)
−100
−50
0
50
100
150
Displacement [mm]


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.013
TOP FIBER - Stress: 0.026
Stress (N/mm^2)
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
Strain (mm/mm)
1e−7


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
2000
4000
6000
8000
10000
12000
14000
0.00
0.05
0.10
0.15
0.20
1
2
3
4
5
6
7
Undeformed
Deformed

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

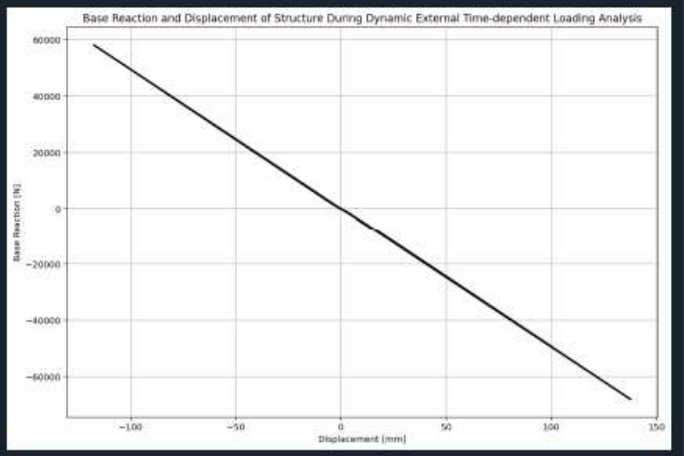
Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

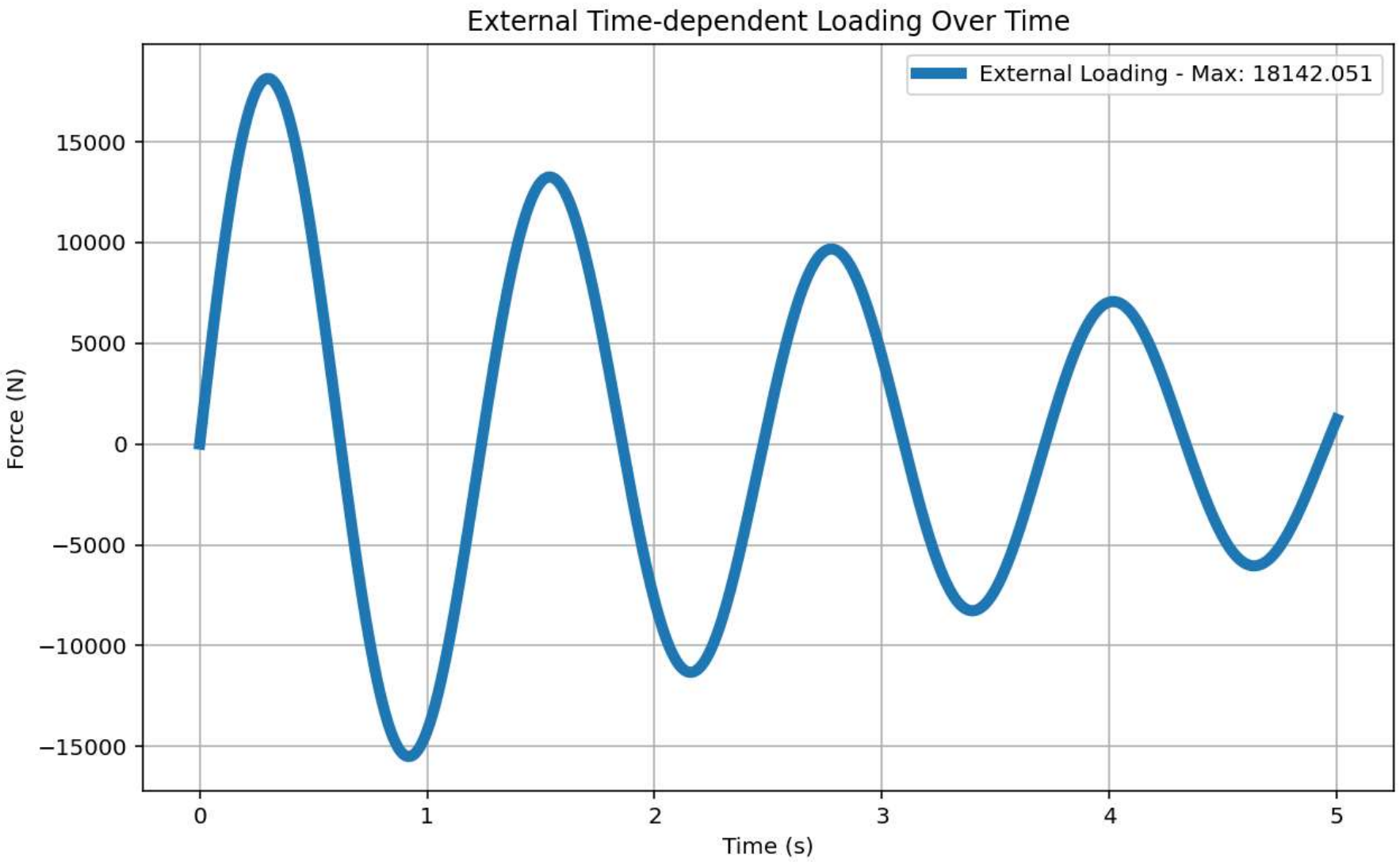
W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

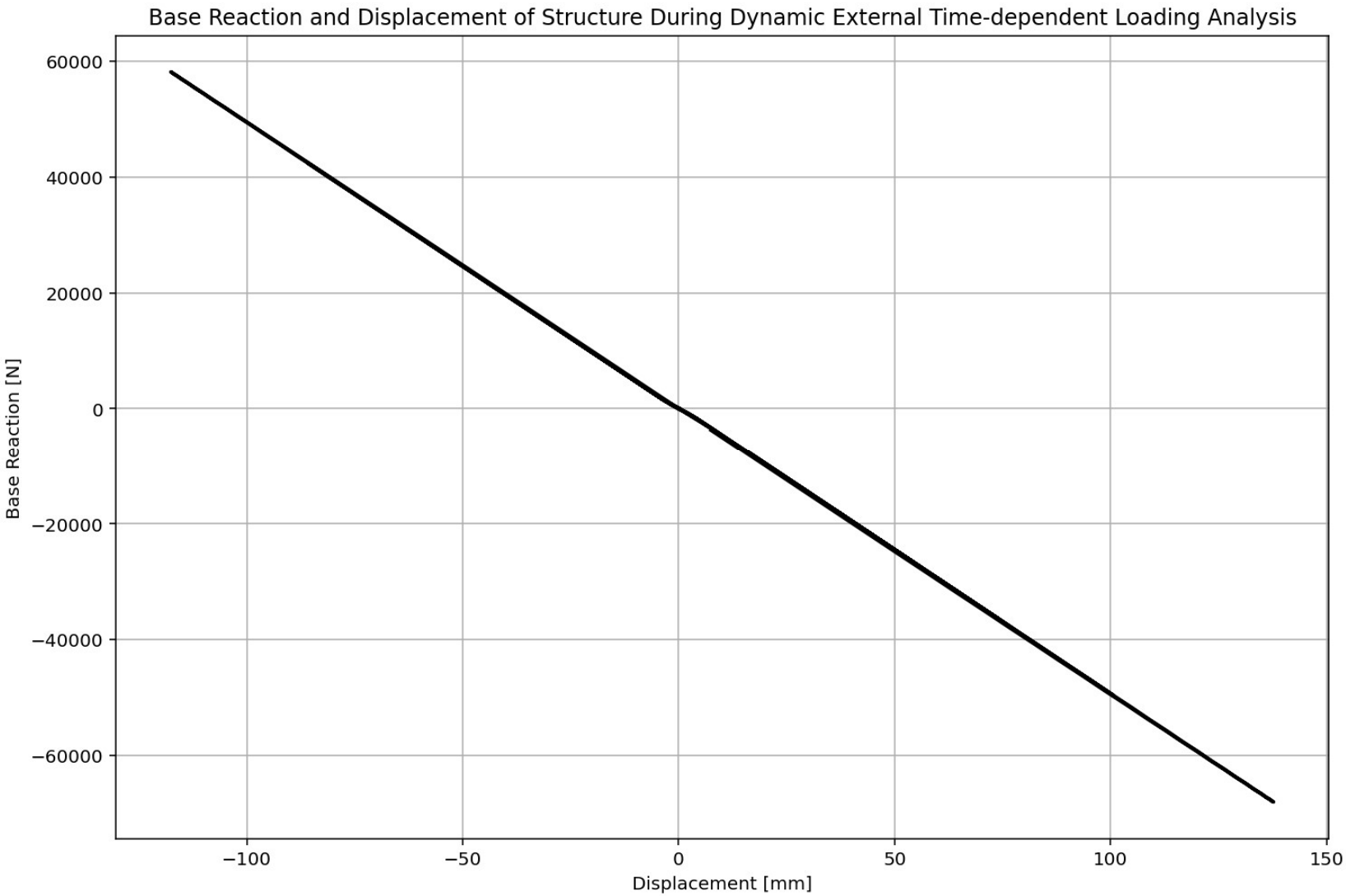
```
736     if ANAL_TYPE == 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING': # DYNAMIC TIME-HISTORY ANALYSI
737         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
738         duration = 20.0             # [s] Analysis duration
739         dt = 0.01                   # [s] Time step
740         DT = dt                     # [s] Time step
741         DT_time = 5.0               # [s] Total external Load Analysis Durations [*******]
742         force_amplitude = 19600.0   # [N] Amplitude Force
743         omega_DT = 5.0715           # [rad/s] Natural angular frequency
744         DR = 0.03                   # Damping Ratio
745
746         # Check function
747         def CHECK_FUN(DT_time ,duration):
748             if DT_time > duration:
749                 print('\n\nAnalysis Duration Must be greater than External Load Duration\n\n'
750                 exit()
751             return -1
752
753         CHECK_FUN(DT_time ,duration)
754         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin(
755             import numpy as np
756             import matplotlib.pyplot as plt
757             # External Load Durations
758             num_steps = int(DT_time / DT)
759             load_time = np.linspace(0, DT_time, num_steps)
760             target_frequency = 1.0 * omega_DT  # Target excitation frequency
761             DT_load = force_amplitude * np.sin(target_frequency * load_time)
762             # Plot External Time-dependent Loading
763             plt.figure(figsize=(10, 6))
764             plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load):.3f
765             plt.title('External Time-dependent Loading Over Time')
766             plt.xlabel('Time (s)')
767             plt.ylabel('Force (N)')
768             plt.grid(True)
769             plt.legend()
```

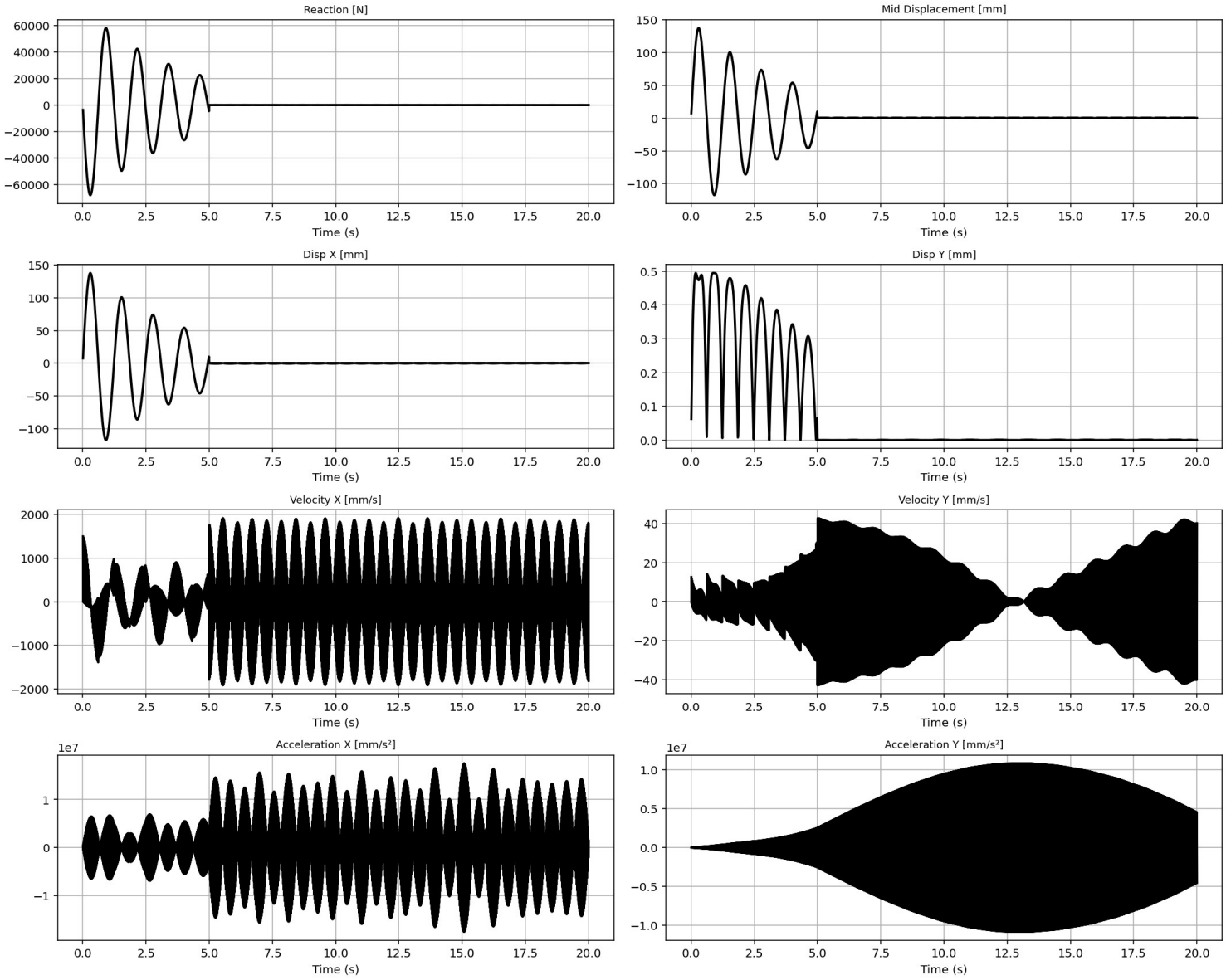


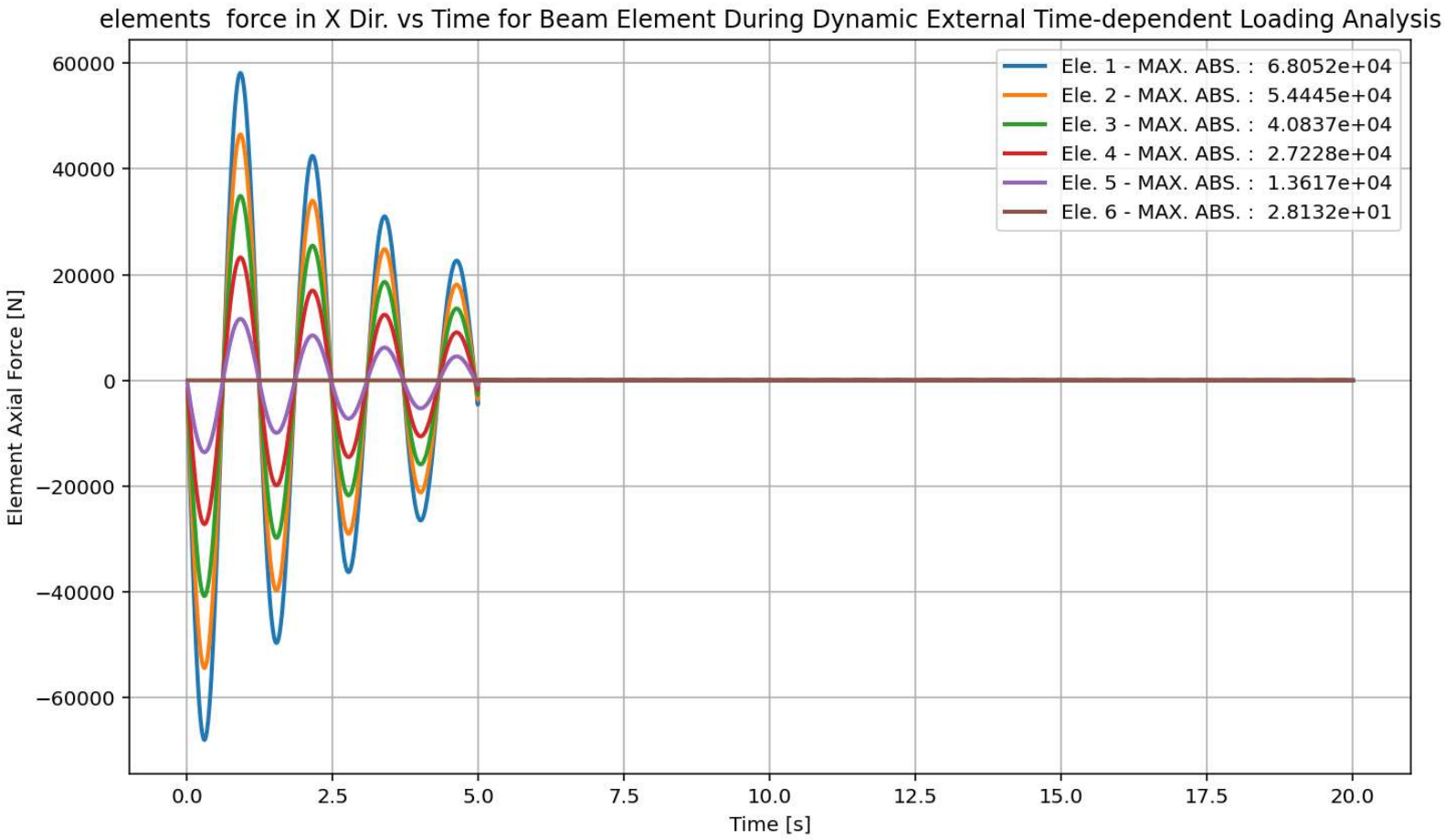
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

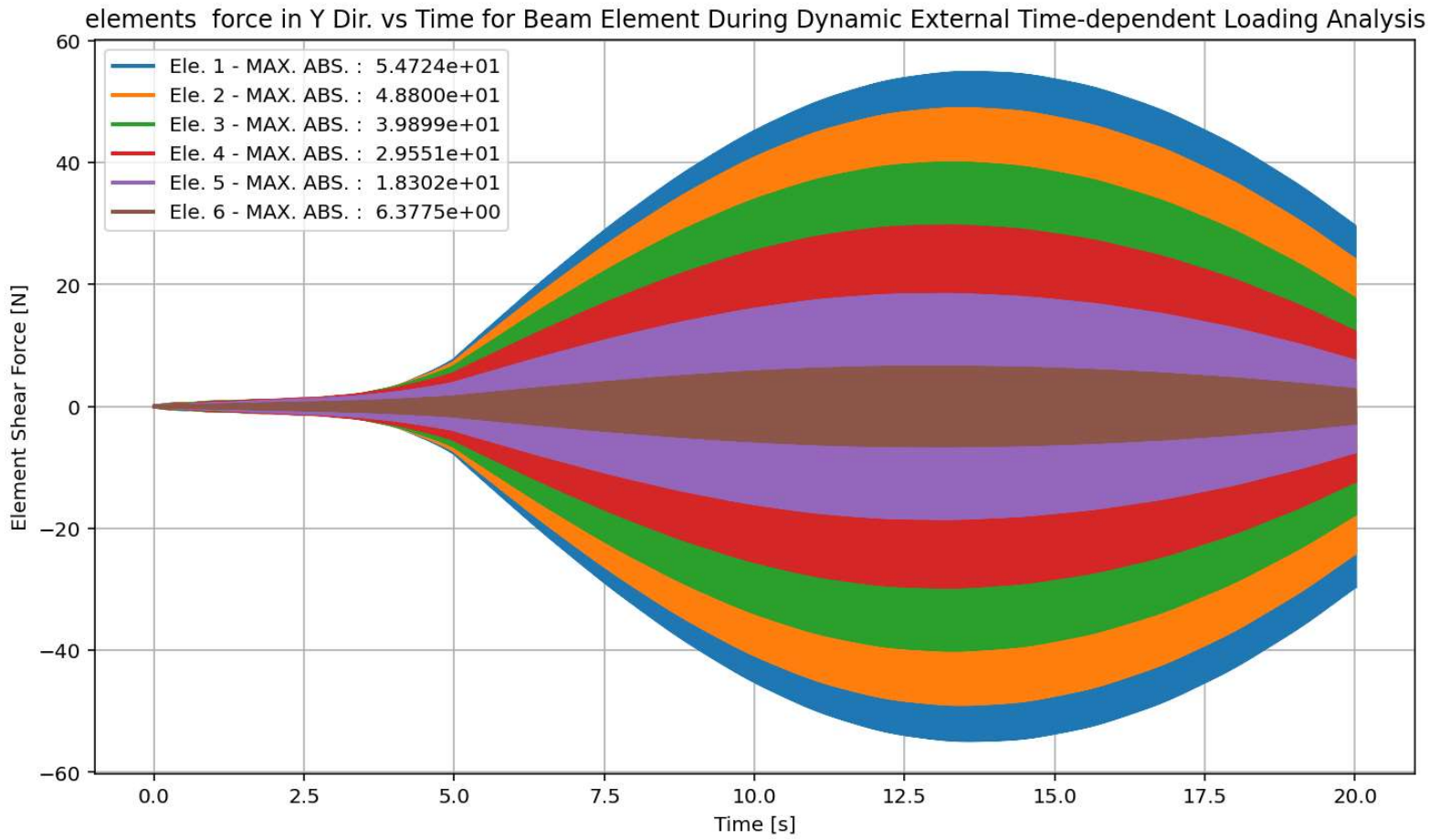
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 112, Col 32 | UTF-8 | CRLF | RW | Mem 56%

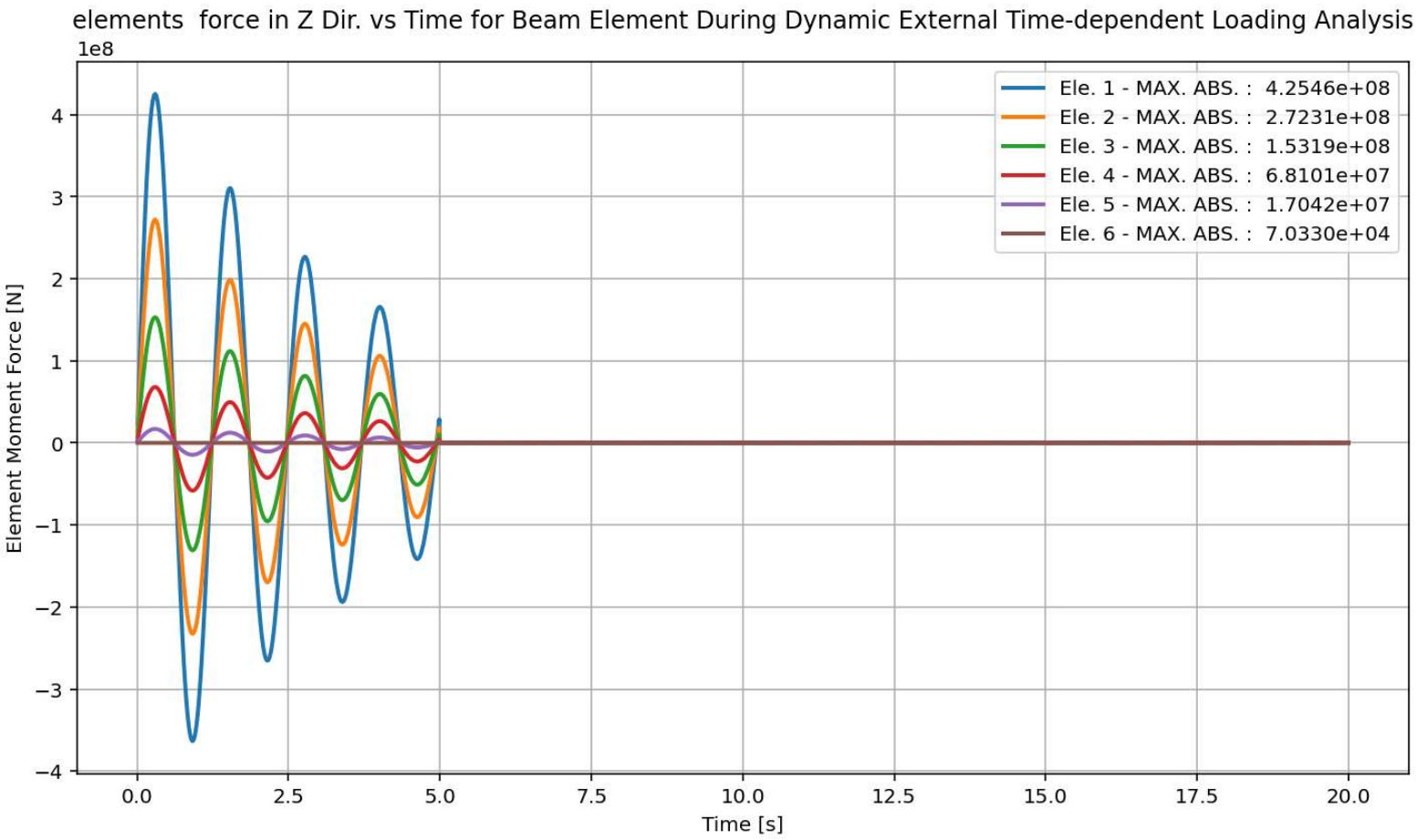

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
15000
10000
5000
0
−5000
−10000
−15000
Force (N)
0
1
2
3
4
5
Time (s)

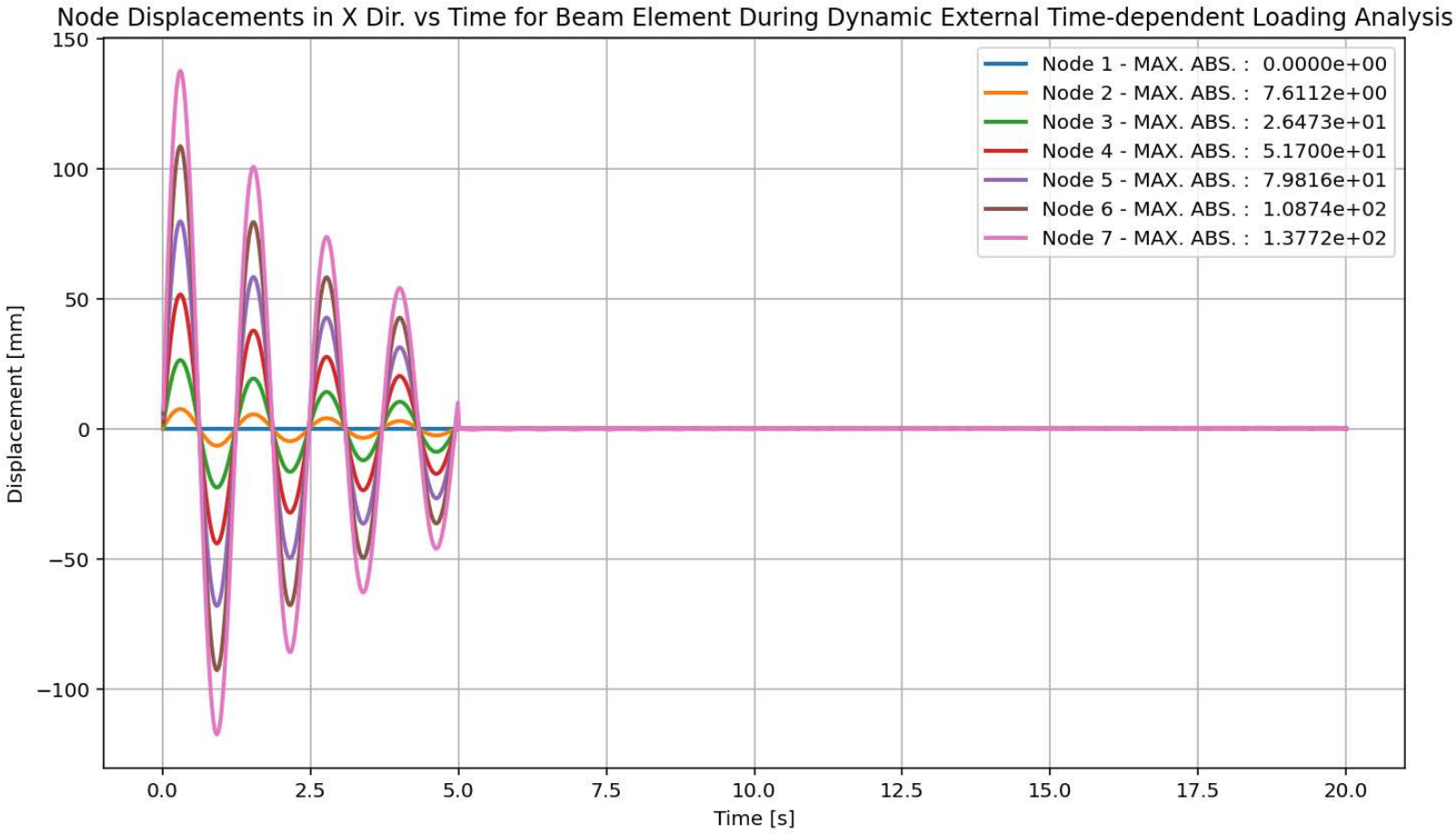

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
60000
40000
20000
0
−20000
−40000
−60000
−100
−50
0
50
100
150
Displacement [mm]

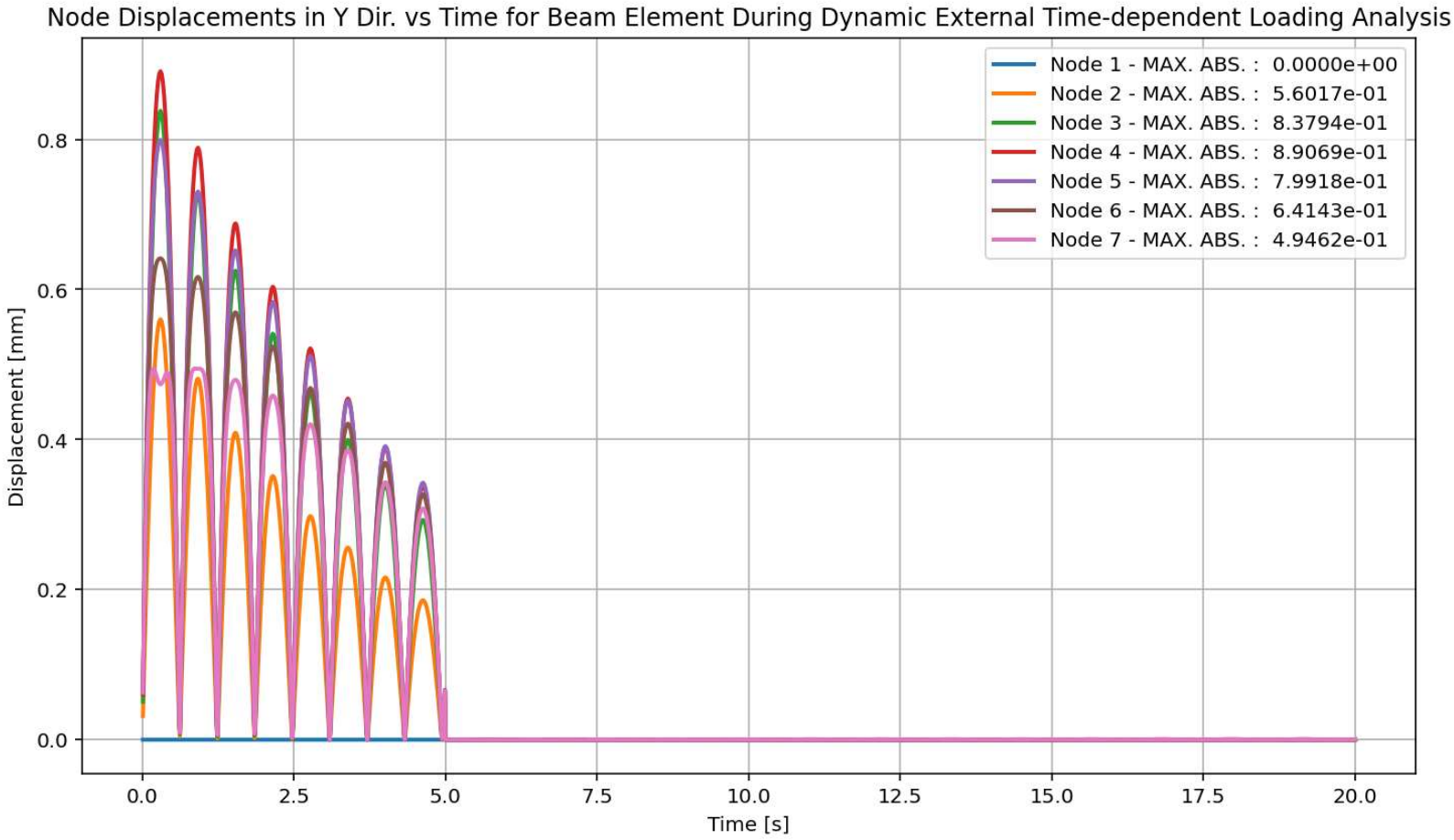

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
1e7
Time (s)
Acceleration Y [mm/s²]
1e7
Time (s)

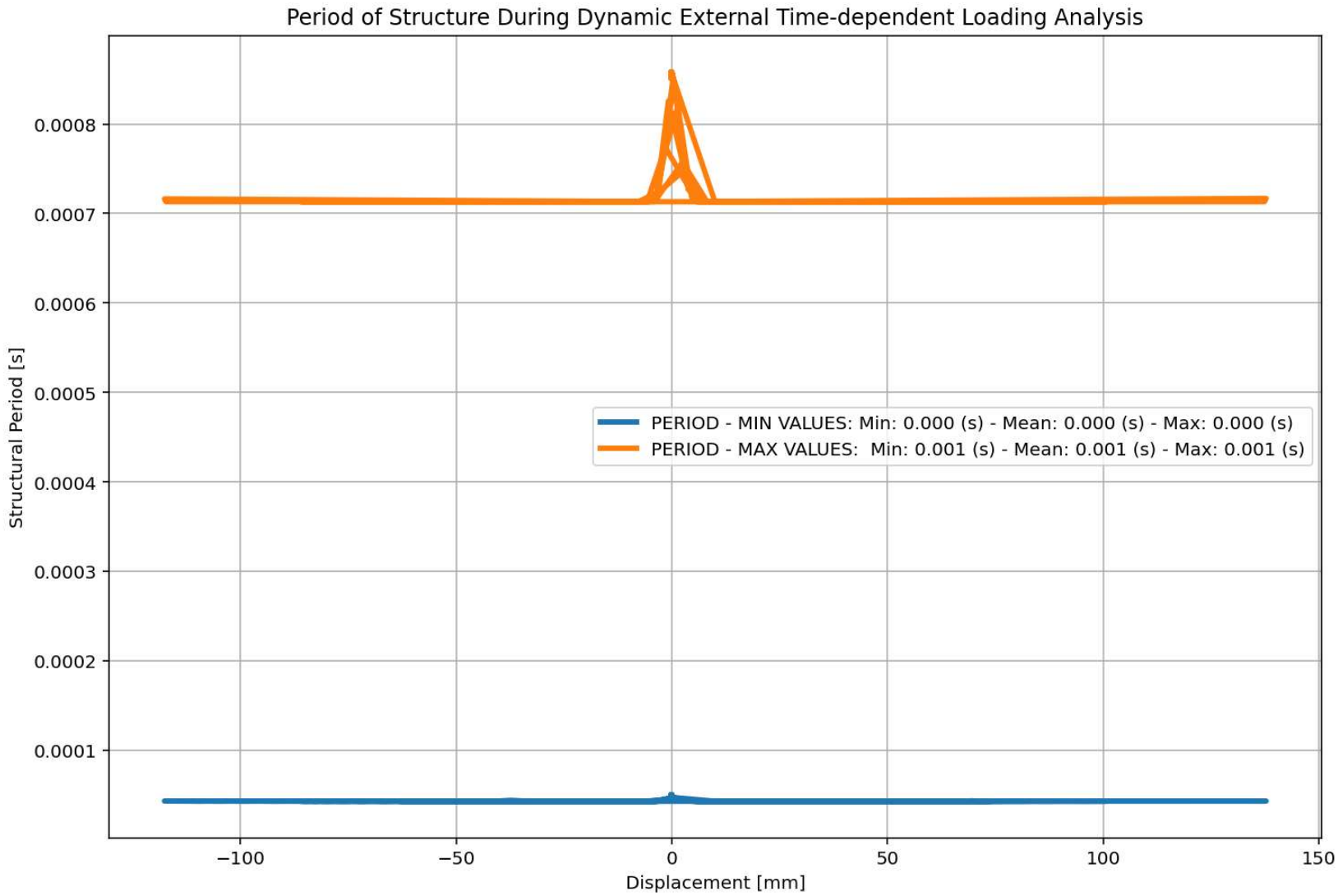

elements force in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 6.8052e+04
Ele. 2 - MAX. ABS. : 5.4445e+04
Ele. 3 - MAX. ABS. : 4.0837e+04
Ele. 4 - MAX. ABS. : 2.7228e+04
Ele. 5 - MAX. ABS. : 1.3617e+04
Ele. 6 - MAX. ABS. : 2.8132e+01
Element Axial Force [N]
60000
40000
20000
0
−20000
−40000
−60000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

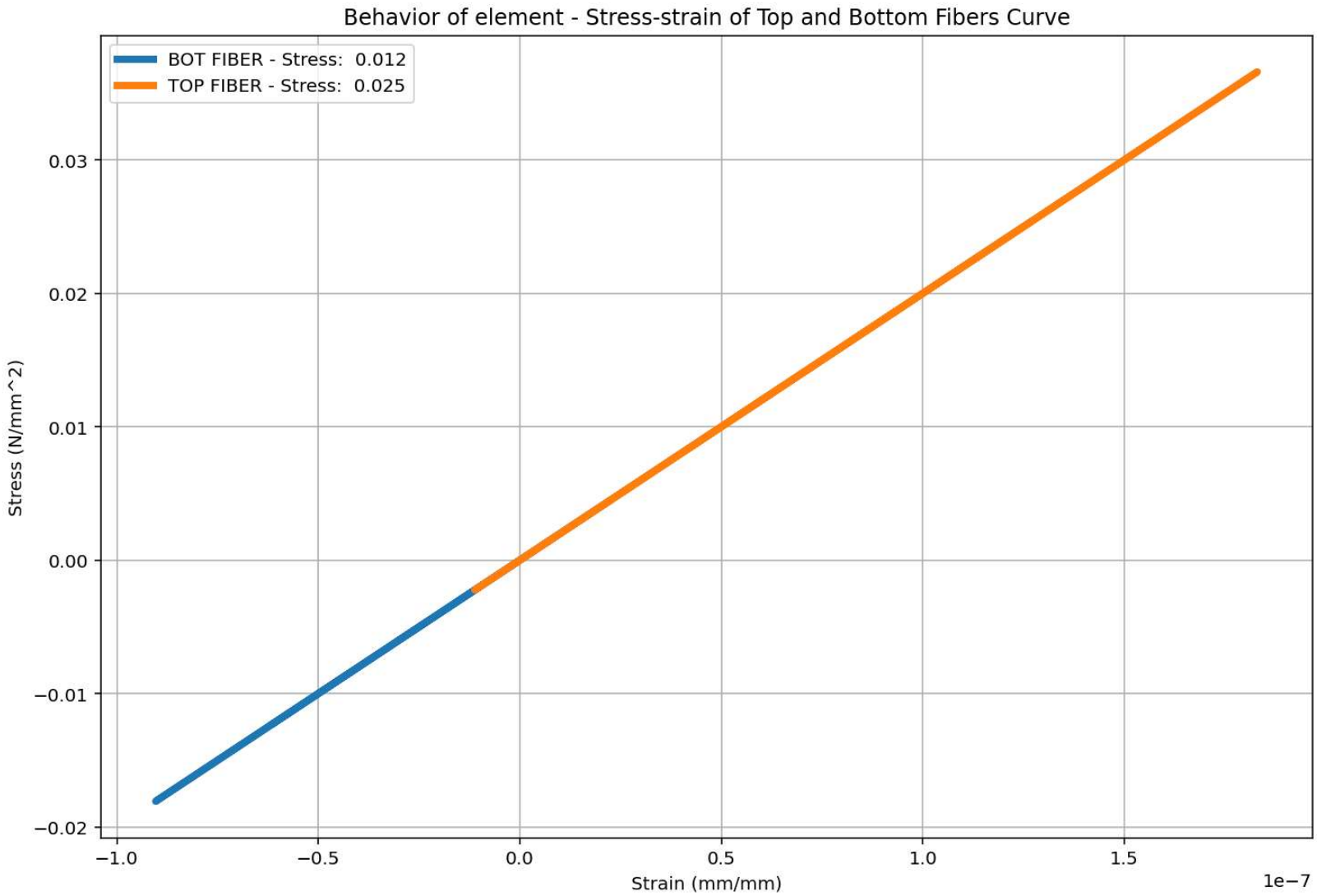

elements force in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 5.4724e+01
Ele. 2 - MAX. ABS. : 4.8800e+01
Ele. 3 - MAX. ABS. : 3.9899e+01
Ele. 4 - MAX. ABS. : 2.9551e+01
Ele. 5 - MAX. ABS. : 1.8302e+01
Ele. 6 - MAX. ABS. : 6.3775e+00
Element Shear Force [N]
60
40
20
0
−20
−40
−60
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

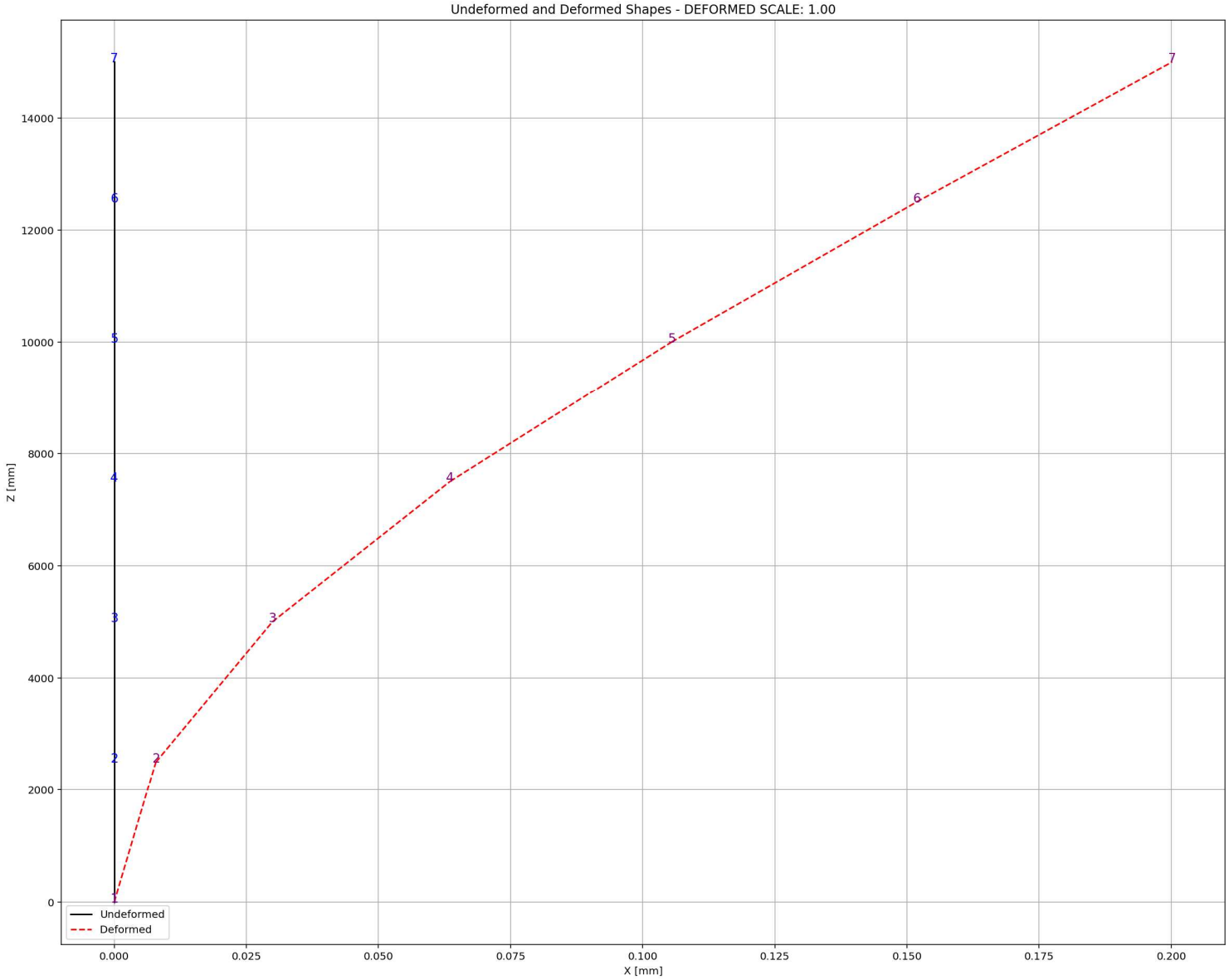

elements force in Z Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
1e8
Element Moment Force [N]
Time [s]
Ele. 1 - MAX. ABS. : 4.2546e+08
Ele. 2 - MAX. ABS. : 2.7231e+08
Ele. 3 - MAX. ABS. : 1.5319e+08
Ele. 4 - MAX. ABS. : 6.8101e+07
Ele. 5 - MAX. ABS. : 1.7042e+07
Ele. 6 - MAX. ABS. : 7.0330e+04
4
3
2
1
0
−1
−2
−3
−4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Node Displacements in X Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 7.6112e+00
Node 3 - MAX. ABS. : 2.6473e+01
Node 4 - MAX. ABS. : 5.1700e+01
Node 5 - MAX. ABS. : 7.9816e+01
Node 6 - MAX. ABS. : 1.0874e+02
Node 7 - MAX. ABS. : 1.3772e+02
Displacement [mm]
150
100
50
0
−50
−100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements in Y Dir. vs Time for Beam Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.6017e-01
Node 3 - MAX. ABS. : 8.3794e-01
Node 4 - MAX. ABS. : 8.9069e-01
Node 5 - MAX. ABS. : 7.9918e-01
Node 6 - MAX. ABS. : 6.4143e-01
Node 7 - MAX. ABS. : 4.9462e-01
Displacement [mm]
0.0
0.2
0.4
0.6
0.8
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
0.0002
0.0001
PERIOD - MIN VALUES: Min: 0.000 (s) - Mean: 0.000 (s) - Max: 0.000 (s)
PERIOD - MAX VALUES: Min: 0.001 (s) - Mean: 0.001 (s) - Max: 0.001 (s)
−100
−50
0
50
100
150
Displacement [mm]


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.012
TOP FIBER - Stress: 0.025
Stress (N/mm^2)
0.03
0.02
0.01
0.00
−0.01
−0.02
−1.0
−0.5
0.0
0.5
1.0
1.5
Strain (mm/mm)
1e−7


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
2000
4000
6000
8000
10000
12000
14000
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
1
2
3
4
5
6
7
Undeformed
Deformed

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

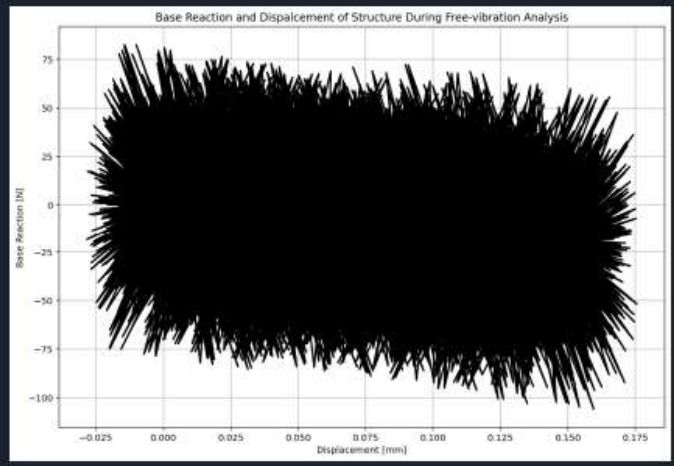
File Edit Search Source Run Debug Consoles Projects Tools View Help

...NERGY_DISSIPATION_CAPACITY_INDEX\14_CONCRETE_COLUMN_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

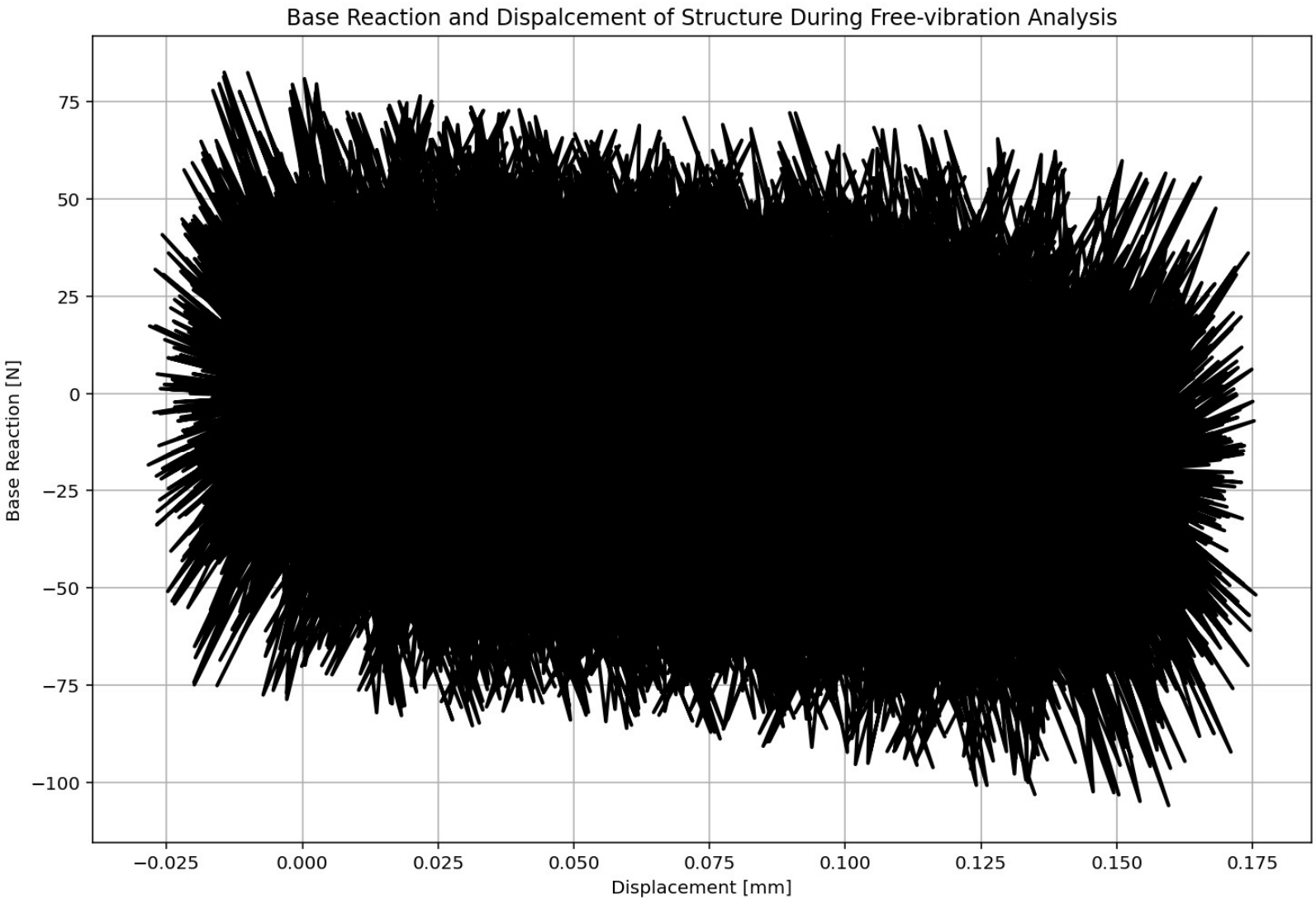
W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

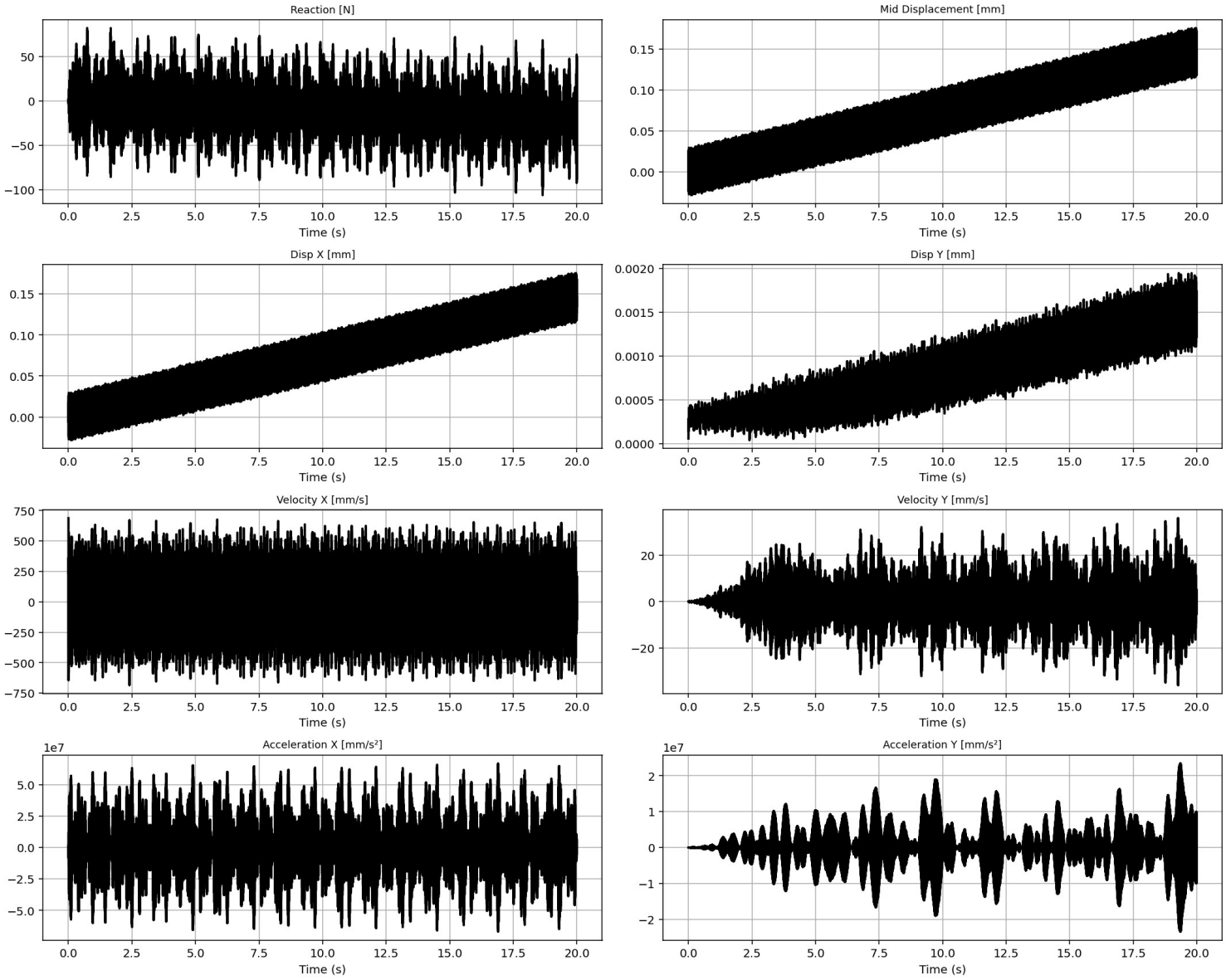
```
902
903     if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
904         #%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
905         u0 = -0.35                          # [mm] Initial displacement
906         v0 = 0.15                           # [mm/s] Initial velocity
907         a0 = 0.065                          # [mm/s^2] Initial acceleration
908         IU = True                           # Free Vibration with Initial Displacement
909         IV = True                           # Free Vibration with Initial Velocity
910         IA = True                           # Free Vibration with Initial Acceleration
911         duration = 20.0                     # [s] Analysis duration
912         dt = 0.001                          # [s] Time step
913         DR = 0.03                           # Damping Ratio
914
915         if IU == True:
916             # Define initial displacment
917             ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: https://openseespydoc
918         if IV == True:
919             # Define initial velocity
920             ops.setNodeVel(center_node, 1, v0, '-commit')  # INFO LINK: https://openseespydoc
921         if IA == True:
922             # Define initial  acceleration
923             ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: https://openseespydo
924
925         ops.constraints('Plain')
926         ops.numberer('Plain')
927         ops.system('BandGeneral')
928         ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK: https://opensees
929         #ops.integrator('CentralDifference')  # JUST FOR LINEAR ANALYSIS - INFO LINK: https:/
930         alpha=0.5; beta=0.25;
931         ops.integrator('Newmark', alpha, beta) # INFO LINK: https://openseespydoc.readthedocs
932         #alpha=2/3;gamma=1.5-alpha; gamma=1.5-alpha;beta=(2-alpha)**2/4;
933         #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://openseespydoc.readthe
934         ops.algorithm('Newton')  # INFO LINK: https://openseespydoc.readthedocs.io/en/latest/
```

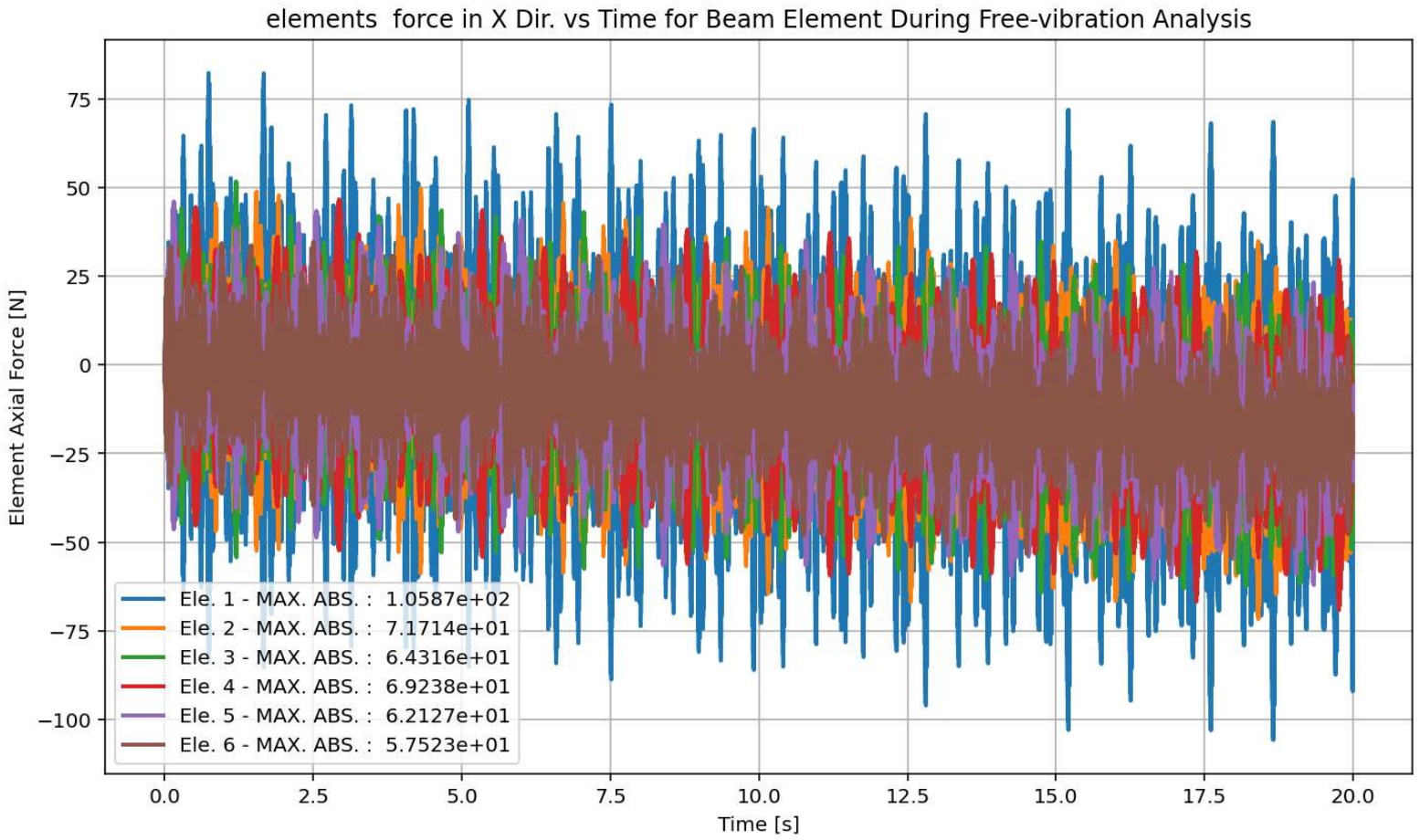


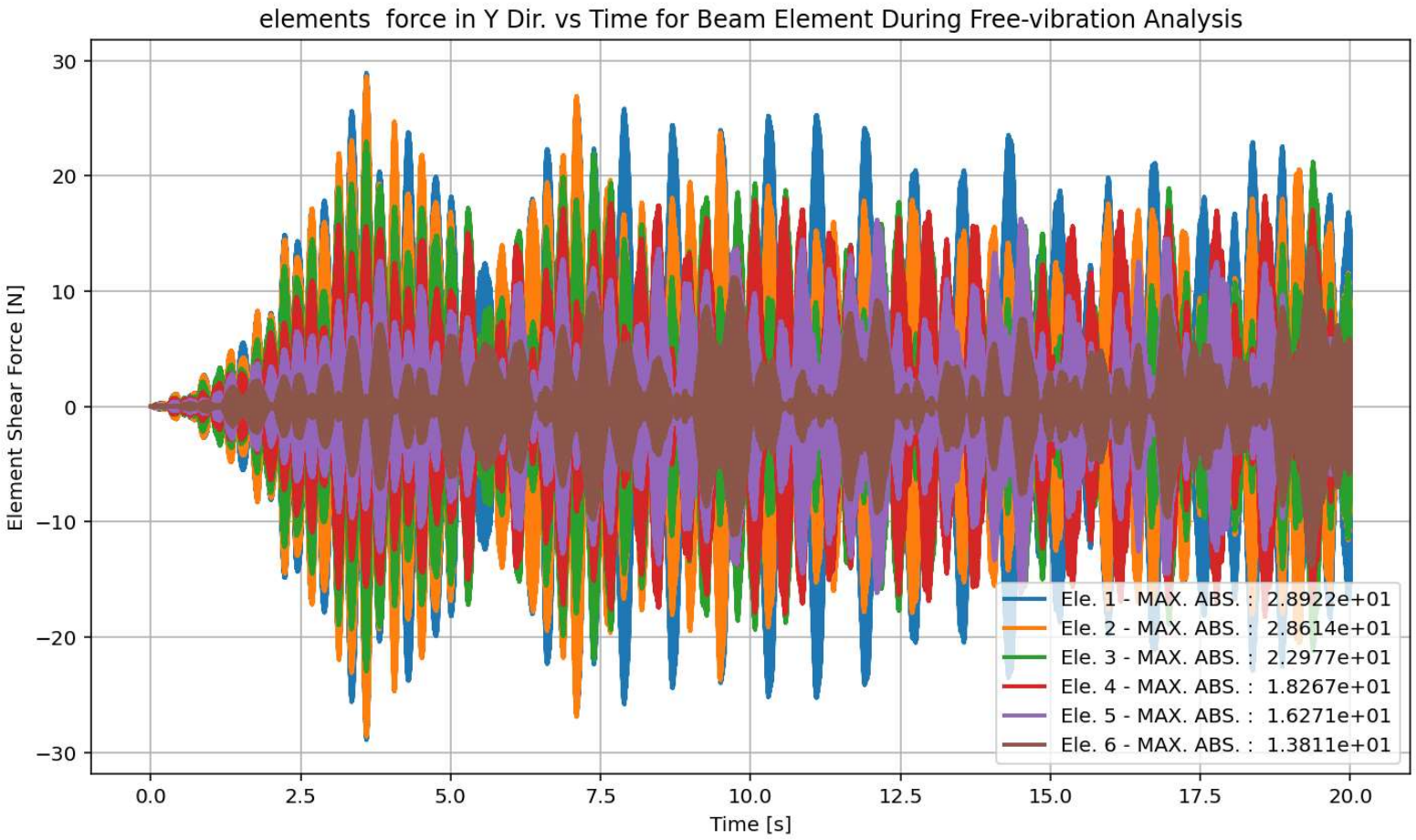
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

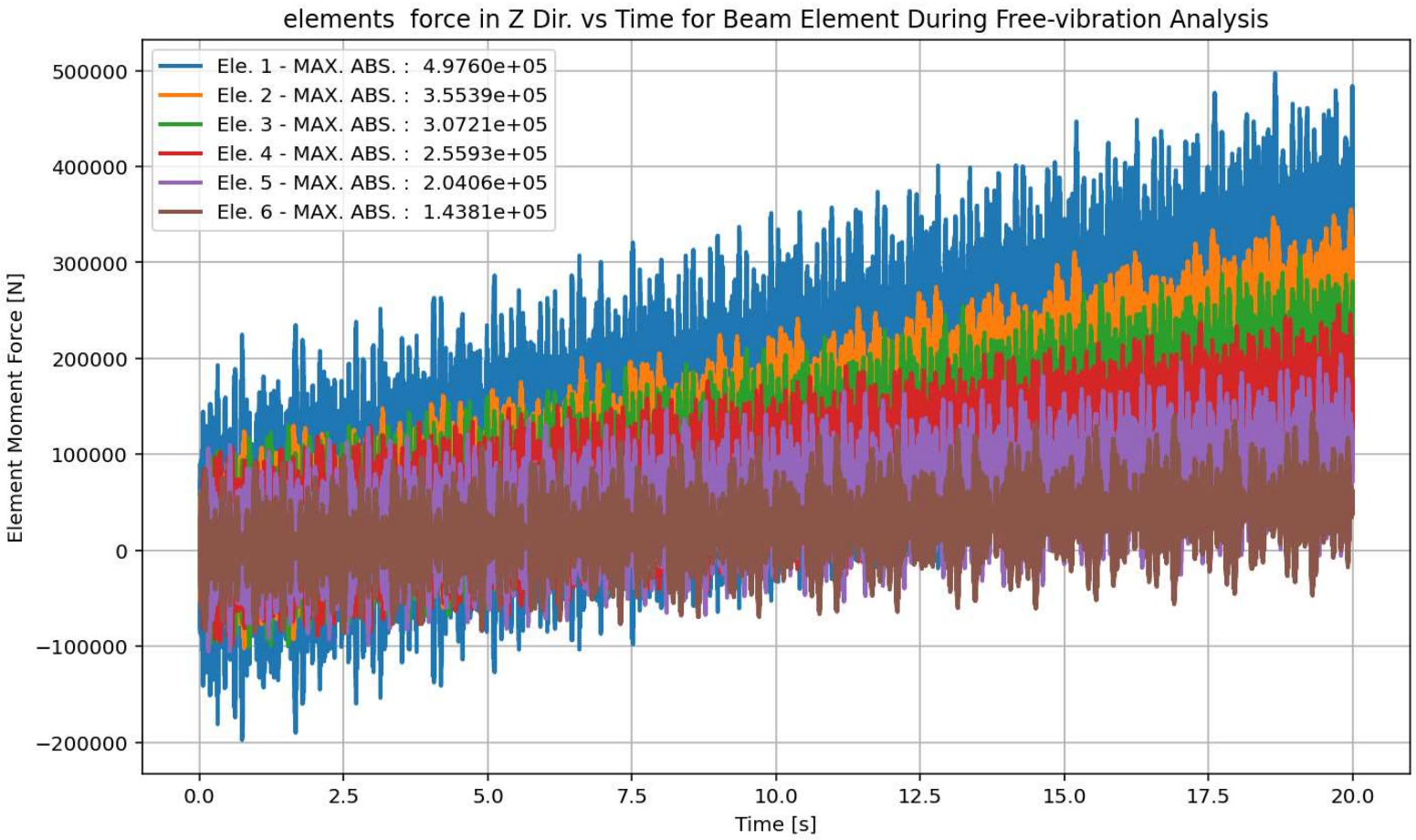
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 112, Col 32 | UTF-8 | CRLF | RW | Mem 47%

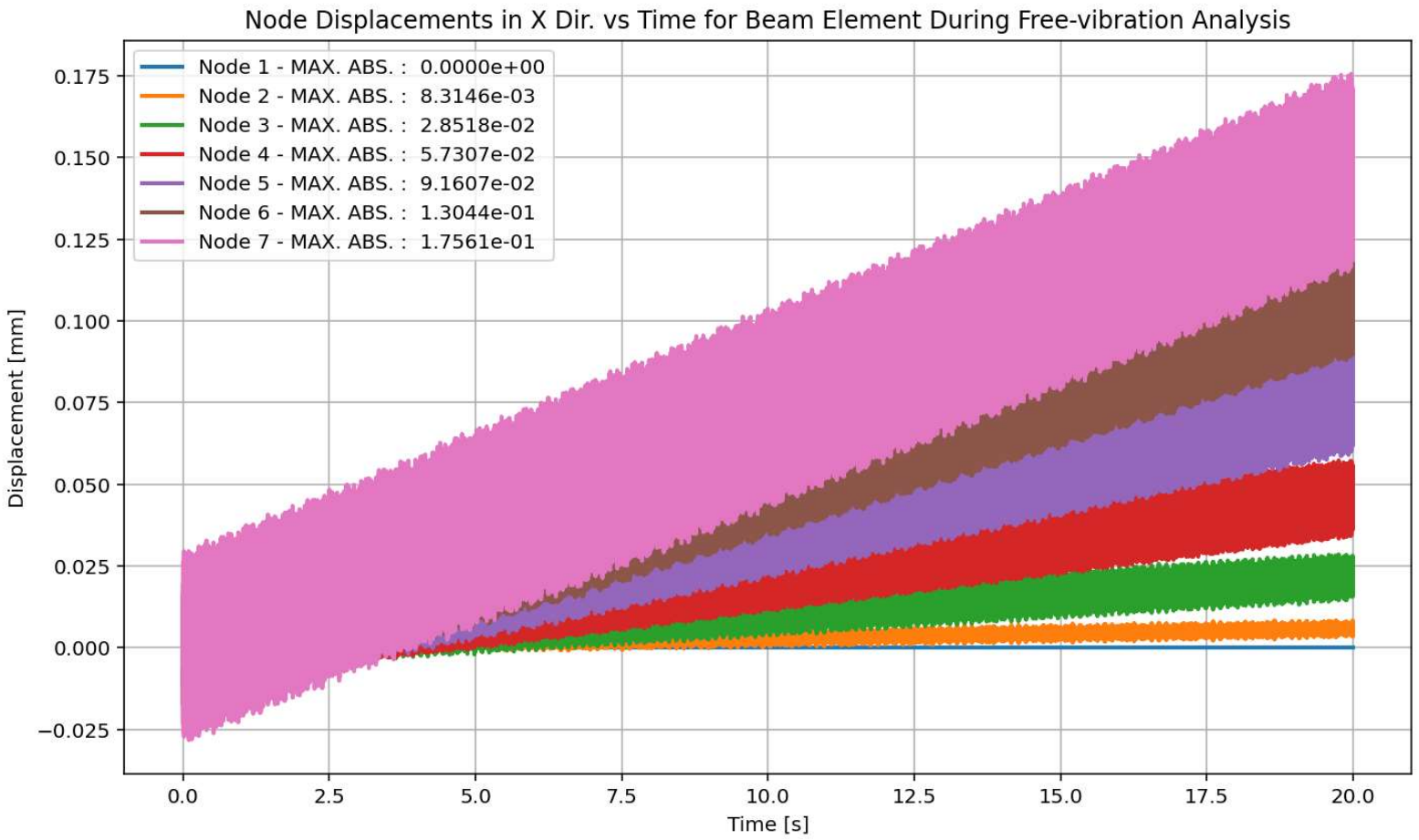

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
75
50
25
0
−25
−50
−75
−100
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Displacement [mm]

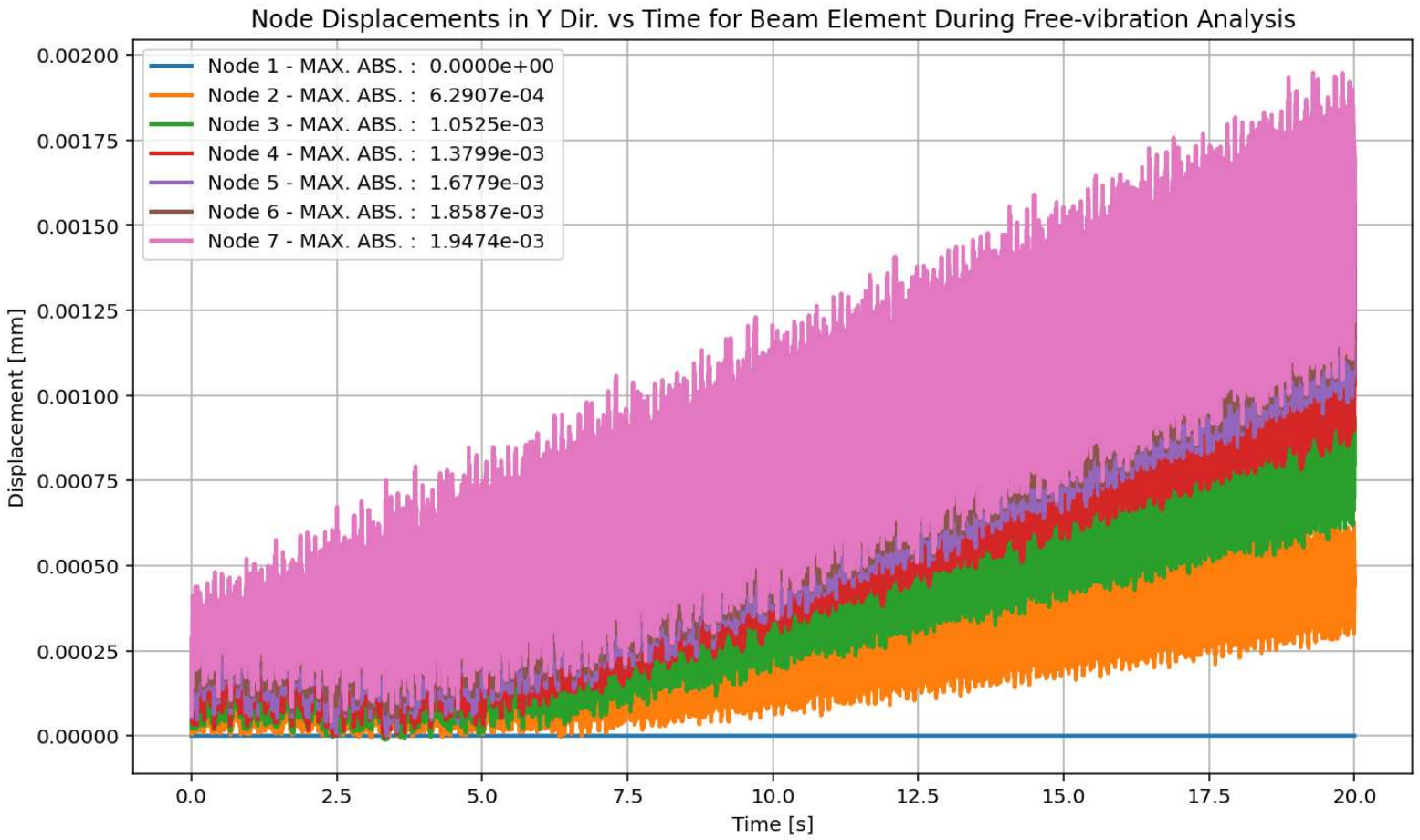

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
1e7
Time (s)
Acceleration Y [mm/s²]
1e7
Time (s)

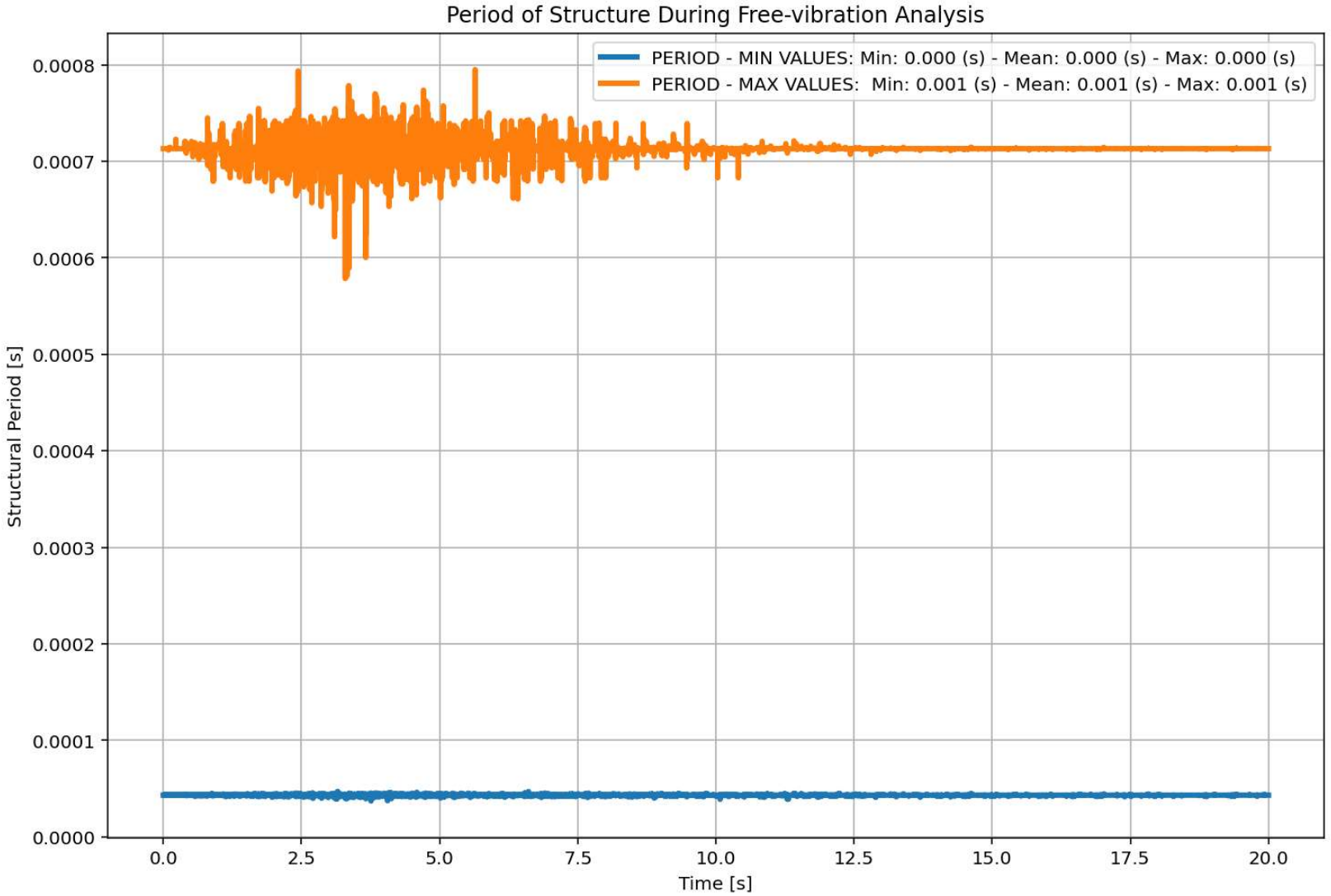

elements force in X Dir. vs Time for Beam Element During Free-vibration Analysis
Element Axial Force [N]
Time [s]
75
50
25
0
−25
−50
−75
−100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 1.0587e+02
Ele. 2 - MAX. ABS. : 7.1714e+01
Ele. 3 - MAX. ABS. : 6.4316e+01
Ele. 4 - MAX. ABS. : 6.9238e+01
Ele. 5 - MAX. ABS. : 6.2127e+01
Ele. 6 - MAX. ABS. : 5.7523e+01

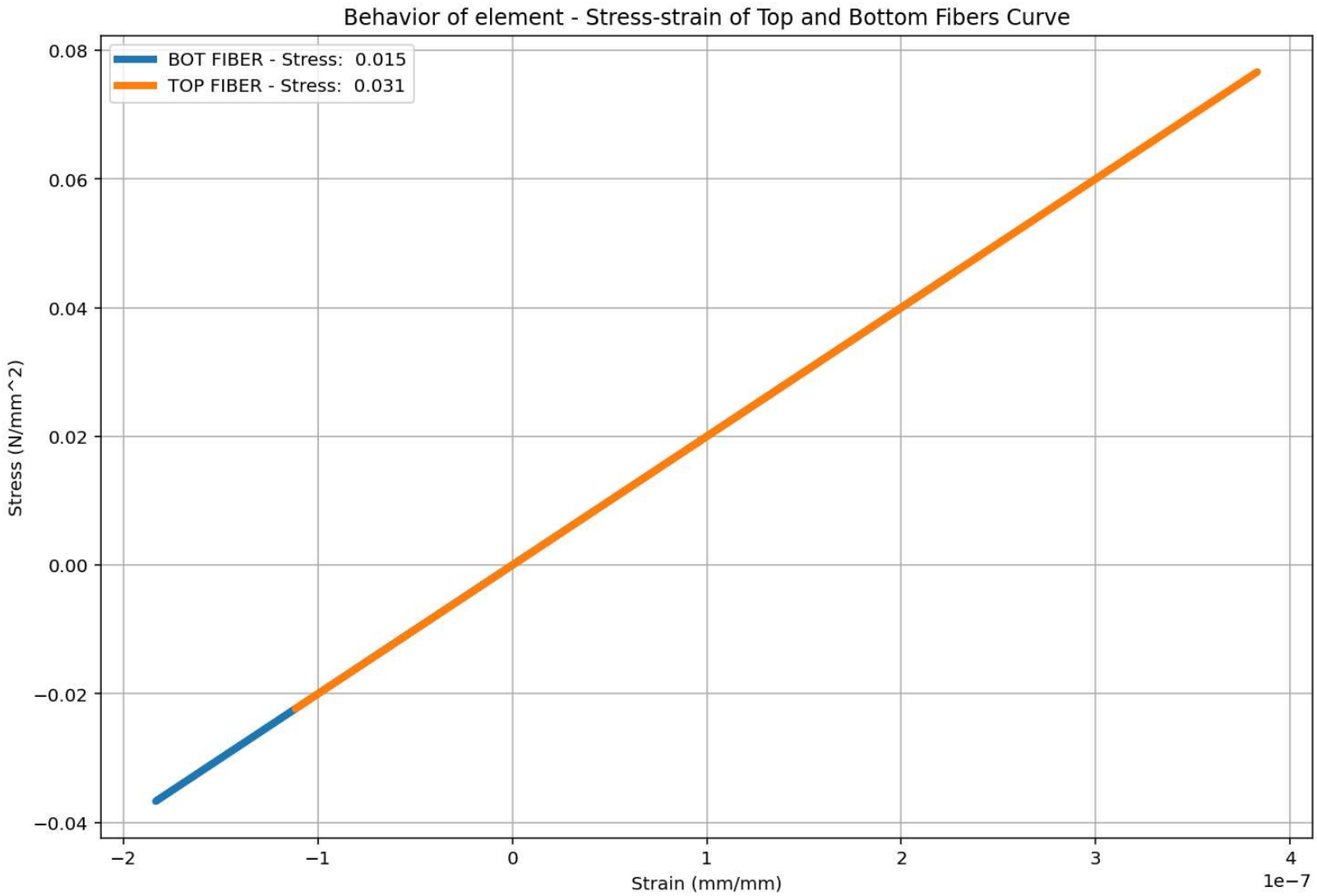

elements force in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Element Shear Force [N]
Time [s]
30
20
10
0
−10
−20
−30
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 2.8922e+01
Ele. 2 - MAX. ABS. : 2.8614e+01
Ele. 3 - MAX. ABS. : 2.2977e+01
Ele. 4 - MAX. ABS. : 1.8267e+01
Ele. 5 - MAX. ABS. : 1.6271e+01
Ele. 6 - MAX. ABS. : 1.3811e+01

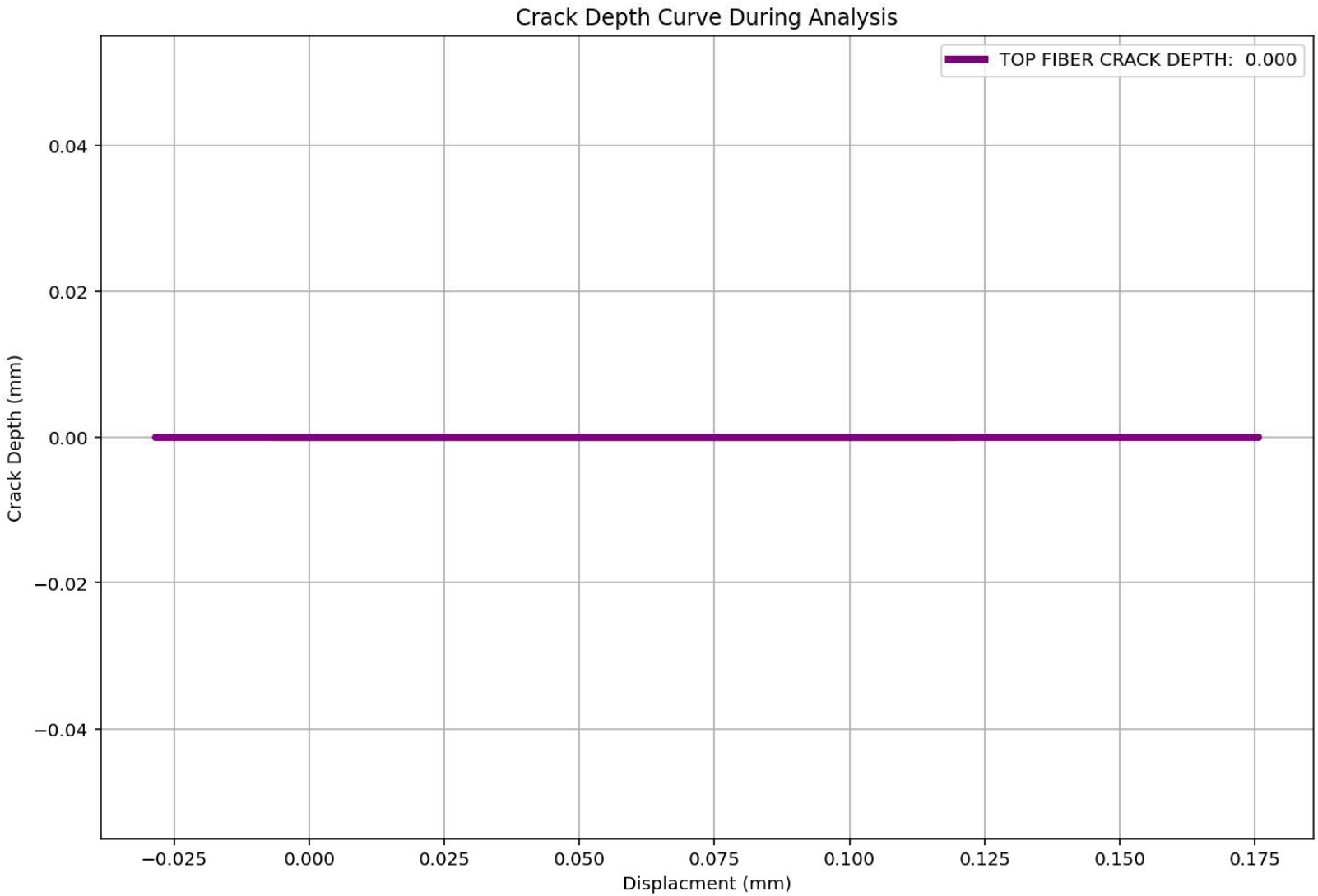

elements force in Z Dir. vs Time for Beam Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 4.9760e+05
Ele. 2 - MAX. ABS. : 3.5539e+05
Ele. 3 - MAX. ABS. : 3.0721e+05
Ele. 4 - MAX. ABS. : 2.5593e+05
Ele. 5 - MAX. ABS. : 2.0406e+05
Ele. 6 - MAX. ABS. : 1.4381e+05
Element Moment Force [N]
500000
400000
300000
200000
100000
0
−100000
−200000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

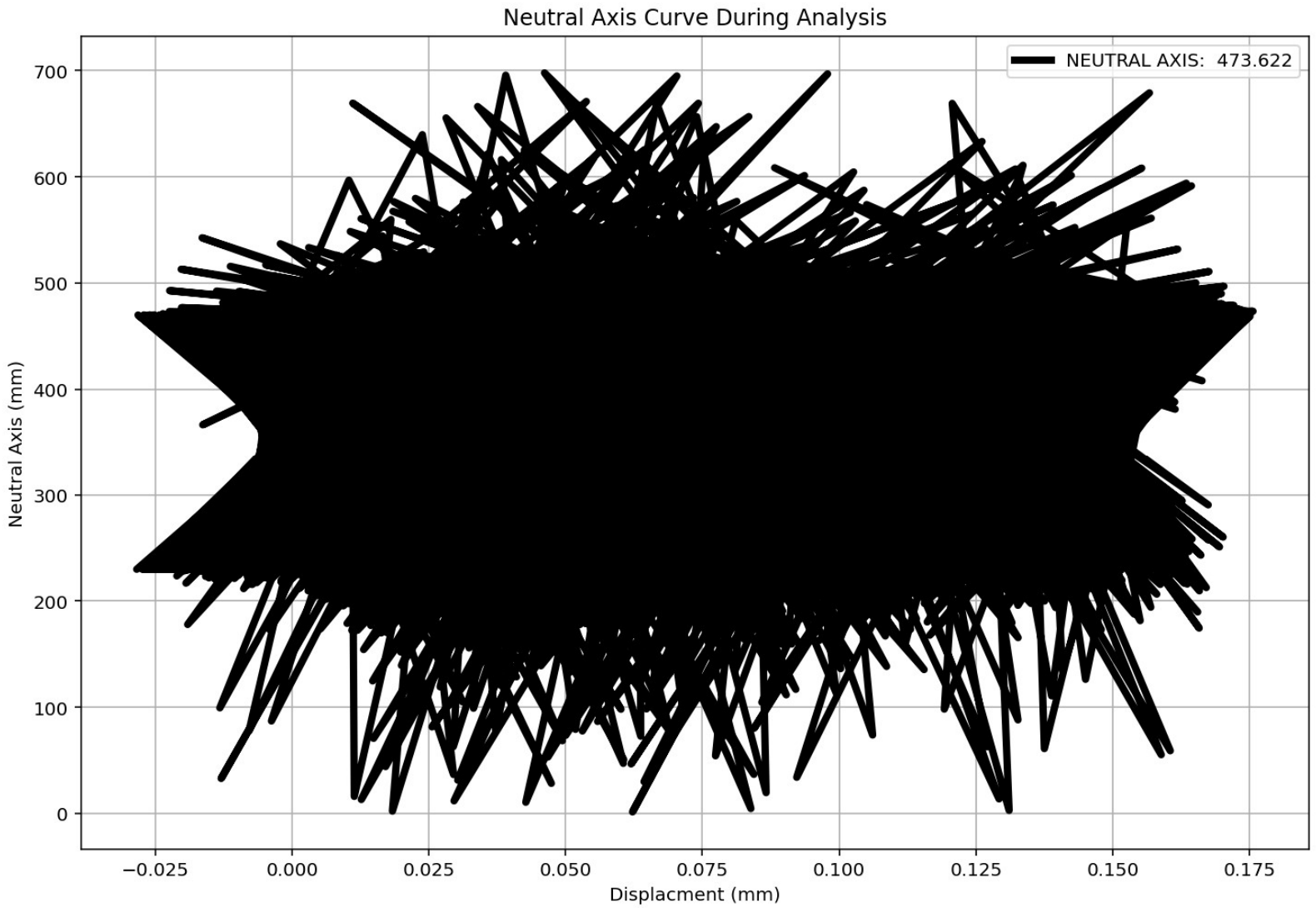

Node Displacements in X Dir. vs Time for Beam Element During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 8.3146e-03
Node 3 - MAX. ABS. : 2.8518e-02
Node 4 - MAX. ABS. : 5.7307e-02
Node 5 - MAX. ABS. : 9.1607e-02
Node 6 - MAX. ABS. : 1.3044e-01
Node 7 - MAX. ABS. : 1.7561e-01
Displacement [mm]
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

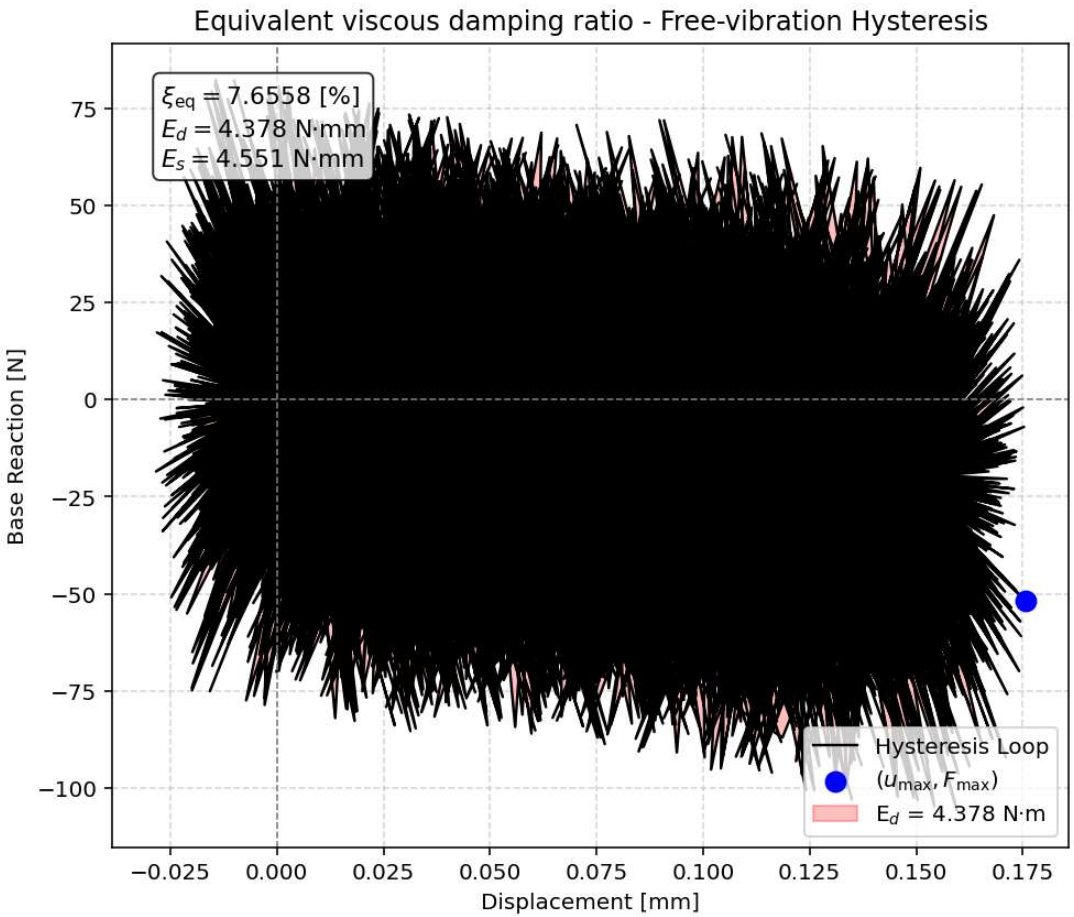

Node Displacements in Y Dir. vs Time for Beam Element During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 6.2907e-04
Node 3 - MAX. ABS. : 1.0525e-03
Node 4 - MAX. ABS. : 1.3799e-03
Node 5 - MAX. ABS. : 1.6779e-03
Node 6 - MAX. ABS. : 1.8587e-03
Node 7 - MAX. ABS. : 1.9474e-03
Displacement [mm]
0.00200
0.00175
0.00150
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

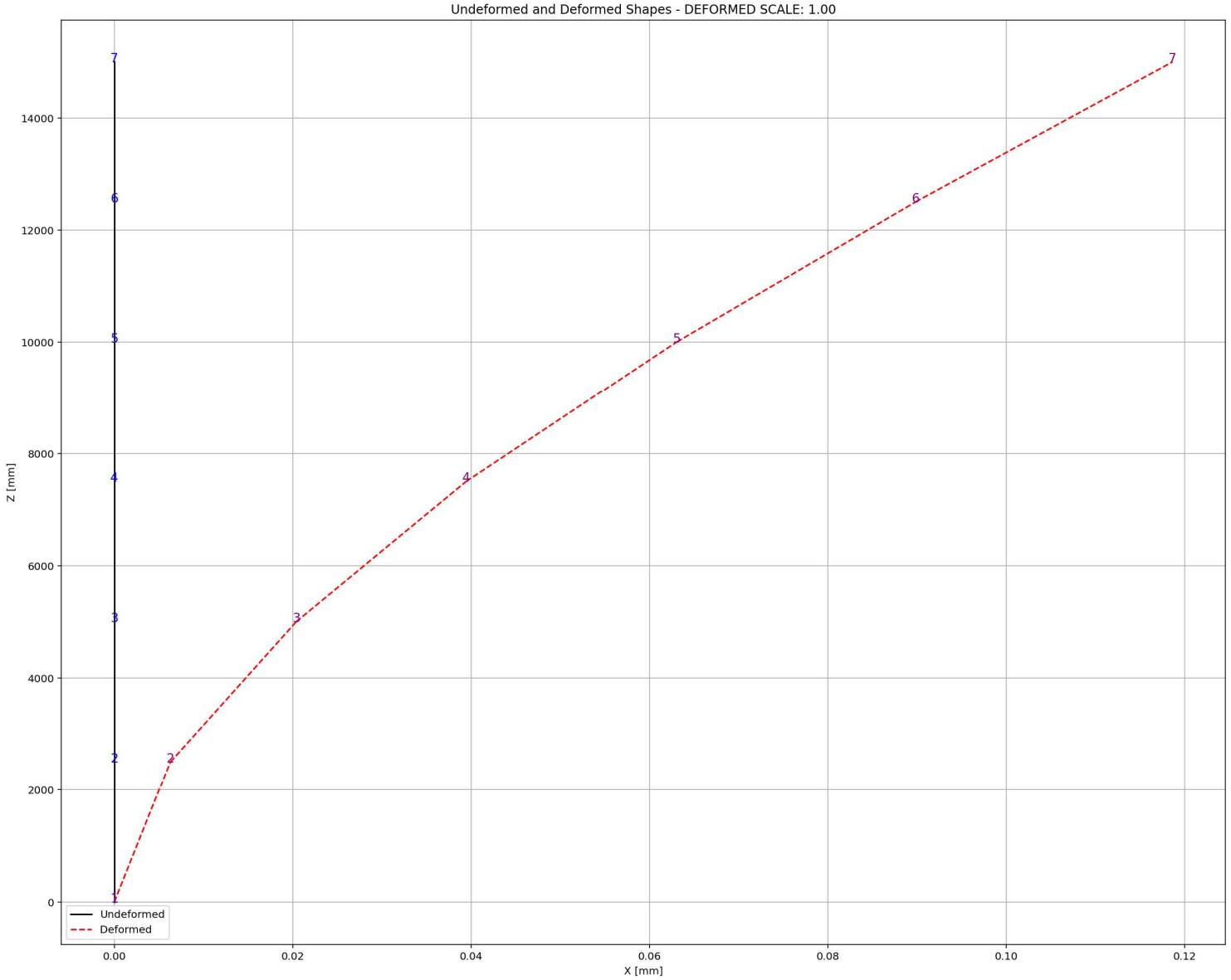

Period of Structure During Free-vibration Analysis
PERIOD - MIN VALUES: Min: 0.000 (s) - Mean: 0.000 (s) - Max: 0.000 (s)
PERIOD - MAX VALUES: Min: 0.001 (s) - Mean: 0.001 (s) - Max: 0.001 (s)
Structural Period [s]
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
0.0006
0.0007
0.0008
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.015
TOP FIBER - Stress: 0.031
Stress (N/mm^2)
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−2
−1
0
1
2
3
4
Strain (mm/mm)
1e−7


Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Displacment (mm)


Neutral Axis Curve During Analysis
NEUTRAL AXIS: 473.622
Neutral Axis (mm)
Displacment (mm)
700
600
500
400
300
200
100
0
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 7.6558$ [%]
$E_d = 4.378$ N·mm
$E_s = 4.551$ N·mm
Base Reaction [N]
Displacement [mm]
75
50
25
0
−25
−50
−75
−100
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Hysteresis Loop
(u_{max}, F_{max})
E_d = 4.378 N·m


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
2000
4000
6000
8000
10000
12000
14000
0.00
0.02
0.04
0.06
0.08
0.10
0.12
1
2
3
4
5
6
7
Undeformed
Deformed

SEISMIC ANALYSIS

Spyder (Python 3.12)

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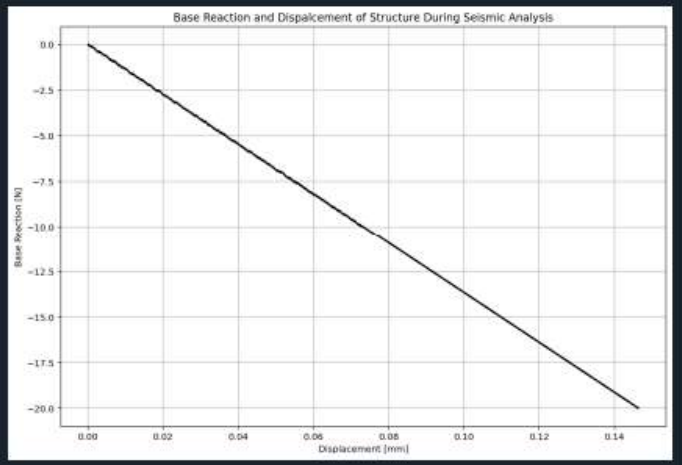
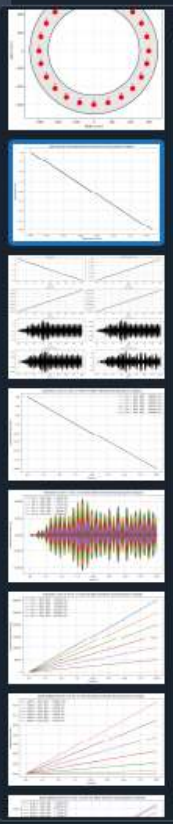
...NERGY_DISSIPATION_CAPACITY_INDEX\14_CONCRETE_COLUMN_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RETE_COLUMN_8_ANA\W(x,t)_CONCRETE_COLUMN_8_ANA.py

W(x,t)_CONCRETE_COLUMN_8_ANA.py | CONCRETE_CIRCULALR...SECTION_FUN_TWO.py

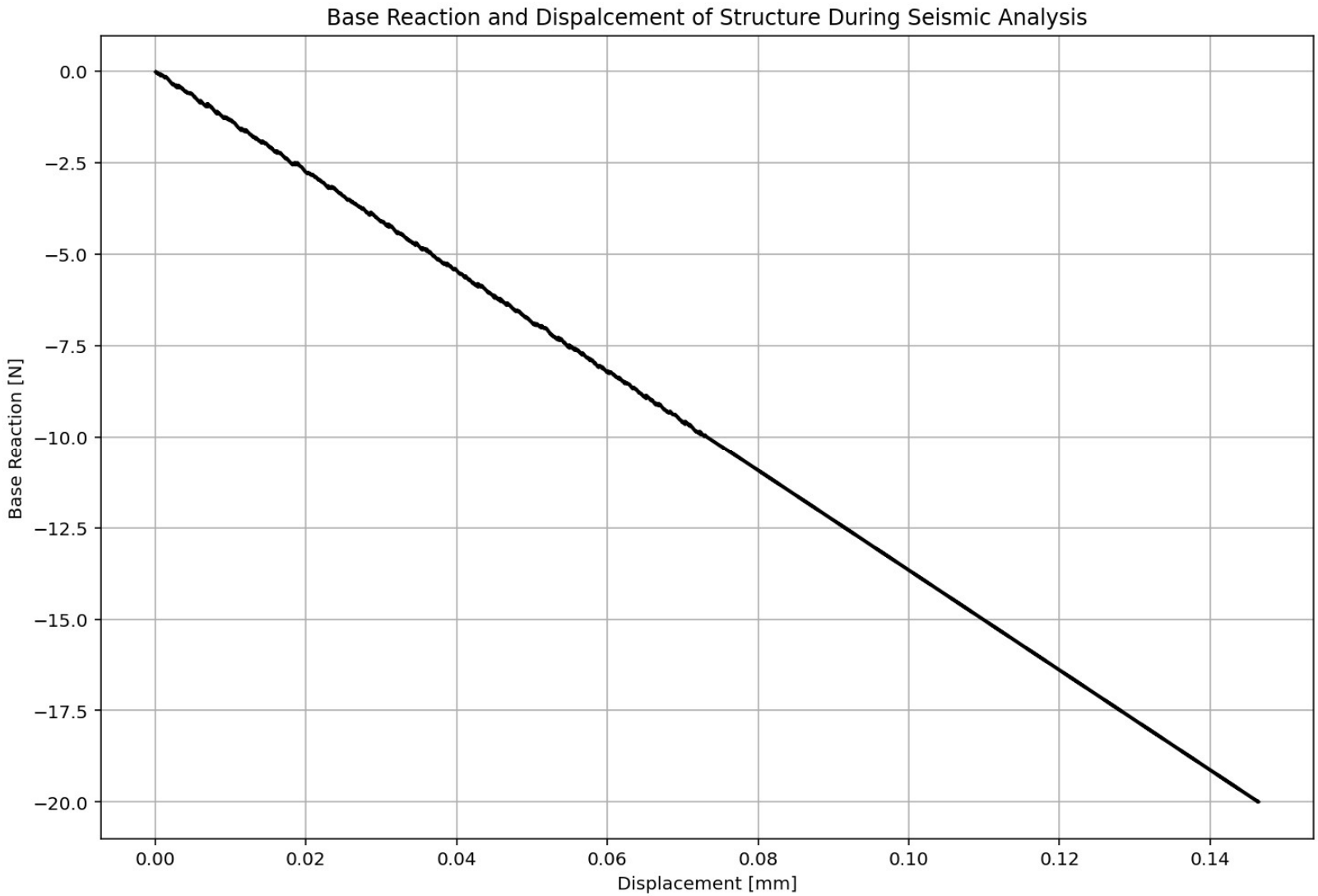
```
1020
1021    if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
1022        #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
1023        duration = 20.0                 # [s] Analysis duration
1024        dt = 0.01                       # [s] Time step
1025        GMfact = 9810                   # [mm/s²]standard acceleration of gravity or stand
1026        SSF_X = 50.0                    # Seismic Acceleration Scale Factor in X Direction
1027        SSF_Y = 50.0                    # Seismic Acceleration Scale Factor in Y Direction
1028        iv0_X = 0.0005                  # [mm/s] Initial velocity applied to the node  in
1029        iv0_Y = 0.0005                  # [mm/s] Initial velocity applied to the node  in
1030        st_iv0 = 0.0                    # [s] Initial velocity applied starting time
1031        SEI = 'X'                       # Seismic Direction
1032        DR = 0.03                       # Damping ratio
1033
1034        # Define time series for input motion (Acceleration time history)
1035        if SEI == 'X':
1036            SEISMIC_TAG_01 = 100
1037            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Accelerati
1038            # Define load patterns
1039            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fact $
1040            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01, '-v
1041        if SEI == 'Y':
1042            SEISMIC_TAG_02 = 200
1043            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Accelerati
1044            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02, '-v
1045        if SEI == 'XY':
1046            SEISMIC_TAG_01 = 100
1047            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Accelerati
1048            # Define load patterns
1049            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fact $
1050            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01, '-v
1051            SEISMIC_TAG_02 = 200
1052            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Accelerati
```

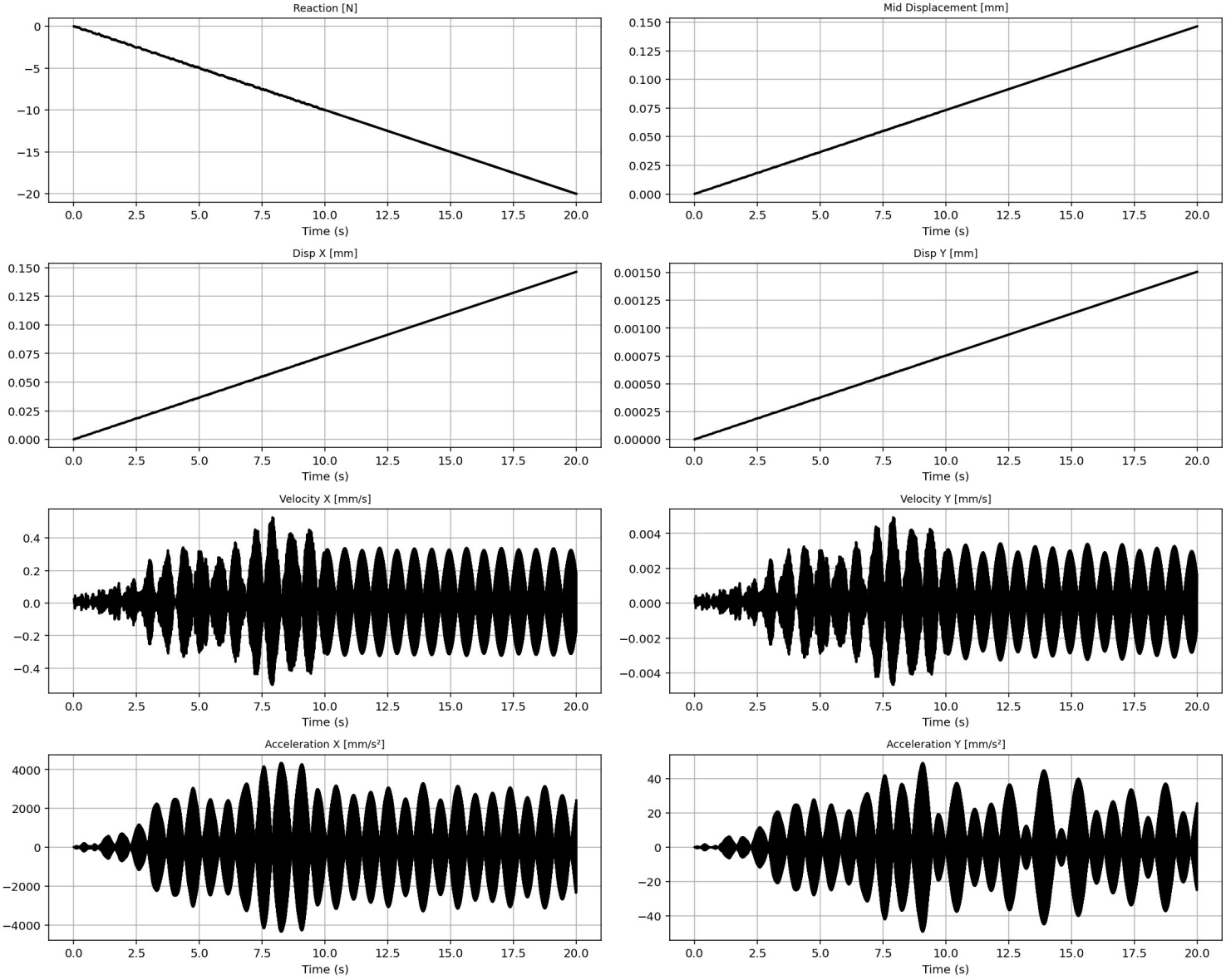


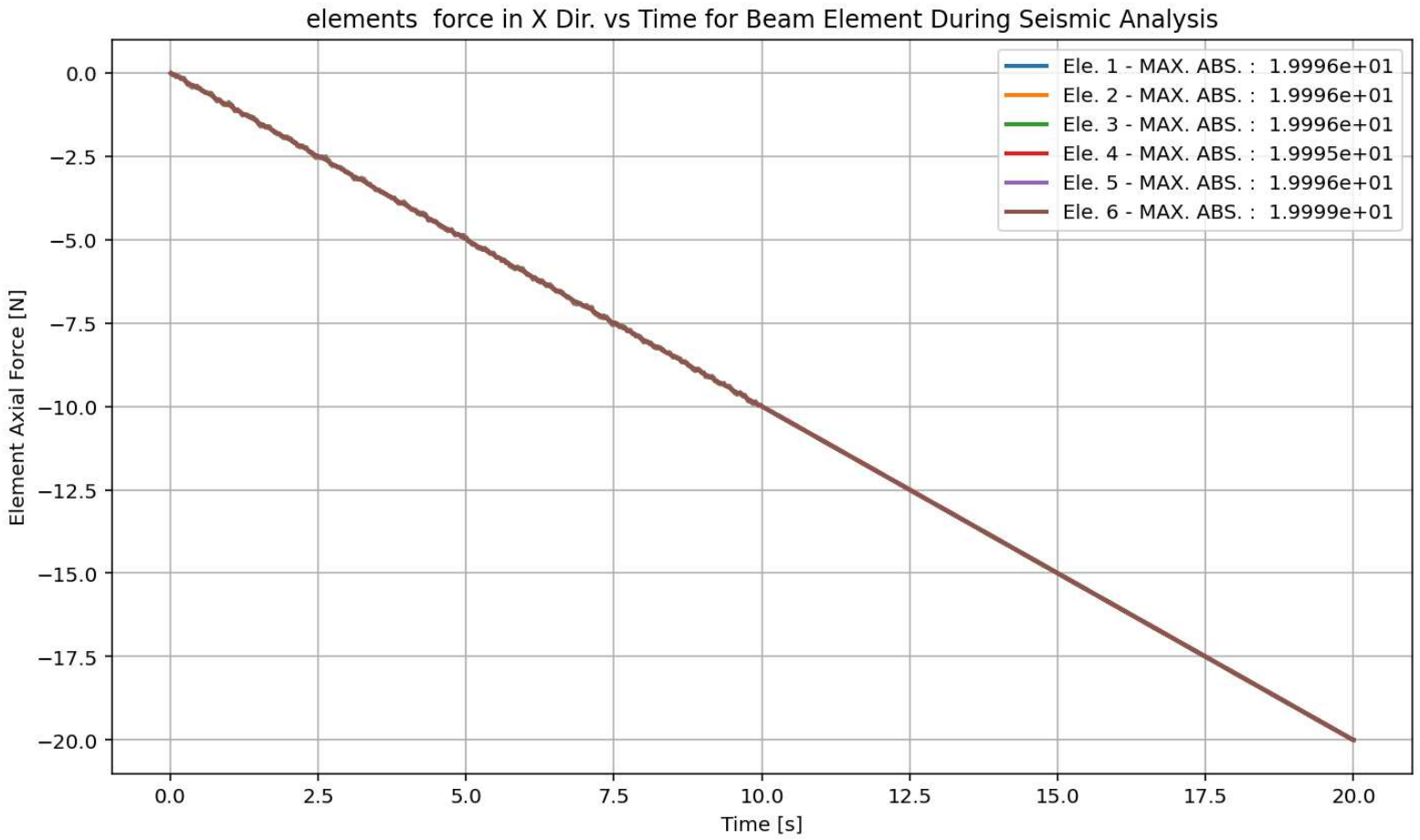


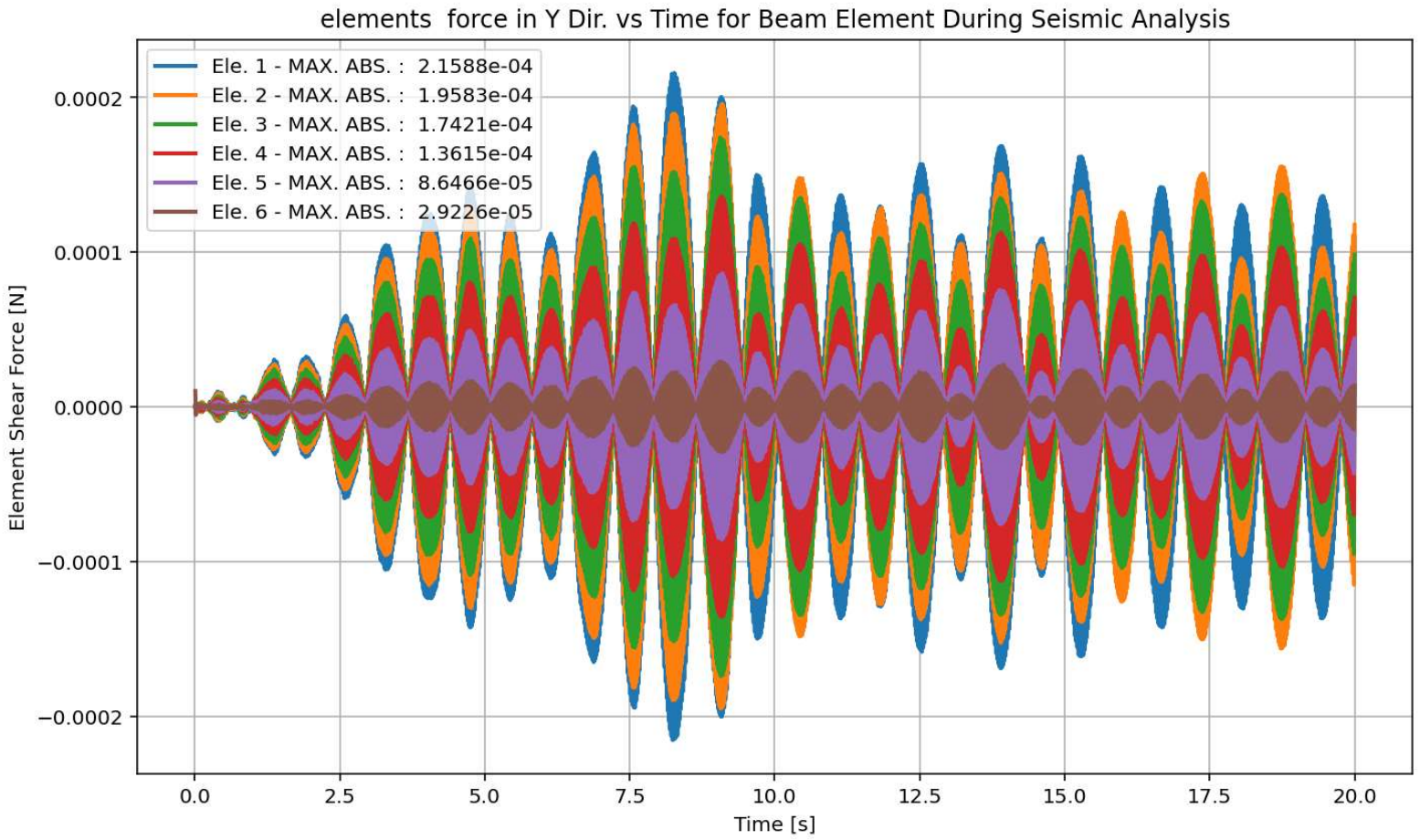
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

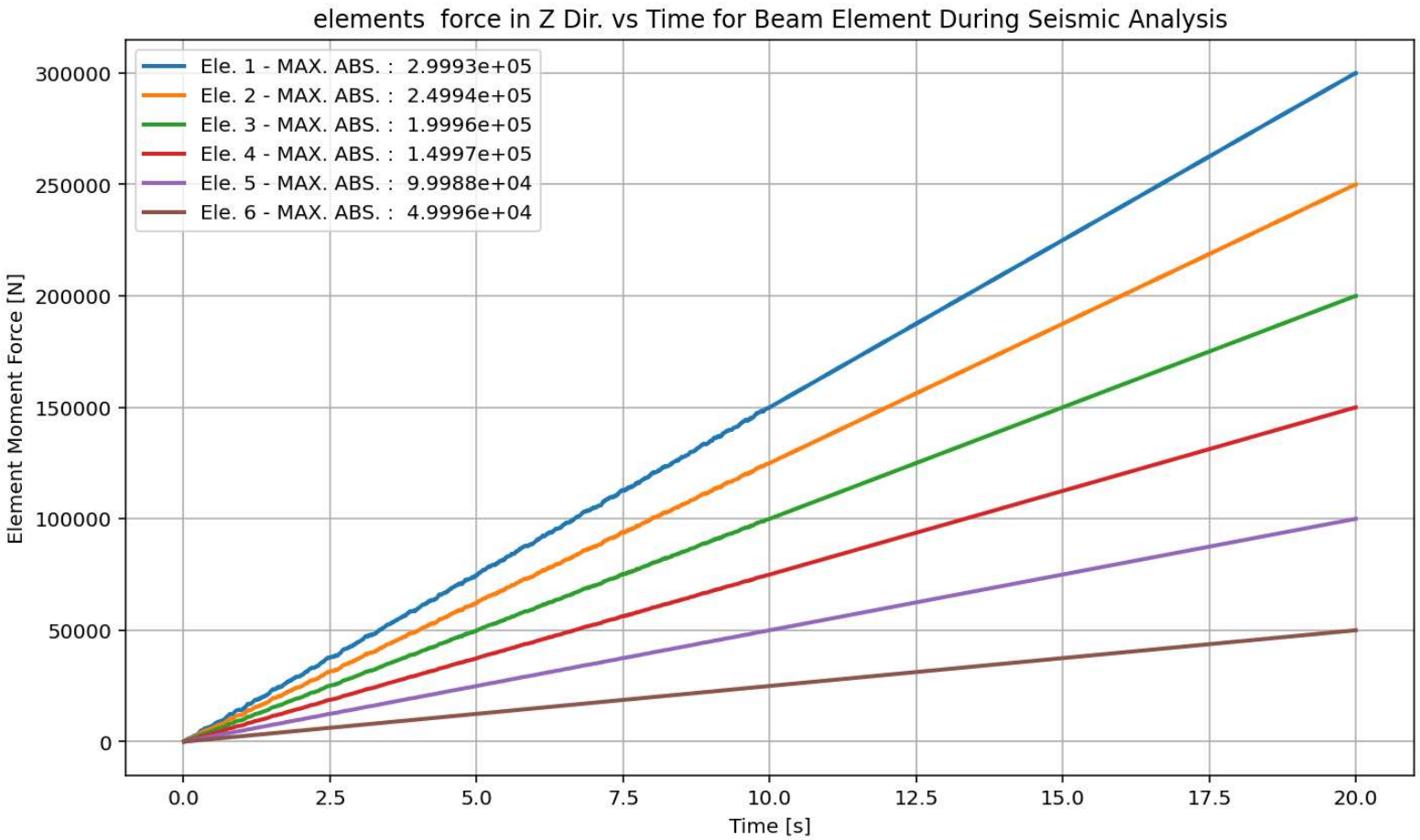
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1020, Col 5 | UTF-8 | CRLF | RW | Mem 48%

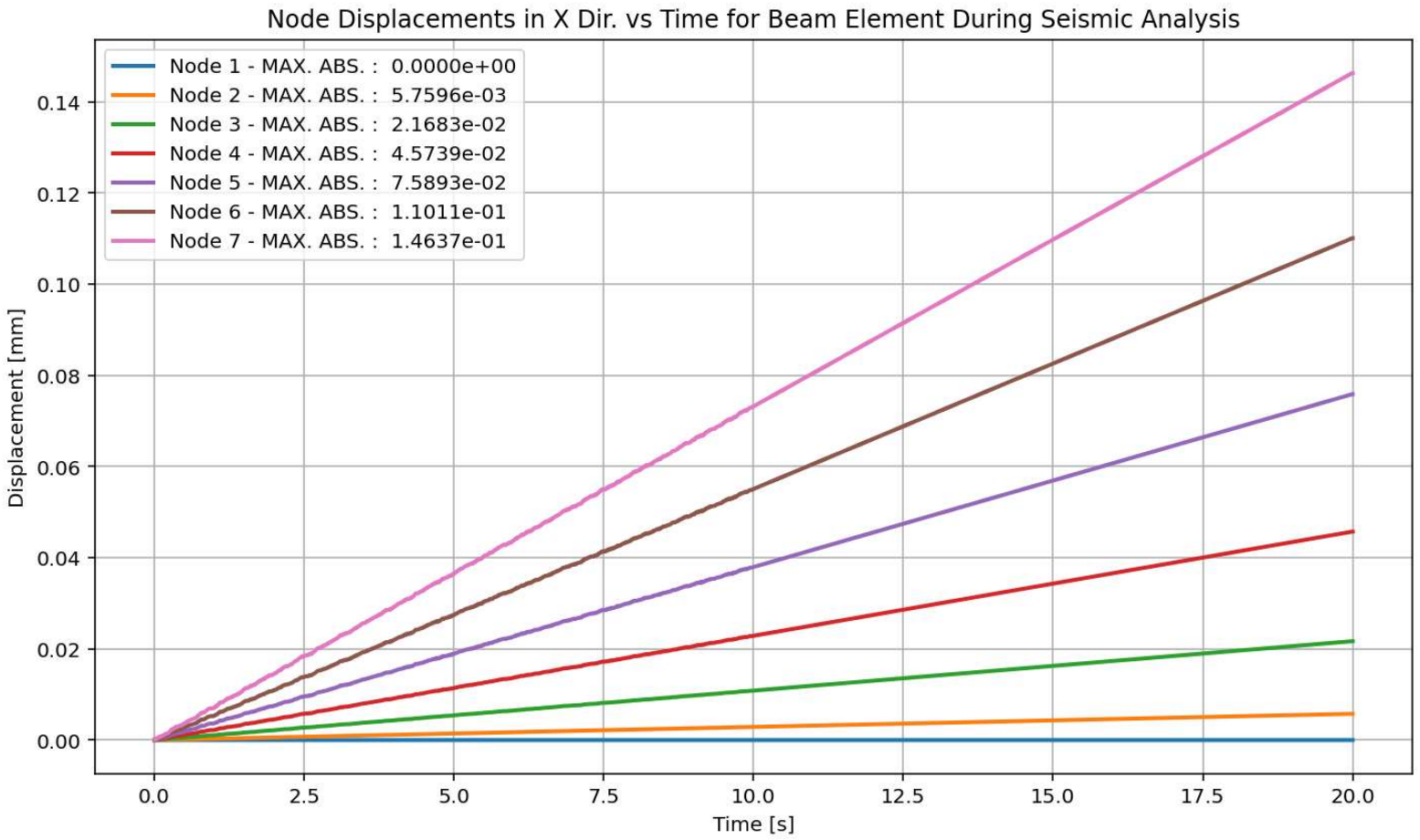

Base Reaction and Dispalcement of Structure During Seismic Analysis
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
Base Reaction [N]
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Displacement [mm]

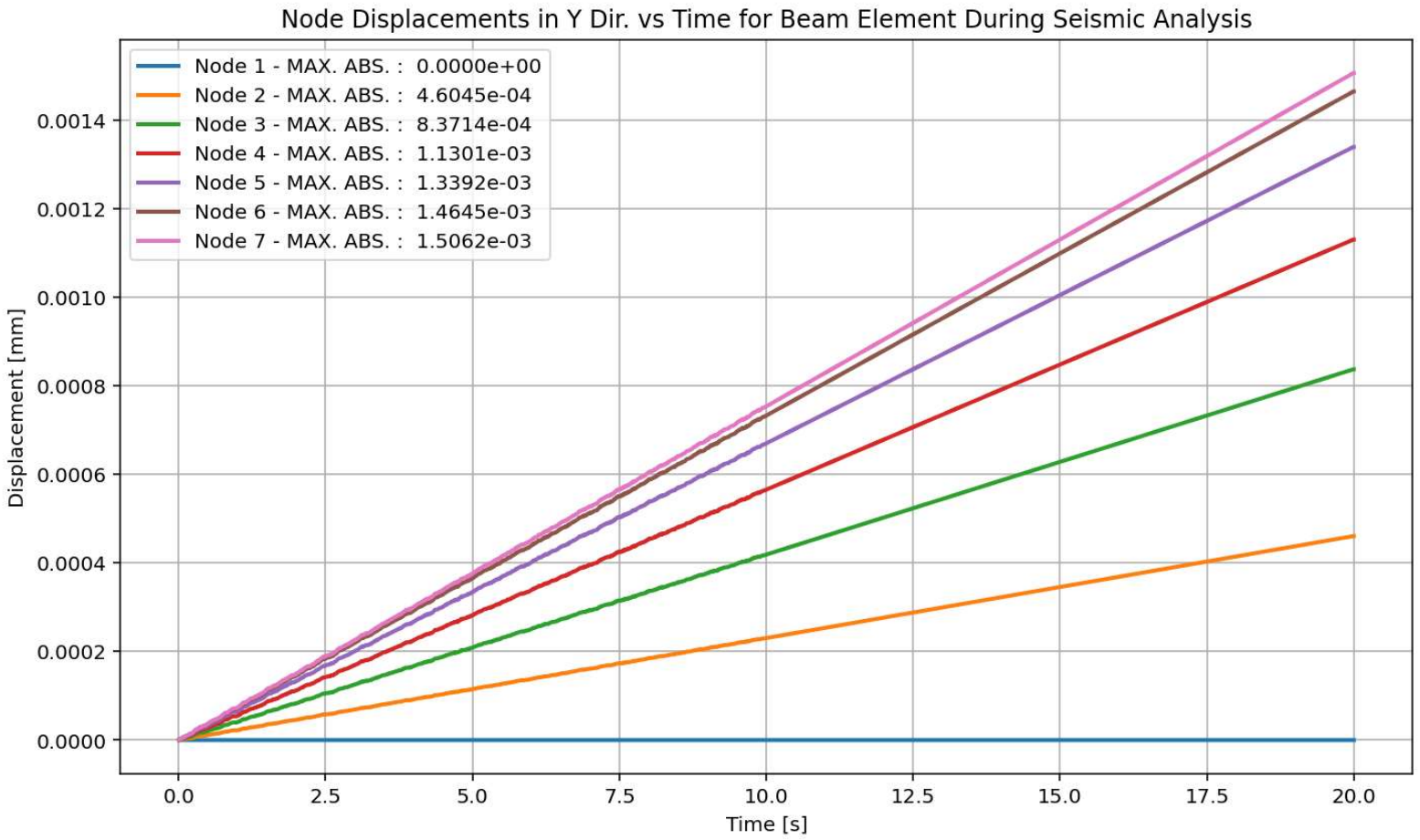

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

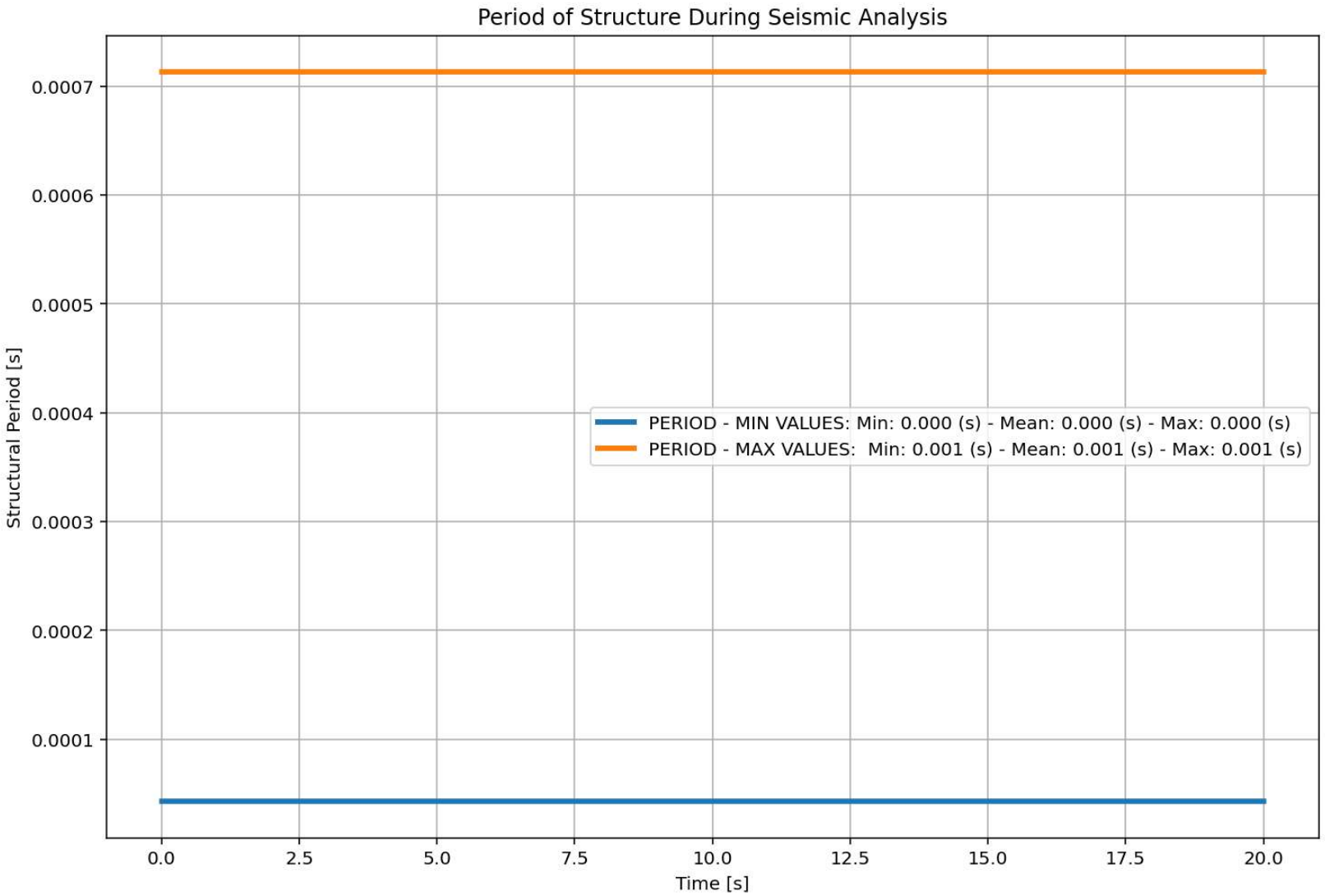

elements force in X Dir. vs Time for Beam Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.9996e+01
Ele. 2 - MAX. ABS. : 1.9996e+01
Ele. 3 - MAX. ABS. : 1.9996e+01
Ele. 4 - MAX. ABS. : 1.9995e+01
Ele. 5 - MAX. ABS. : 1.9996e+01
Ele. 6 - MAX. ABS. : 1.9999e+01
Element Axial Force [N]
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

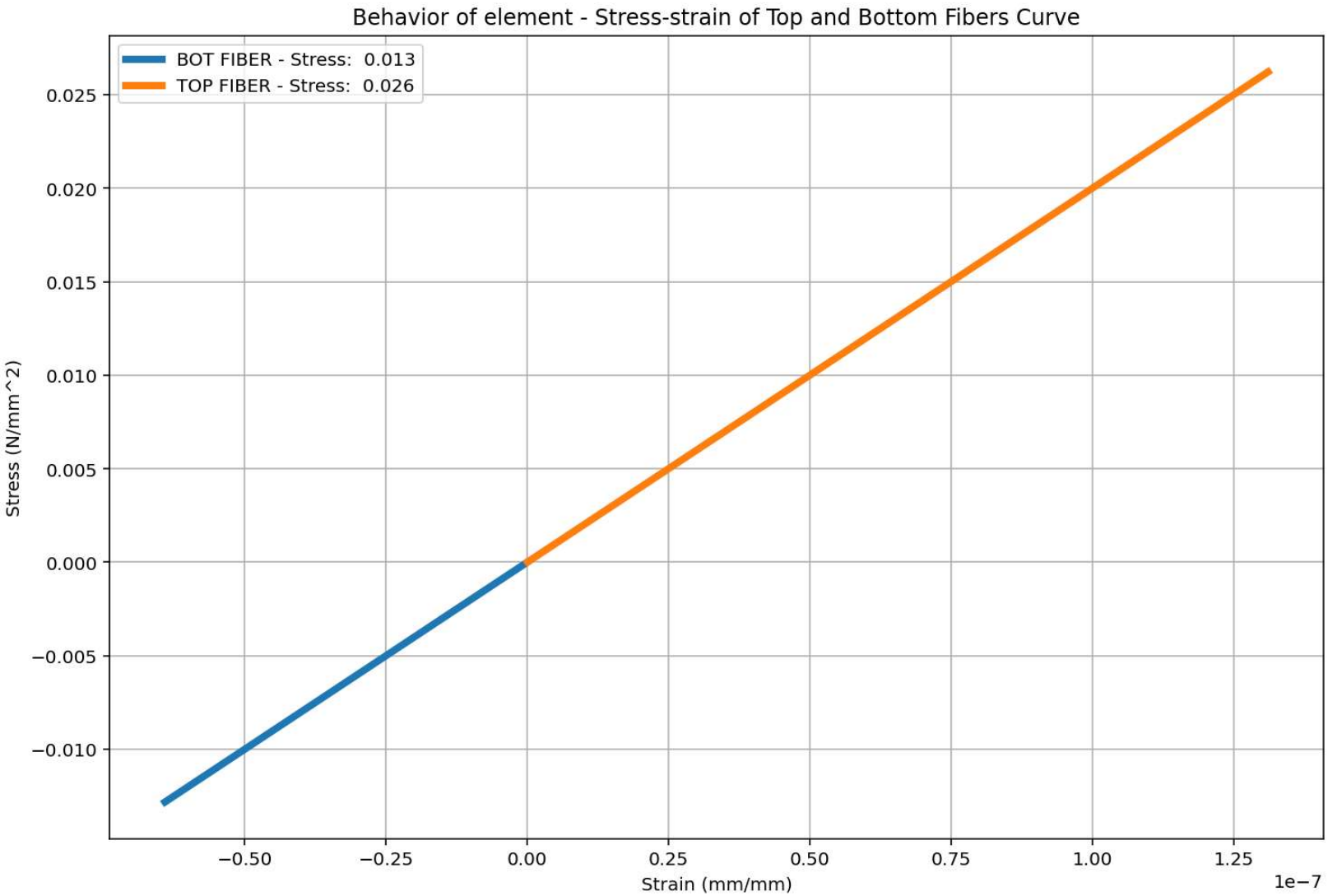

elements force in Y Dir. vs Time for Beam Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 2.1588e-04
Ele. 2 - MAX. ABS. : 1.9583e-04
Ele. 3 - MAX. ABS. : 1.7421e-04
Ele. 4 - MAX. ABS. : 1.3615e-04
Ele. 5 - MAX. ABS. : 8.6466e-05
Ele. 6 - MAX. ABS. : 2.9226e-05
Element Shear Force [N]
0.0002
0.0001
0.0000
−0.0001
−0.0002
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

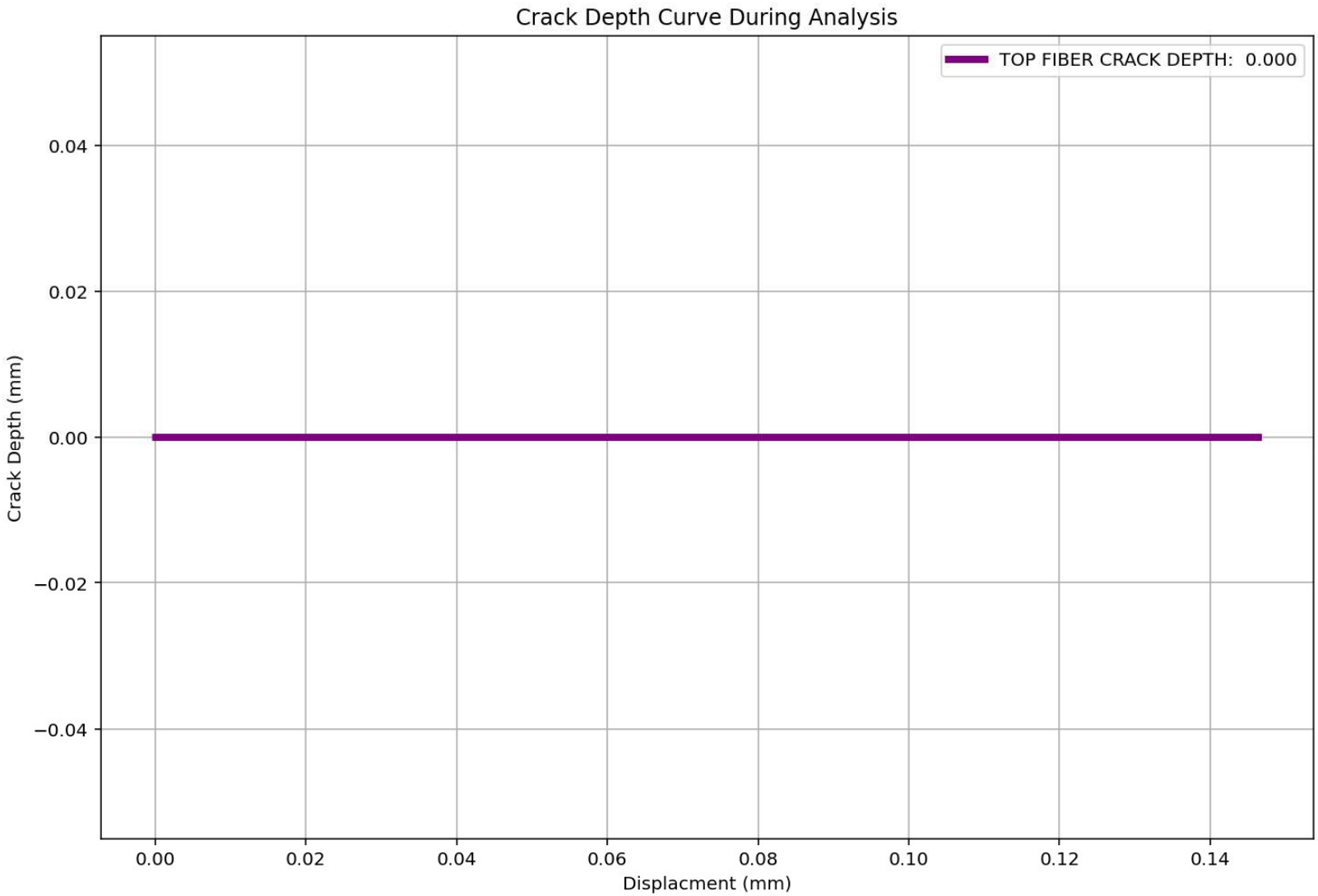

elements force in Z Dir. vs Time for Beam Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 2.9993e+05
Ele. 2 - MAX. ABS. : 2.4994e+05
Ele. 3 - MAX. ABS. : 1.9996e+05
Ele. 4 - MAX. ABS. : 1.4997e+05
Ele. 5 - MAX. ABS. : 9.9988e+04
Ele. 6 - MAX. ABS. : 4.9996e+04
Element Moment Force [N]
300000
250000
200000
150000
100000
50000
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

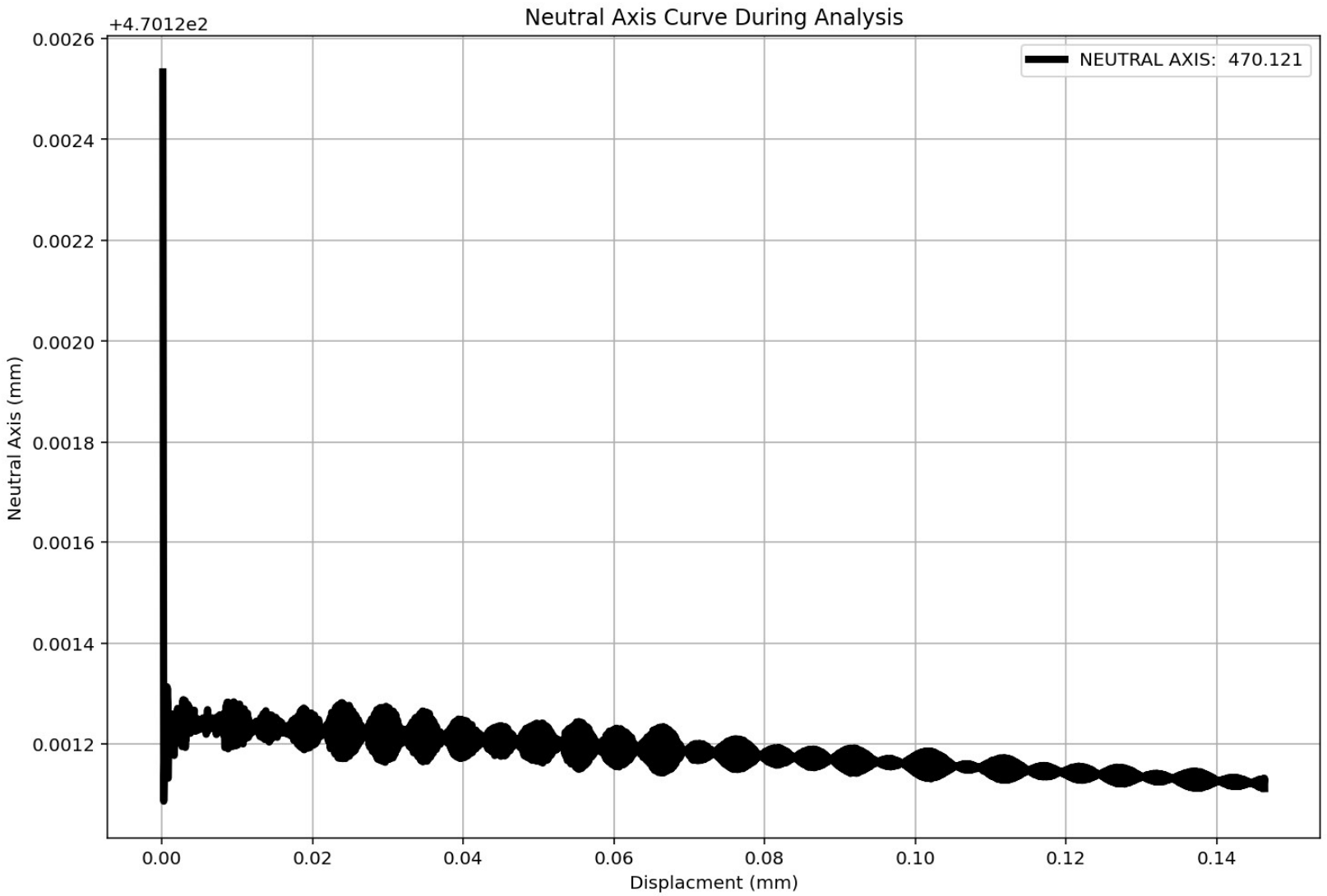

Node Displacements in X Dir. vs Time for Beam Element During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.7596e-03
Node 3 - MAX. ABS. : 2.1683e-02
Node 4 - MAX. ABS. : 4.5739e-02
Node 5 - MAX. ABS. : 7.5893e-02
Node 6 - MAX. ABS. : 1.1011e-01
Node 7 - MAX. ABS. : 1.4637e-01
Displacement [mm]
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Time [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

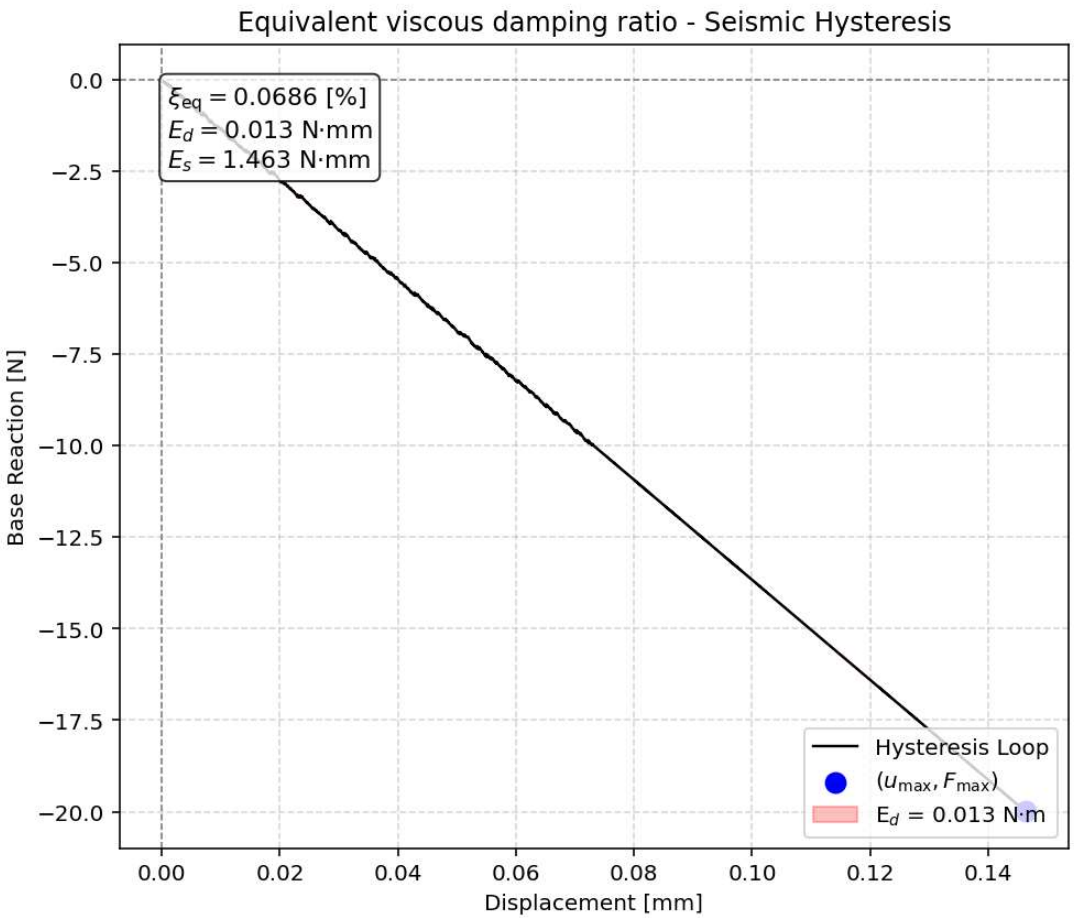

Node Displacements in Y Dir. vs Time for Beam Element During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 4.6045e-04
Node 3 - MAX. ABS. : 8.3714e-04
Node 4 - MAX. ABS. : 1.1301e-03
Node 5 - MAX. ABS. : 1.3392e-03
Node 6 - MAX. ABS. : 1.4645e-03
Node 7 - MAX. ABS. : 1.5062e-03
Displacement [mm]
0.0000
0.0002
0.0004
0.0006
0.0008
0.0010
0.0012
0.0014
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

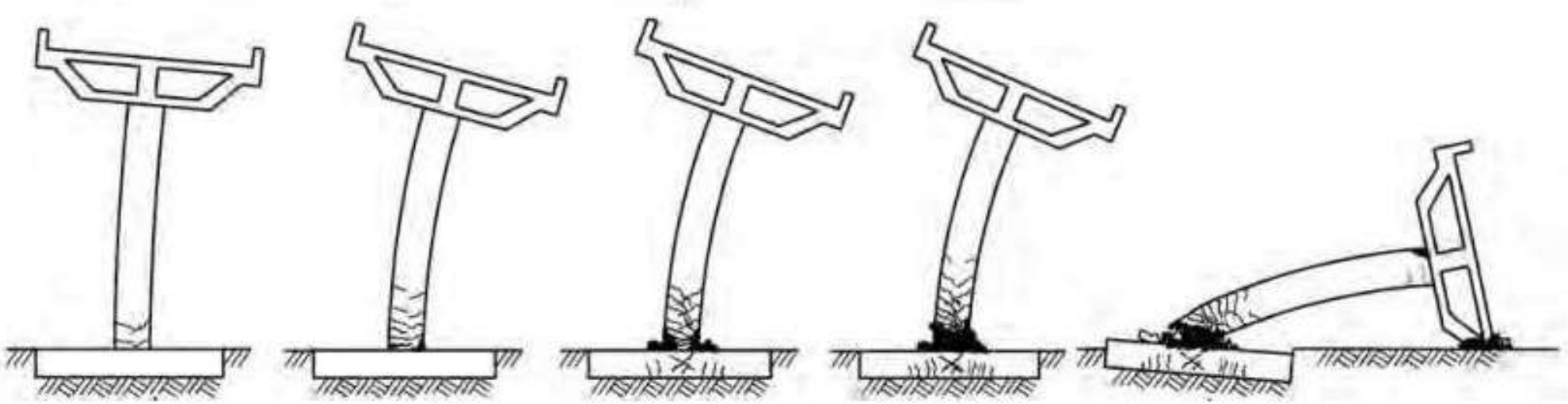

Period of Structure During Seismic Analysis
Structural Period [s]
0.0007
0.0006
0.0005
0.0004
0.0003
0.0002
0.0001
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.000 (s) - Mean: 0.000 (s) - Max: 0.000 (s)
PERIOD - MAX VALUES: Min: 0.001 (s) - Mean: 0.001 (s) - Max: 0.001 (s)

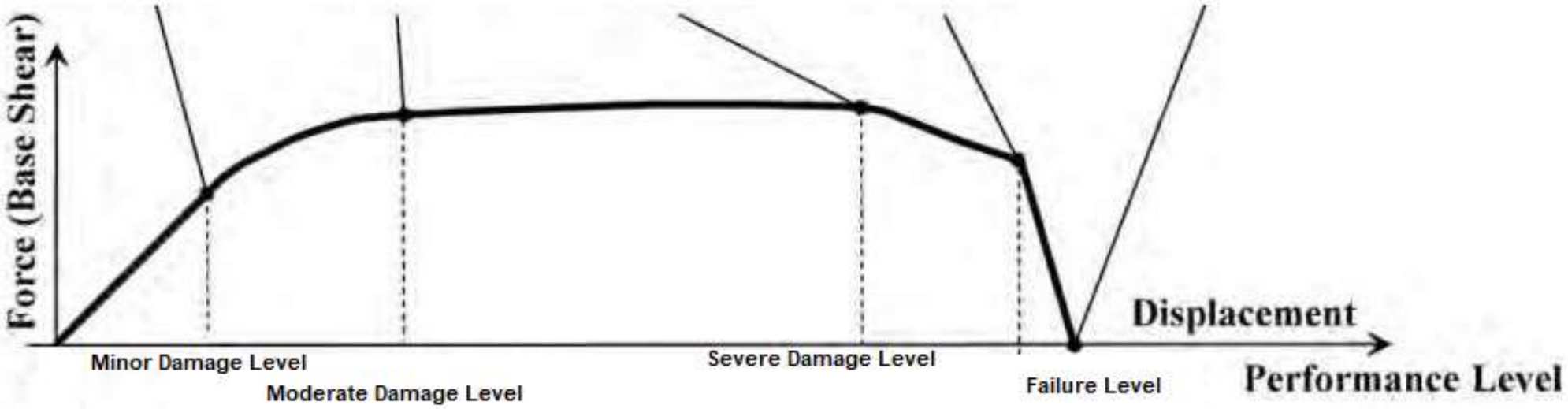

Behavior of element - Stress-strain of Top and Bottom Fibers Curve
BOT FIBER - Stress: 0.013
TOP FIBER - Stress: 0.026
Stress (N/mm^2)
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
Strain (mm/mm)
1e−7

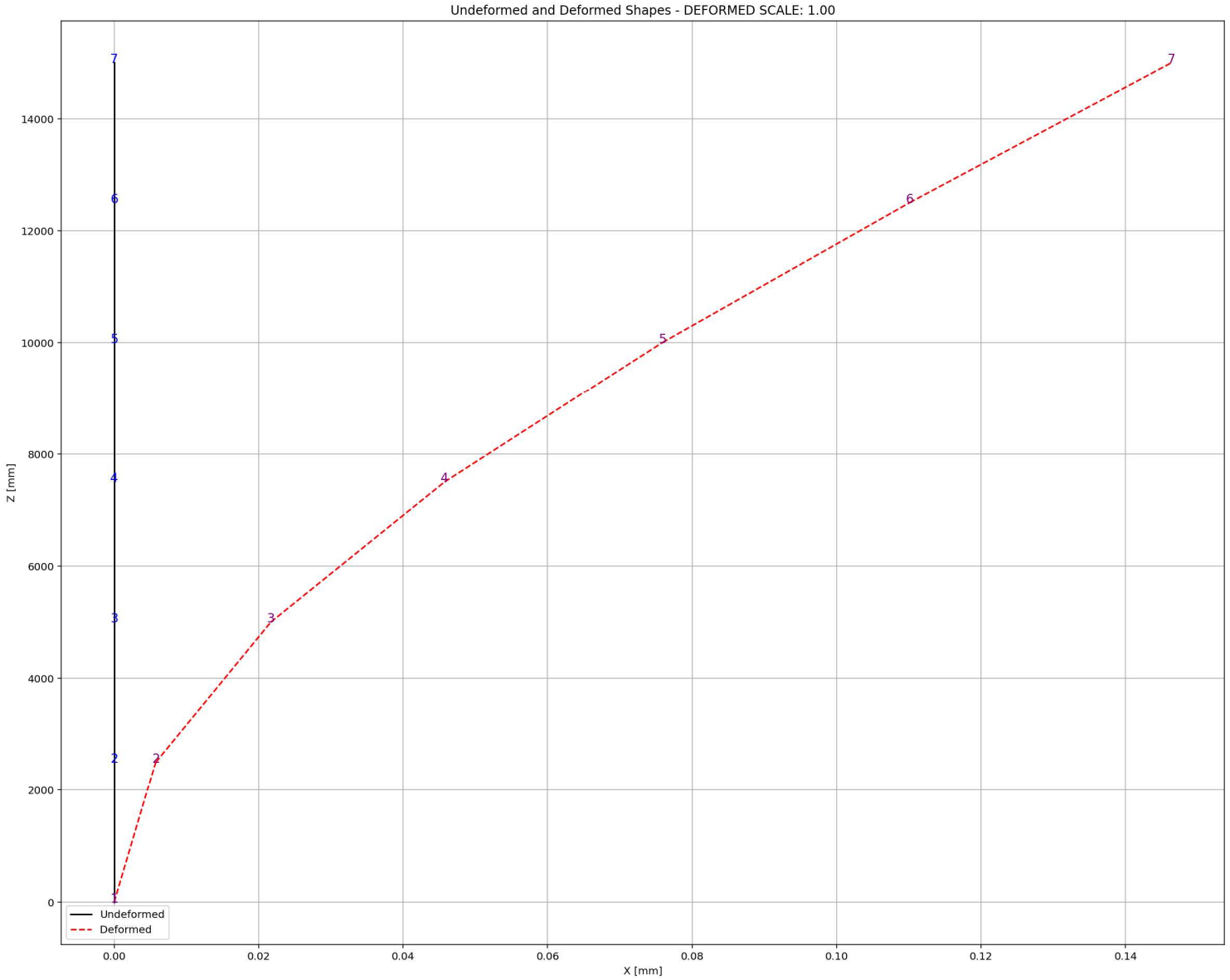

Crack Depth Curve During Analysis
TOP FIBER CRACK DEPTH: 0.000
0.04
0.02
0.00
−0.02
−0.04
Crack Depth (mm)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Displacment (mm)

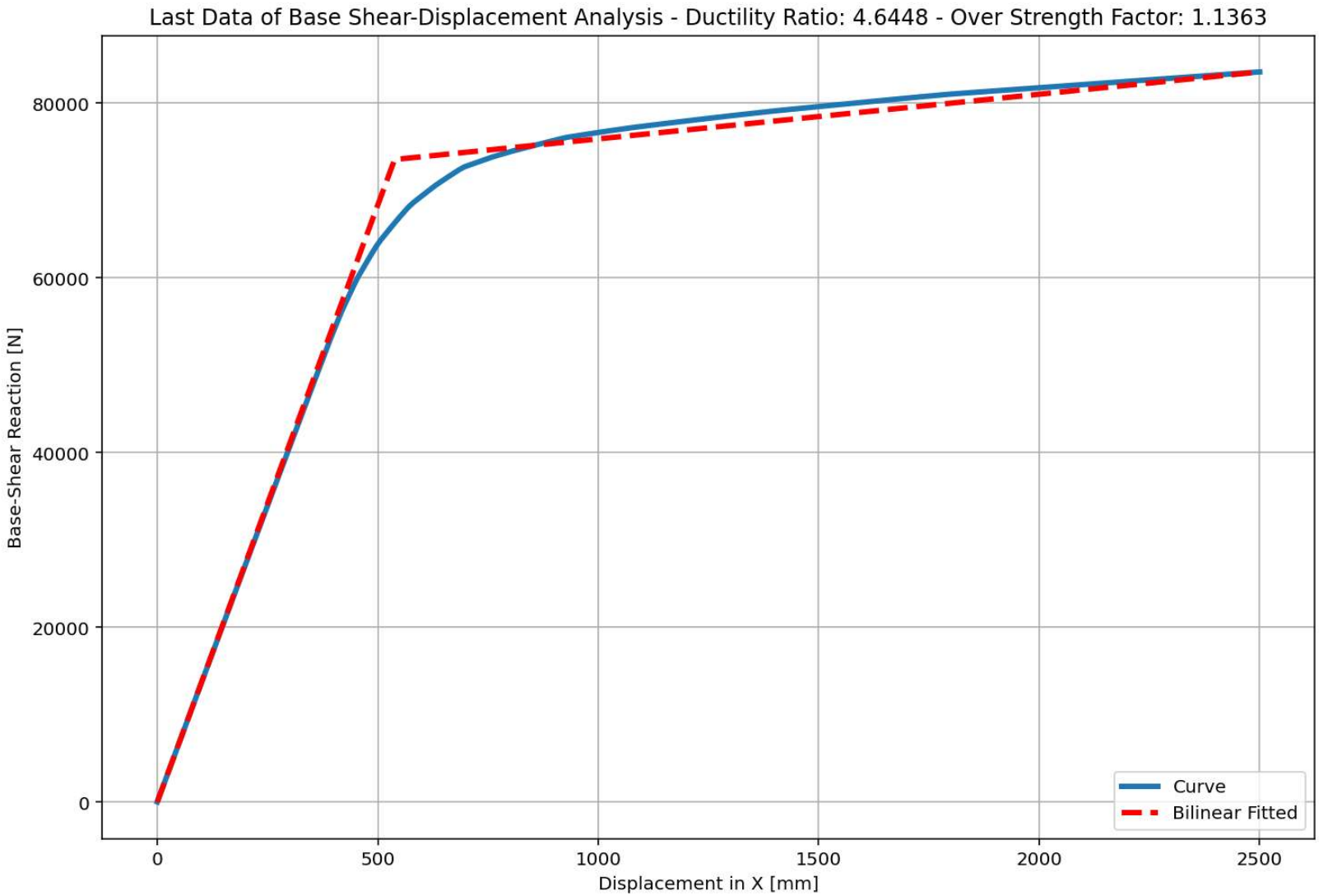

Neutral Axis Curve During Analysis
+4.7012e2
NEUTRAL AXIS: 470.121
Neutral Axis (mm)
0.0026
0.0024
0.0022
0.0020
0.0018
0.0016
0.0014
0.0012
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Displacment (mm)


Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 0.0686$ [%]
$E_d = 0.013$ N·mm
$E_s = 1.463$ N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.013 N·m
Base Reaction [N]
Displacement [mm]
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14

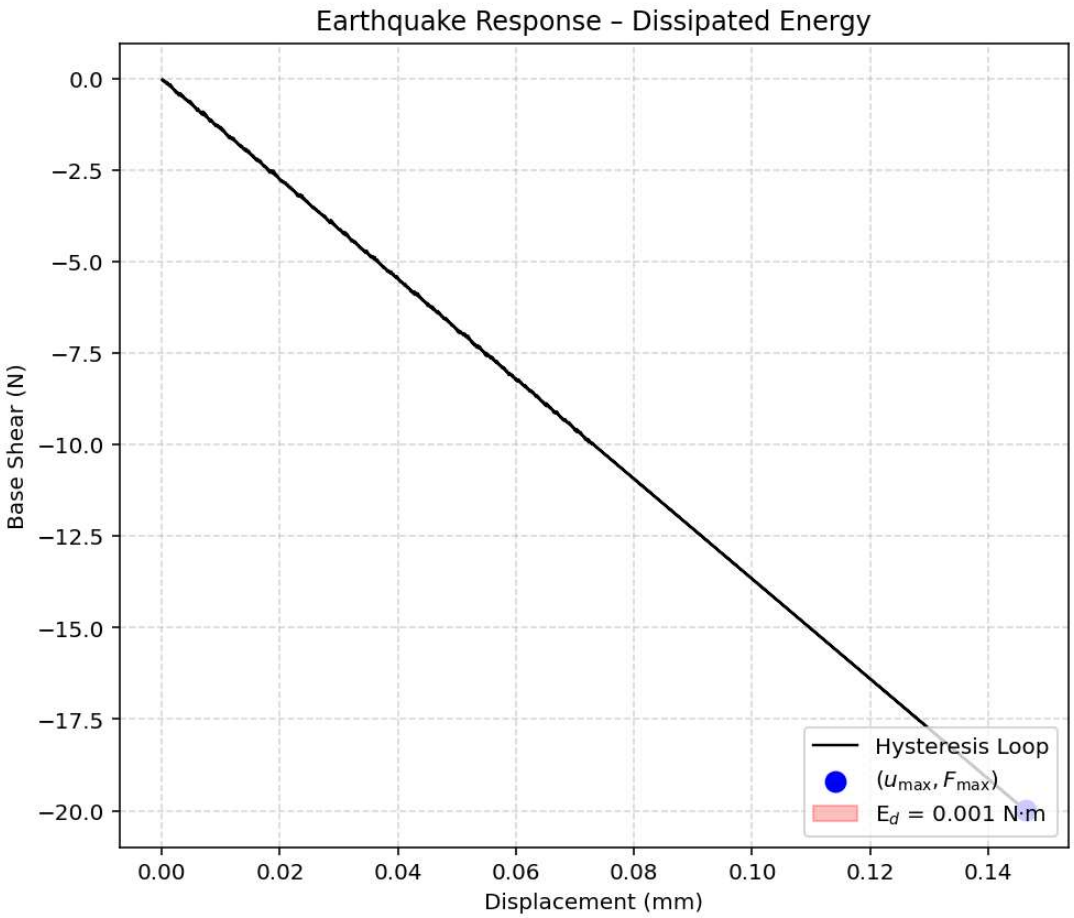


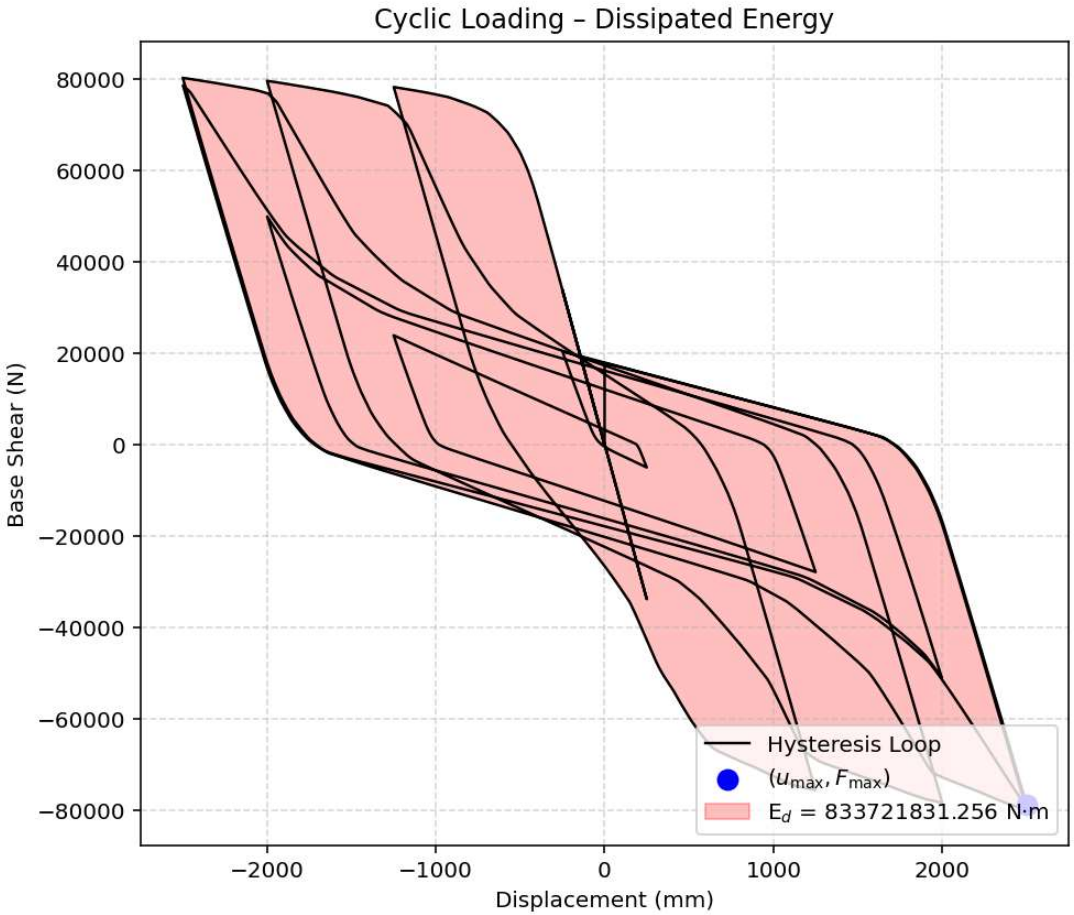

Force (Base Shear)
Displacement
Performance Level
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
2000
4000
6000
8000
10000
12000
14000
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
1
2
3
4
5
6
7
Undeformed
Deformed


Last Data of Base Shear-Displacement Analysis - Ductility Ratio: 4.6448 - Over Strength Factor: 1.1363
Base-Shear Reaction [N]
80000
60000
40000
20000
0
0
500
1000
1500
2000
2500
Displacement in X [mm]
Curve
Bilinear Fitted

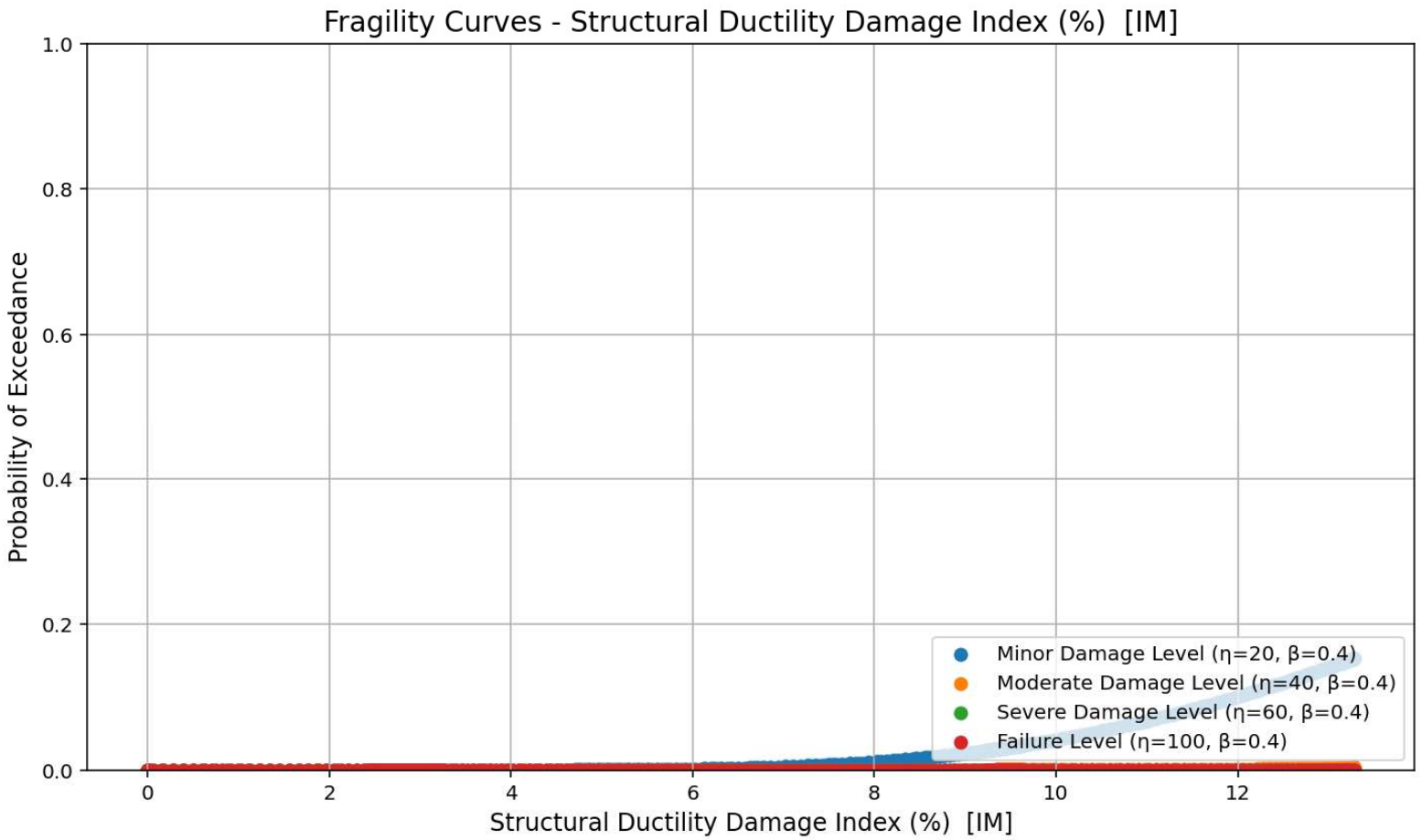
```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
----------------------------
[[   0.          0.       ]
 [  538.23218671 73528.95054126]
 [ 2500.        83552.50874925]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    136.61
 Structure Plastic Stiffness :    33.42
 Structure Tangent Stiffness :    5.11
 Structure Ductility Ratio :      4.64
 Structure Over Strength Factor:  1.14
+-------------------------------------------------+
Over Strength Coefficient (Ω0):     1.1363
Displacement Ductility Ratio (μ):   4.6448
Ductility Coefficient (Rμ):         4.6448
Structural Behavior Coefficient (R): 5.2780
Structural Ductility Damage Index in X Direction: -27.4286 (%)
```


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.001 N·m


Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (mm)
80000
60000
40000
20000
0
−20000
−40000
−60000
−80000
−2000
−1000
0
1000
2000
Hysteresis Loop
(u_{max}, F_{max})
E_d = 833721831.256 N·m

Dissipated Energy from Earthquake= 0.00 N·mm
Dissipated Energy from Cyclic Displacement= 833721831.26 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.000 [%]
ZONE 1: VERY MINOR DAMAGE


Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
Structural Ductility Damage Index (%) [IM]
Minor Damage Level (η=20, β=0.4)
Moderate Damage Level (η=40, β=0.4)
Severe Damage Level (η=60, β=0.4)
Failure Level (η=100, β=0.4)